



Installation Document

Oracle Database 19C and Forms 12C



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SYSTEM REQUIREMENTS

Prerequisite for Operating System

- 64bit Windows Server 2008, 2012, 2016
- 64bit Windows 7, 8.1, 10 Professional.

Minimum RAM required: 8,192 MB (8GB).

- **Required Software:** - (Download the required software from Oracle Website.)
- Database Setup (WINDOWS.X64_193000_db_home).
- Java Development Kit (jdk-8u281-windows-x64.exe) Latest version can be used.
- Fusion Middleware Infrastructure Installer (fmw_12.2.1.4.0_infrastructure.jar)
- Oracle Forms and Reports 12c (12.2.4.0) (setup_fmw_12.2.1.4.0_fr_win64)

PRE INSTALLATION SETUP

HOST FILE

Make sure the "C:\Windows\System32\drivers\etc\hosts" file contains correct entries for both the "local host" and real host names.

127.1.1.1 Localhost

10.0.2.15 DEMO12C /*IP of Machine Name of Machine*/



```
hosts - Notepad
File Edit Format View Help
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#       102.54.94.97       rhino.acme.com       # source server
#       38.25.63.10       x.acme.com           # x client host
#
# localhost name resolution is handled within DNS itself.
#       127.0.0.1         localhost
#       ::1               localhost
#
10.0.2.15 DEMO12C
127.0.0.1 DEMO12C
```

ORACLE_HOSTNAME Environment Variable (Important)

Display System in the Windows Control Panel. In the System Properties dialog box, click Advanced. In Environment Variables dialog box, under System Variables, click New.

In the New System Variable dialog box, enter the following information:

Variable name: ORACLE_HOSTNAME

Variable value: The host name of the computer to use.

Click OK, then in the Environment Variables dialog box, click OK.

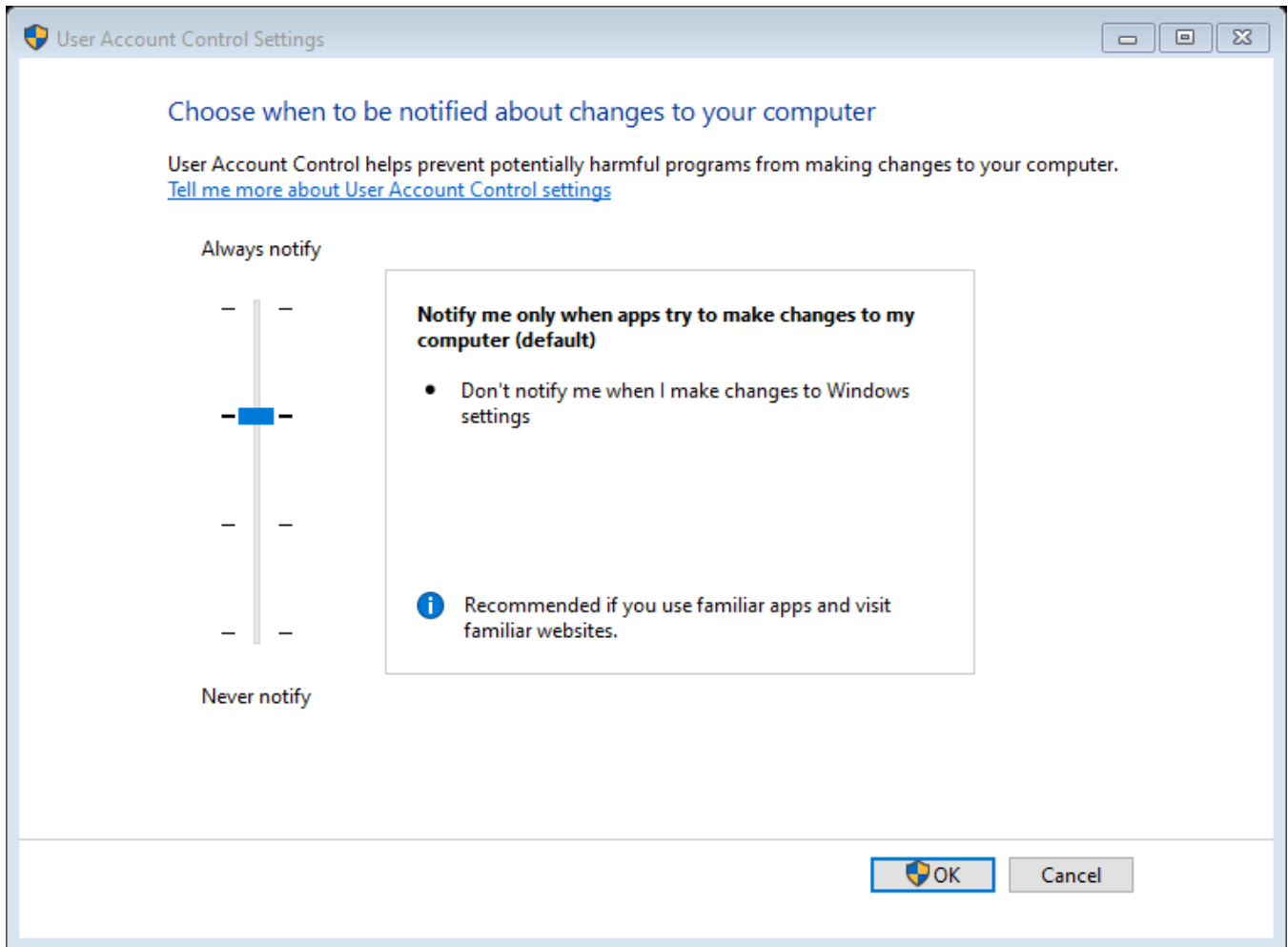
Click OK in the Environment Variables dialog box, then in the System Properties dialog box, click OK.

Reboot machine.



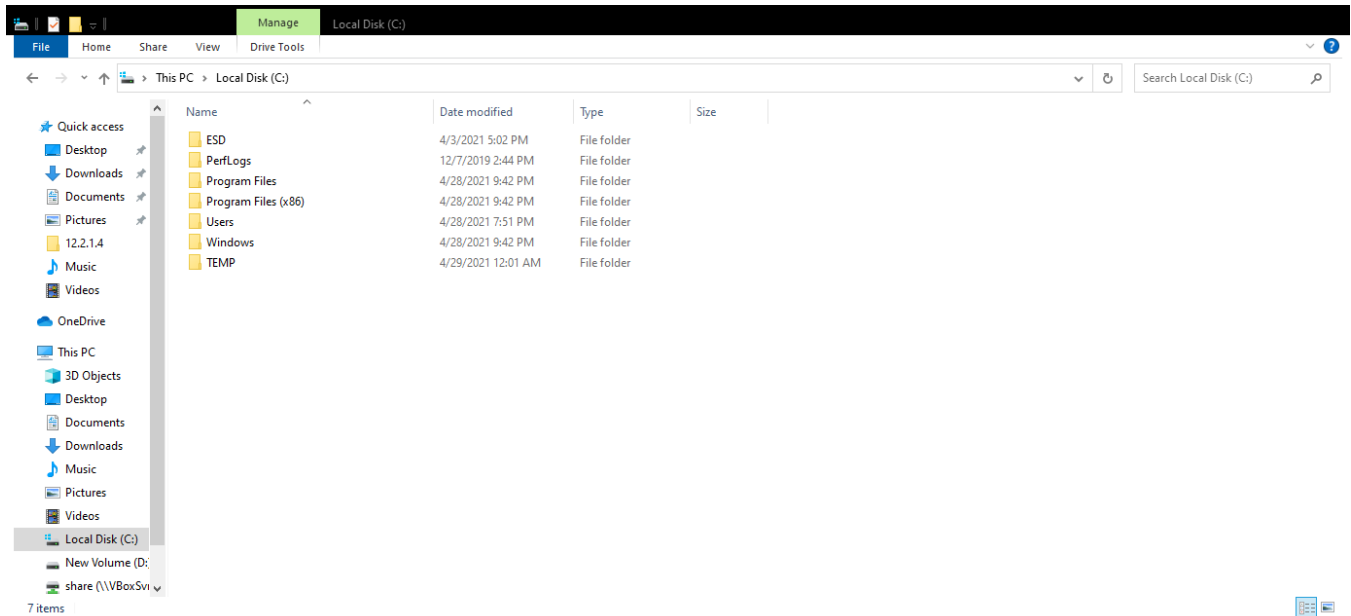
ENVIRONMENT VARIABLES

Disable notifications from Control Panel > User Accounts > User Account Settings.





Create directory “C:\temp”.



Change default temp location to the c:\temp

Control Panel -> System -> Advanced system settings -> Environment Variables

Edit both TEMP and TMP and change the value to c:\temp



System Properties

Computer Name Hardware **Advanced** System Protection Remote

You must be logged on as an Administrator to make most of these changes.

Performance
Visual effects, processor scheduling, memory usage, and virtual memory
Settings...

User Profiles
Desktop settings related to your sign-in
Settings...

Startup and Recovery
System startup, system failure, and debugging information
Settings...

Environment Variables...

OK Cancel Apply

Environment Variables

User variables for Administrator

Variable	Value
OneDrive	C:\Users\Administrator\OneDrive
Path	C:\Users\Administrator\AppData\Local\Microsoft\WindowsApps;
TEMP	C:\TEMP
TMP	C:\TEMP

New... Edit... Delete

System variables

Variable	Value
ComSpec	C:\WINDOWS\system32\cmd.exe
DriverData	C:\Windows\System32\Drivers\DriverData
FP_NO_HOST_CHECK	NO
NUMBER_OF_PROCESSORS	3
OS	Windows_NT
Path	C:\WINDOWS\system32;C:\WINDOWS;C:\WINDOWS\System32\Wb...
PATHEXT	.COM;.EXE;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;.WSH;.MSC

New... Edit... Delete

OK Cancel



DATABASE INSTALLATION

Step 1

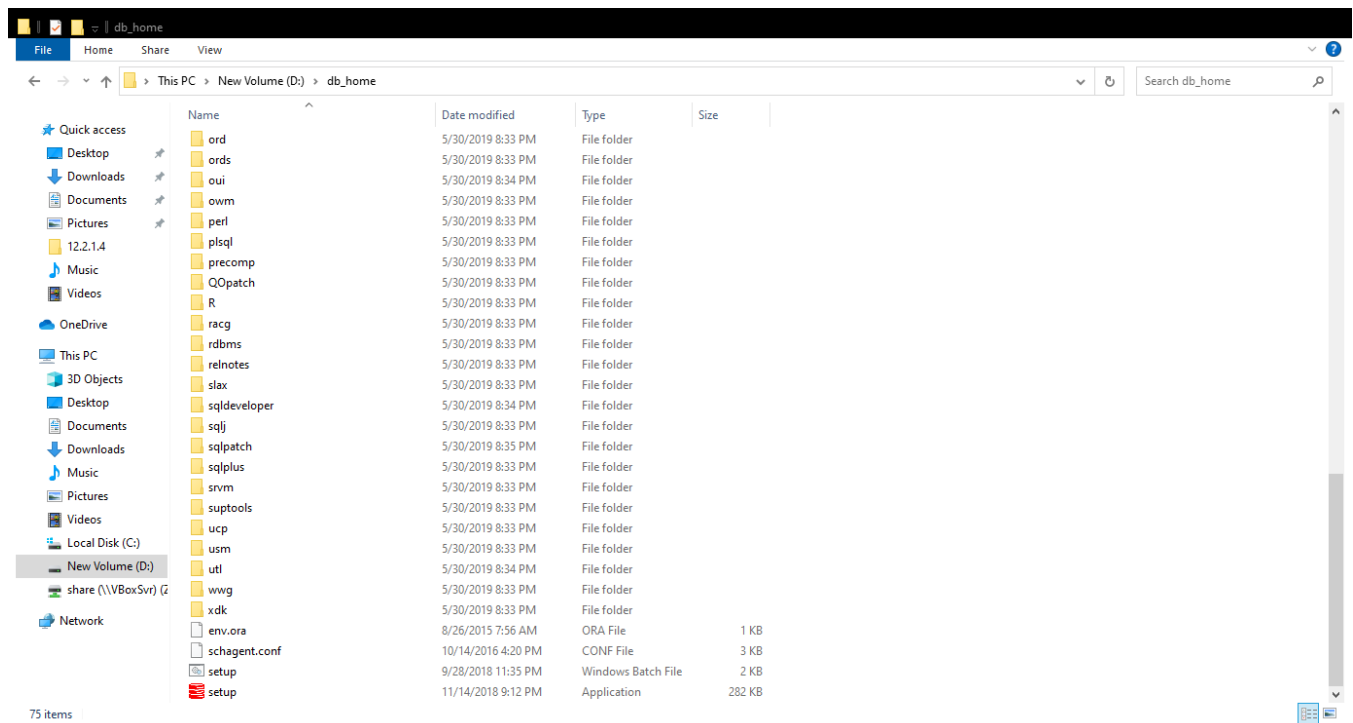
Extract “WINDOWS.X64_193000_db_home” to d:\

Rename “WINDOWS.X64_193000_db_home” as “db_home”.

It should show as “d:\db_home”

Step 2

Get inside the folder “d:\db_home”. Run the Setup.exe





STEP 3 – Installation Wizard

Select “Create and configure a single instance database” option and click on the Next button to move on to System Class selection page:

Select Configuration Option

Configuration Option

Database Installation Options

Install Type

Oracle Home User

Installation Location

Configuration Type

Database Identifiers

Configuration Options

Database Storage

Management Options

Recovery Options

Schema Passwords

Prerequisite Checks

Summary

Install Product

Finish

19^c ORACLE[®]
Database

Select any of the following install options.

☒ Create and configure a single instance database.
This option creates a starter database.

☐ Set Up Software Only

Note 1: For RAC install, do 'Set Up Software Only' and then execute DBCA (Database Configuration Assistant) from the oracle home.
Note 2: To upgrade an Oracle Database, do 'Set Up Software Only' and then execute DBUA (Database Upgrade Assistant) from the oracle home.

Help

< Back

Next >

Install

Cancel



Select Server class option and click on the Next button to move on to Install Type step.

Oracle Database 19c Installer - Step 2 of 17

Select System Class

19^c ORACLE Database

Configuration Option

System Class

Install Type

Database Edition

Oracle Home User

Installation Location

Configuration Type

Database Identifiers

Configuration Options

Database Storage

Management Options

Recovery Options

Schema Passwords

Prerequisite Checks

Summary

Install Product

Finish

☐ Desktop class

Choose this option if you are installing on a laptop or desktop class system. This option includes a starter database and allows minimal configuration.

☒ Server class

Choose this option if you are installing on a server class system, which Oracle defines as a system used in a production data center. This option allows for more advanced configuration options.

Help

< Back Next > Install Cancel



Chose Advance Install and click on the Next button to move on to Database Edition step.

Oracle Database 19c Installer - Step 3 of 17

Select Install Type

19^c ORACLE[®] Database

[Configuration Option](#)
[System Class](#)
[Install Type](#)
[Database Edition](#)
Oracle Home User
Installation Location
Configuration Type
Database Identifiers
Configuration Options
Database Storage
Management Options
Recovery Options
Schema Passwords
Prerequisite Checks
Summary
Install Product
Finish

☐ Typical install
Perform full Oracle Database installation with basic configuration.

☒ Advanced install
Allows advanced selections such as different passwords for the SYS, SYSMAN, SYSTEM and DBSNMP accounts, database character set, product languages, automated backups, custom installation, and alternative storage options such as Oracle Automatic Storage Management.

[Help](#) [< Back](#) [Next >](#) [Install](#) [Cancel](#)



The version of the database to install depends on your needs. For an application developer or and medium-size companies, the standard version covers practically all needs. To take full advantage of Oracle 19c, you can install the Enterprise version. Obviously, the Enterprise version requires more resources (storage, RAM, CPU).

Oracle Database 19c Installer - Step 4 of 17

Select Database Edition

19^c ORACLE Database

Which database edition do you want to install?

☒ **Enterprise Edition**

Oracle Database 19c Enterprise Edition is a self-managing database that has the scalability, performance, high availability, and security features required to run the most demanding, mission-critical applications.

☐ **Standard Edition 2**

Oracle Database 19c Standard Edition 2 is a full-featured data management solution ideally suited to the needs of medium-sized businesses.

[Configuration Option](#)

[System Class](#)

[Install Type](#)

Database Edition

[Oracle Home User](#)

Installation Location

Configuration Type

Database Identifiers

Configuration Options

Database Storage

Management Options

Recovery Options

Schema Passwords

Prerequisite Checks

Summary

Install Product

Finish

[Help](#)

[< Back](#) [Next >](#) [Install](#) [Cancel](#)



Choose this to use one of those user accounts which Windows 10 has created for you during its installation.

Oracle Database 19c Installer - Step 5 of 17

Specify Oracle Home User

19^c ORACLE[®] Database

For enhanced security, you may choose to run Windows Services from this Oracle home with a non-administrator account. Oracle recommends that you choose a Virtual Account or specify a standard Windows User Account for this purpose.

☐ Use Virtual Account

☐ Use Existing Windows User

User Name:

Password:

☐ Create New Windows User

User Name:

Password:

Confirm Password:

The newly created user is denied Windows logon privileges.

☒ **Use Windows Built-in Account**

A new Message will popup. Select Yes and continue.



In a default Windows installation, the Oracle base directory appears as follows: DRIVE_LETTER:\app\username where username is the Oracle installation user if you choose Windows built-in account, else it is the Oracle Home user (standard Windows user account). Leave it as default.

Oracle Database 19c Installer - Step 7 of 17

Select Configuration Type

19^c ORACLE[®] Database

Select the type of database that you want to create.

☒ **General Purpose / Transaction Processing**
A starter database designed for general purpose use, or for transaction-heavy applications.

☐ **Data Warehousing**
A starter database optimized for data warehousing applications.

Configuration Option
System Class
Install Type
Database Edition
Oracle Home User
Installation Location
Configuration Type
Database Identifiers
Configuration Options
Database Storage
Management Options
Recovery Options
Schema Passwords
Prerequisite Checks
Summary
Install Product
Finish

Help **< Back** **Next >** **Install** **Cancel**



Select General Purpose and Press Next

Oracle Database 19c Installer - Step 7 of 17

Select Configuration Type

19^c ORACLE[®] Database

Select the type of database that you want to create.

☒ **General Purpose / Transaction Processing**
A starter database designed for general purpose use, or for transaction-heavy applications.

☐ **Data Warehousing**
A starter database optimized for data warehousing applications.

Configuration Option
System Class
Install Type
Database Edition
Oracle Home User
Installation Location
Configuration Type
Database Identifiers
Configuration Options
Database Storage
Management Options
Recovery Options
Schema Passwords
Prerequisite Checks
Summary
Install Product
Finish

Help **< Back** **Next >** **Install** **Cancel**



Select Instance name. **DO UNCHECK THE OPTION “CREATE AS CONTAINER DATABASE”**

Oracle Database 19c Installer - Step 8 of 17

Specify Database Identifiers

19c ORACLE Database

Provide the identifier information required to access the database uniquely. An Oracle database is uniquely identified by a Global database name, typically of the form "name.domain". A database is referenced by at least one Oracle instance which is uniquely identified from any other instance on this computer by an Oracle system identifier (SID).

Global database name:

Oracle system identifier (SID):

☐ **Create as Container database**

Creates a database container for consolidating multiple databases into a single database and enables database virtualization.

Pluggable database name:

[Configuration Option](#)
[System Class](#)
[Install Type](#)
[Database Edition](#)
[Oracle Home User](#)
[Installation Location](#)
[Configuration Type](#)
Database Identifiers
[Configuration Options](#)
Database Storage
Management Options
Recovery Options
Schema Passwords
Prerequisite Checks
Summary
Install Product
Finish

[Help](#) [< Back](#) [Next >](#) [Install](#) [Cancel](#)



Continue with the default option. **Do not check “Automatic Memory Management” checkbox.**

Oracle Database 19c Installer - Step 9 of 17

Specify Configuration Options

19^c ORACLE[®] Database

- Configuration Option
- System Class
- Install Type
- Database Edition
- Oracle Home User
- Installation Location
- Configuration Type
- Database Identifiers
- Configuration Options**
- Database Storage
- Management Options
- Recovery Options
- Schema Passwords
- Prerequisite Checks
- Summary
- Install Product
- Finish

Memory | Character sets | Sample schemas

Enabling Automatic Memory Management allows the database to distribute memory automatically between the system global area (SGA) and the program global area (PGA), based on user-specified overall database memory target size. If automatic memory management is not enabled, then the SGA and PGA must be sized manually.

☐ **Enable Automatic Memory Management**

Allocate memory: 256 3277 8191 3,277 40 %

SGA target:	2457	MB
PGA aggregate target:	820	MB
Target database memory:	3277	MB

[Help](#) [< Back](#) [Next >](#) [Install](#) [Cancel](#)



Chose File System and press next.

Oracle Database 19c Installer - Step 10 of 17

Specify Database Storage Options

19c ORACLE Database

[Configuration Option](#)
[System Class](#)
[Install Type](#)
[Database Edition](#)
[Oracle Home User](#)
[Installation Location](#)
[Configuration Type](#)
[Database Identifiers](#)
[Configuration Options](#)
Database Storage
[Management Options](#)
Recovery Options
Schema Passwords
Prerequisite Checks
Summary
Install Product
Finish

☒ **File system**

Use a file system for database storage. For optimal database organization and performance, Oracle recommends that you install data files and the Oracle database software on different disks.

Specify database file location:

☐ **Oracle Automatic Storage Management**

Oracle Automatic Storage Management (Oracle ASM) simplifies database storage administration, and places database files for optimal I/O performance. Select this option if you intend to use Oracle ASM.



Press no Next. No Action Required.

Oracle Database 19c Installer - Step 11 of 17

Specify Management Options

19c ORACLE Database

Configuration Option
System Class
Install Type
Database Edition
Oracle Home User
Installation Location
Configuration Type
Database Identifiers
Configuration Options
Database Storage
Management Options
Recovery Options
Schema Passwords
Prerequisite Checks
Summary
Install Product
Finish

Oracle Database 12c is managed by Oracle Database Express by default. You can use Oracle Enterprise Manager 12c Cloud Control to manage each Oracle Database 19c centrally. Specify the details of the Cloud Control configuration to manage your database.

☐ Register with Enterprise Manager (EM) Cloud Control

OMS Host:

OMS Port:

EM Admin User Name:

EM Admin Password:

Help < Back Next > Install Cancel



Press Next. No Action Required.

Oracle Database 19c Installer - Step 12 of 17

Specify Recovery Options

19^c ORACLE[®] Database

Enable or disable recovery for your database. If you choose to enable recovery, then the location specified will be used as recovery area storage.

☐ Enable Recovery

☒ File system

Recovery area location:

☐ Oracle Automatic Storage Management

Recovery Options

[Configuration Option](#)

[System Class](#)

[Install Type](#)

[Database Edition](#)

[Oracle Home User](#)

[Installation Location](#)

[Configuration Type](#)

[Database Identifiers](#)

[Configuration Options](#)

[Database Storage](#)

[Management Options](#)

[Schema Passwords](#)

Prerequisite Checks

Summary

Install Product

Finish



Chose Option “Use the same password for all accounts”. Enter the password as “Manager1” and press next.

Oracle Database 19c Installer - Step 13 of 17

Specify Schema Passwords

19c ORACLE Database

The starter database contains pre-loaded schemas, most of which have passwords that are expired and locked at the end of installation. After installation is complete, you must unlock and set new passwords for those accounts you want to use. Schemas used for database management and postinstallation functions are left unlocked, and passwords for these accounts will not expire. Specify the passwords for these accounts.

☐ Use different passwords for these accounts

	Password	Confirm password
SYS		
SYSTEM		

☒ Use the same password for all accounts

Password: Confirm password:

[Help](#) [< Back](#) [Next >](#) [Install](#) [Cancel](#)



Press Next No action required. Press Install

Oracle Database 19c Installer - Step 15 of 17

Summary

19c ORACLE Database

[Configuration Option](#)
[System Class](#)
[Install Type](#)
[Database Edition](#)
[Oracle Home User](#)
[Installation Location](#)
[Configuration Type](#)
[Database Identifiers](#)
[Configuration Options](#)
[Database Storage](#)
[Management Options](#)
[Recovery Options](#)
[Schema Passwords](#)
[Prerequisite Checks](#)
Summary
[Install Product](#)
[Finish](#)

Oracle Database 19c Installer

Global settings

- Oracle Home User: NT AUTHORITY\SYSTEM [\[Edit\]](#)
- Install method: Advanced installation [\[Edit\]](#)
- Database edition: Enterprise Edition (Create and configure a database) [\[Edit\]](#)
- Oracle base: D:\app\Administrator [\[Edit\]](#)
- Software location: D:\db_home
- OraMTS Port Number: 49152

Database information

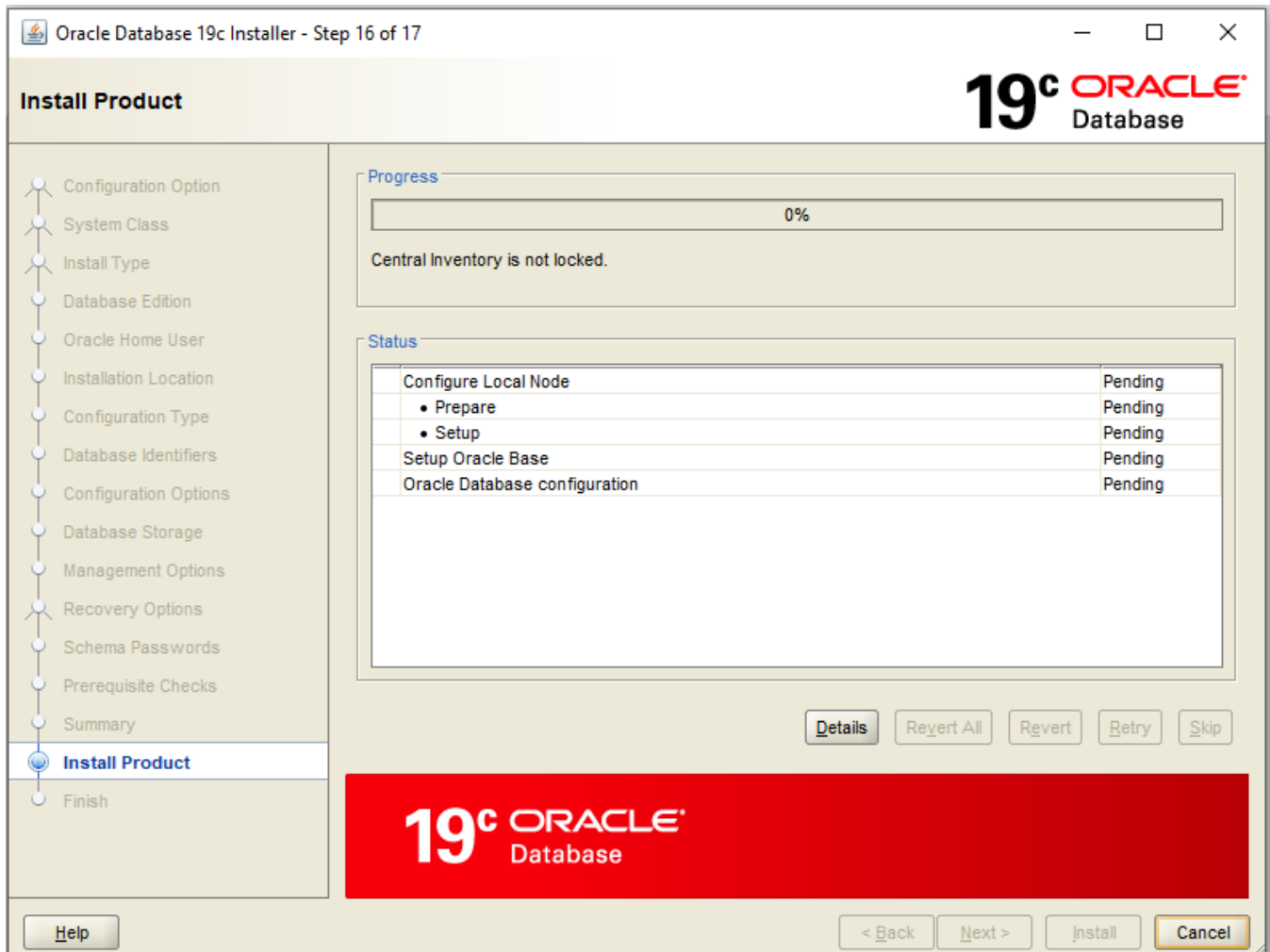
- Configuration: General Purpose / Transaction Processing [\[Edit\]](#)
- Global database name: orcl [\[Edit\]](#)
- Oracle system identifier (SID): orcl [\[Edit\]](#)
- Allocated memory: 3277 MB [\[Edit\]](#)
- Automatic memory management option: FALSE [\[Edit\]](#)
- Database character set : AL32UTF8 (Unicode UTF-8 Universal character set) [\[Edit\]](#)
- Management method: Database express [\[Edit\]](#)
- Database storage mechanism: File system [\[Edit\]](#)
- Database file location: D:\app\Administrator\oradata [\[Edit\]](#)
- Recovery: Disabled [\[Edit\]](#)

[Save Response File...](#)

[Help](#) [< Back](#) [Next >](#) [Install](#) [Cancel](#)

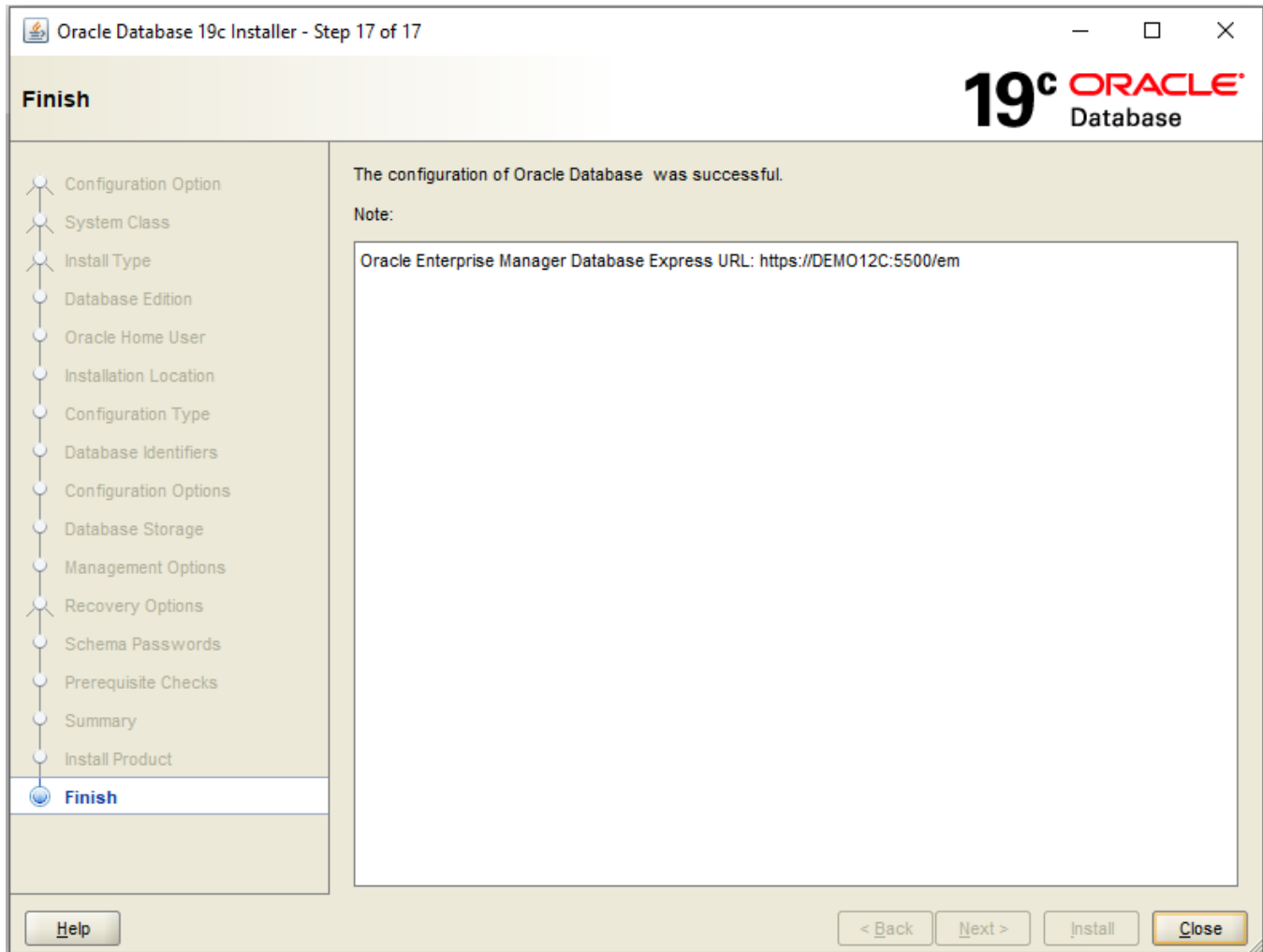


Installation Starts.





STEP 4 – Installation completed.



Note the “Oracle Enterprise Manager Database path” and press close. < <https://machine name:5500/em/login>>



STEP 5 – Disable Case sensitive and Password Life.

- a. Login as sys as sysdba/Manager1 on sqlplus.
- b. `ALTER SYSTEM SET SEC_CASE_SENSITIVE_LOGON = FALSE;`
- c. `ALTER PROFILE DEFAULT LIMIT PASSWORD_LIFE_TIME UNLIMITED;`
- d. Edit the SQLNET.ORA file to add `SQLNET.ALLOWED_LOGON_VERSION_SERVER` setting
`SQLNET.ALLOWED_LOGON_VERSION_SERVER=11`
- e. Restart the server.
- f. Login as sys as sysdba/Manager1
- g. `alter user system identified by Manager1;`
- h. Restart the machine.

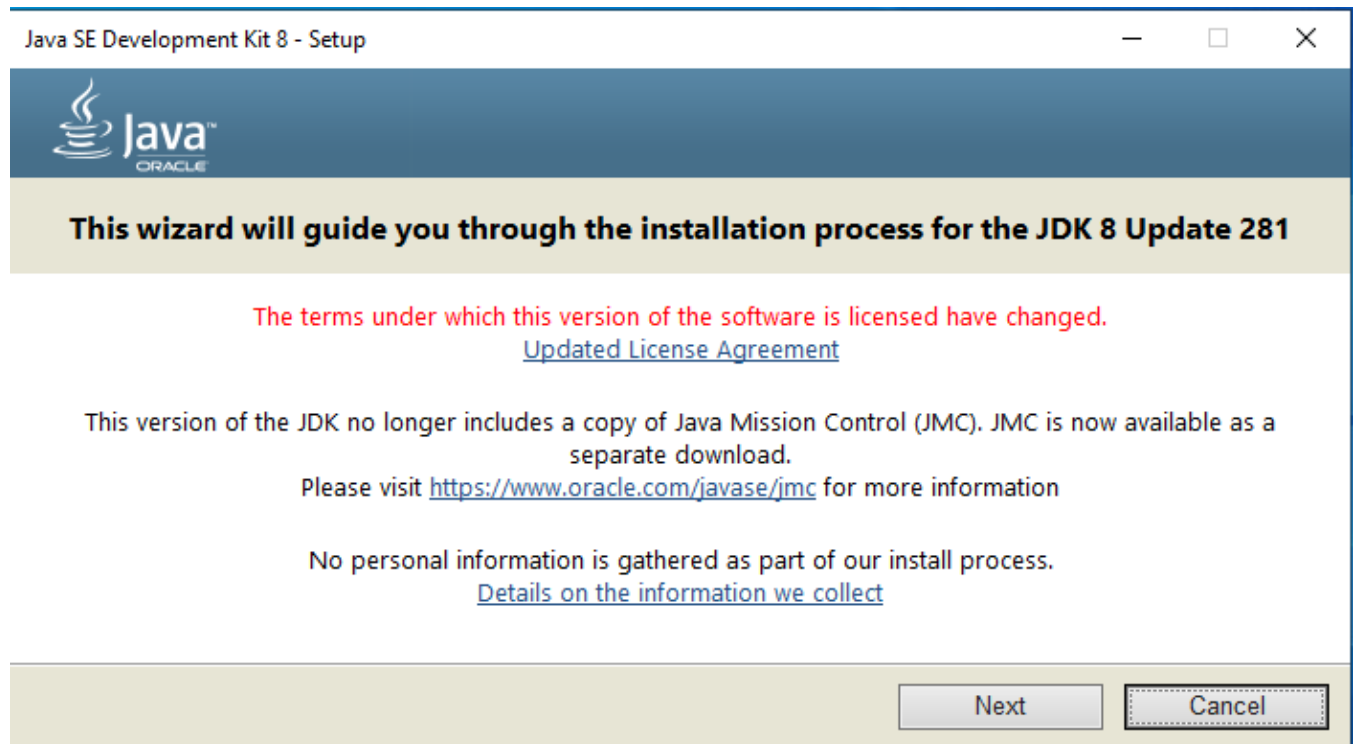


Installation Forms 12C

JDK Installation

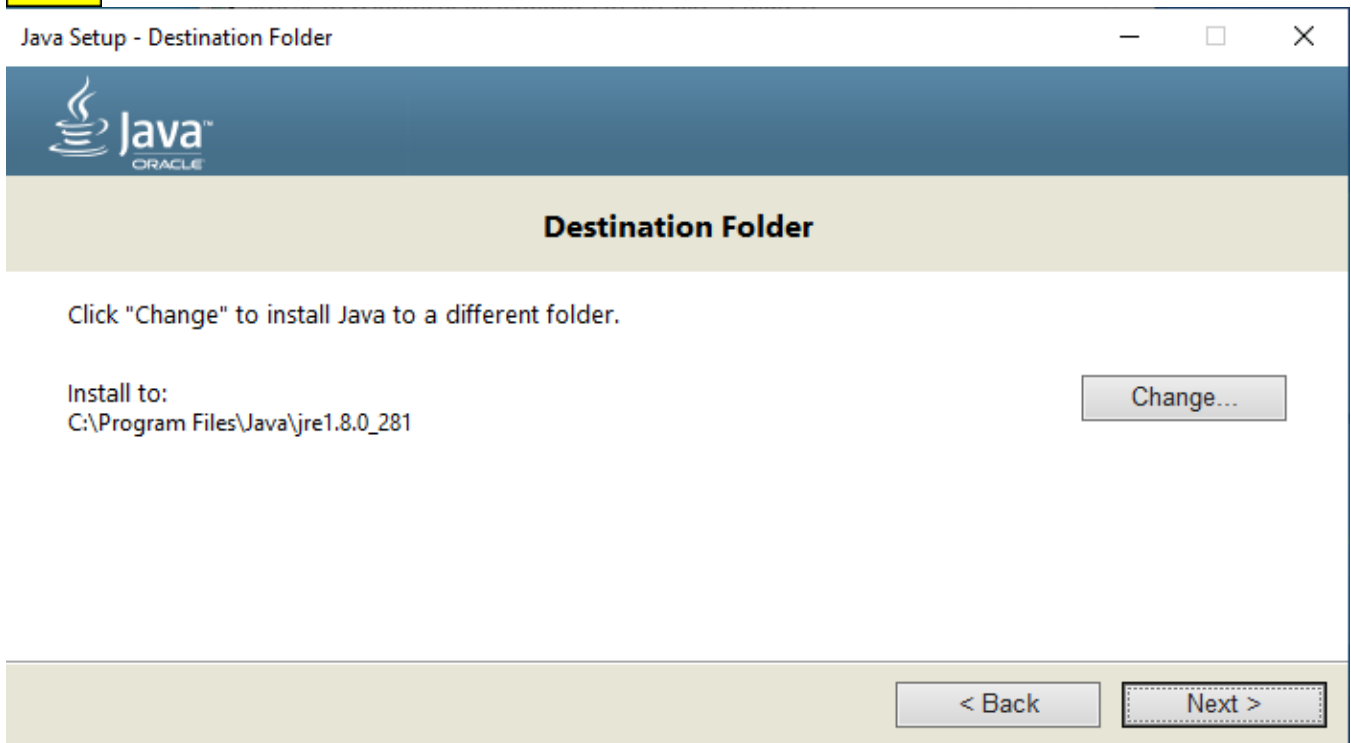
Double click jdk-8u281-windows-x64 (The one you are using)

Click “Next” in the welcome screen

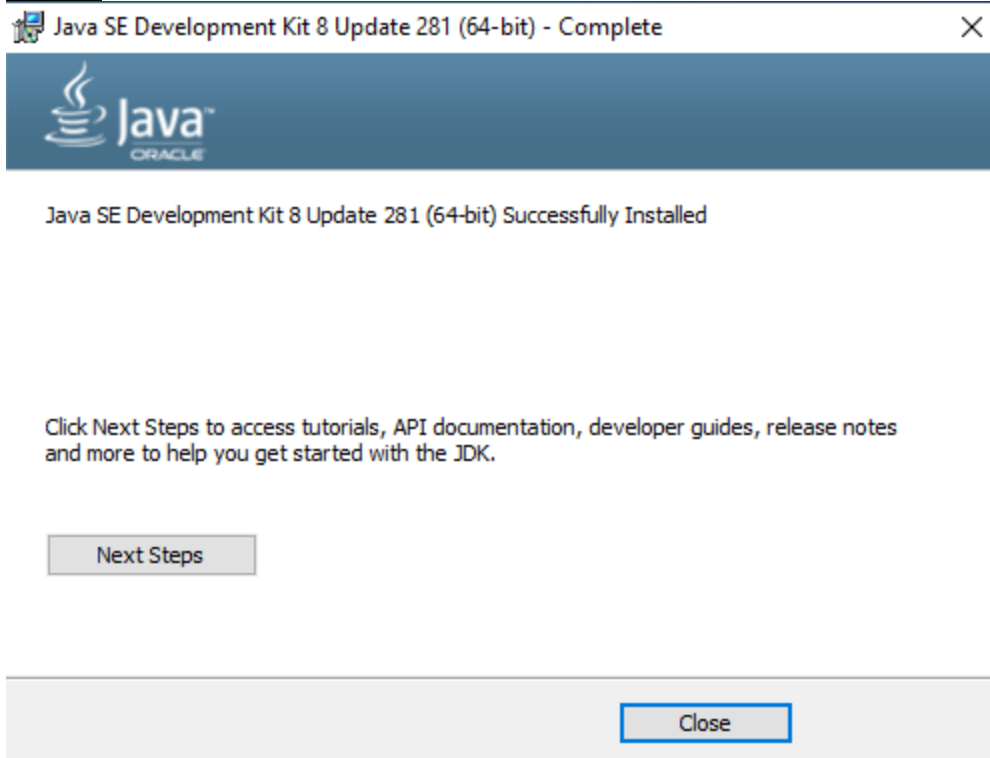


Keep the defaults, click “Next” and continue with the default installation and complete the setup.

Click “Next”



Click "Close"

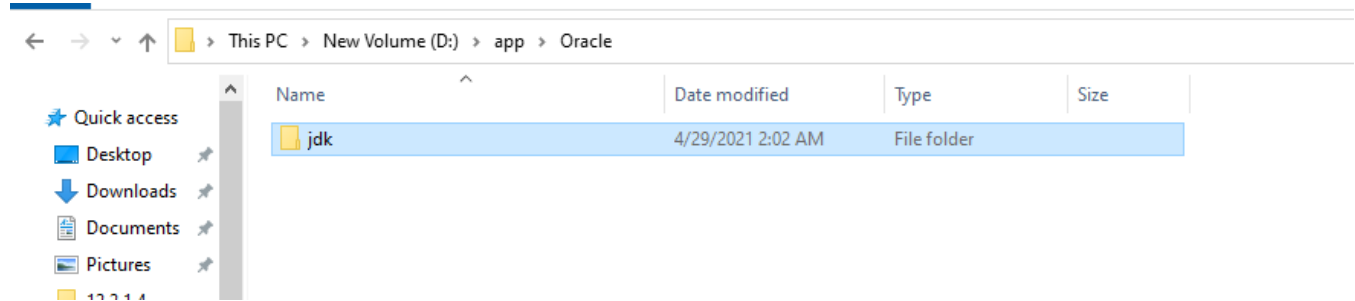




Navigate to “C:\Program Files\Java”

Copy “jdk1.8.0_281” to “D:\app\oracle” and rename the folder name to jdk.

(If Oracle folder is not present then create new “ORACLE”)





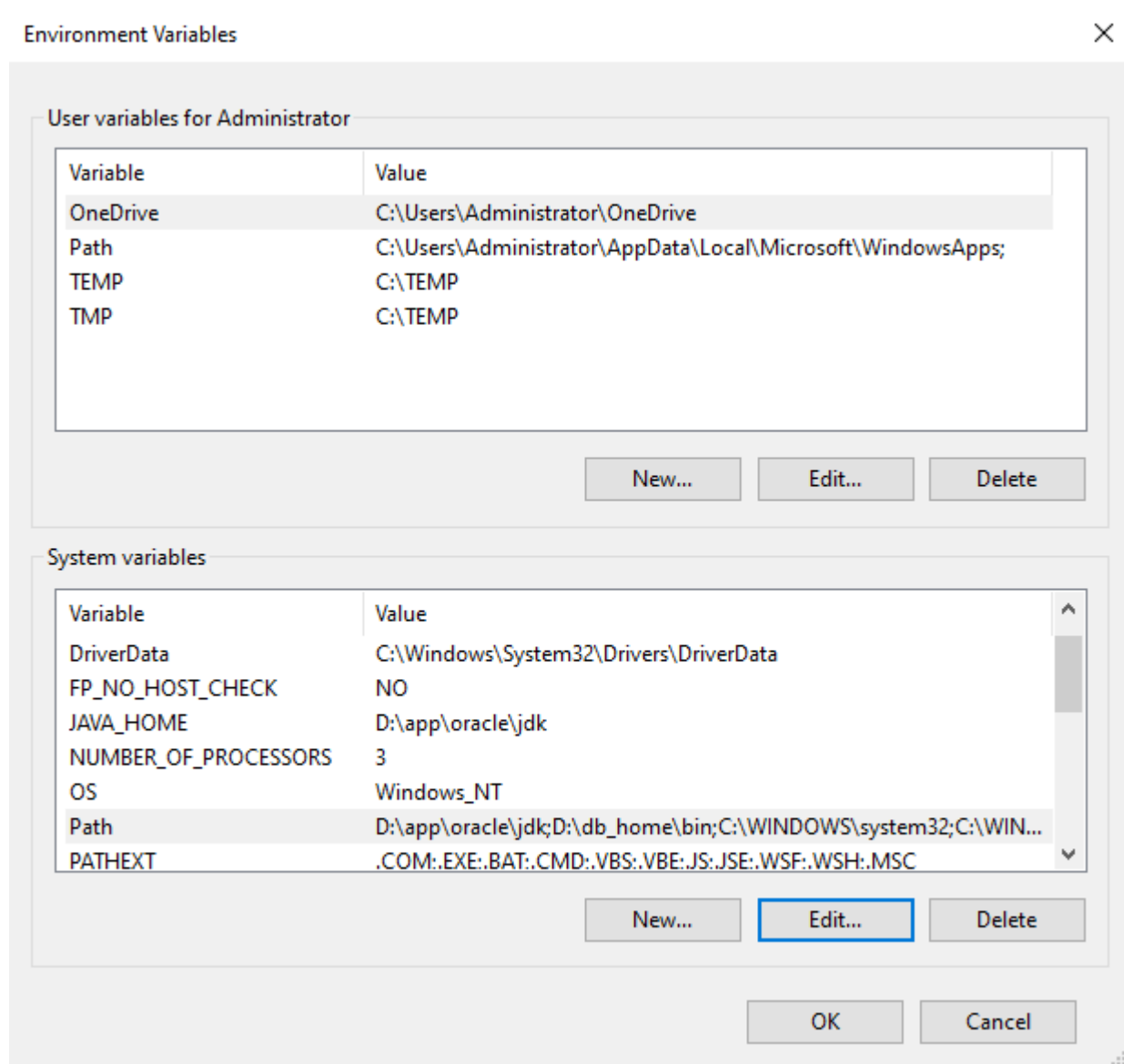
Add the environment variable "JAVA_HOME"

Control Panel -> System -> Advanced system settings -> Environment Variables Click "New" on the System variables.

Enter "JAVA_HOME" in the variable name, and "D:\app\oracle\jdk" in the variable value and click "OK"

Find the system environment "Path" and click "Edit"

Add "D:\app\oracle\jdk\bin;" to the beginning of variable value.





Restart to take effect of the system update on path variables.



Weblogic Installation

Run the installer

Open command and go to

```
D:\app\Oracle\jdk\bin> java -jar D:\12C\fmw_12.2.1.4.0_infrastructure.jar
```

(Note that all the installation file are in 12C folder)

```
Administrator: Command Prompt - java -jar D:\12.2.1.4\fmw_12.2.1.4.0_infrastructure.jar
Microsoft Windows [Version 10.0.19042.906]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>cd\

C:\>d:

D:\>cd app

D:\app>cd Oracle

D:\app\Oracle>cd jdk

D:\app\Oracle\jdk>cd bin

D:\app\Oracle\jdk\bin>java -jar D:\12C\ fmw_12.2.1.4.0_infrastructure.jar
Error: Unable to access jarfile D:\12C\
```



Click the "Next" button on the welcome screen



Select the "Skip Auto Updates" options, then click the "Next" button.



Oracle Fusion Middleware 12c Infrastructure Installation - Step 2 of 8

Auto Updates

ORACLE
FUSION MIDDLEWARE

- Welcome
- Auto Updates**
- Installation Location
- Installation Type
- Prerequisite Checks
- Installation Summary
- Installation Progress
- Installation Complete

☒ Skip Auto Updates

☐ Select patches from directory

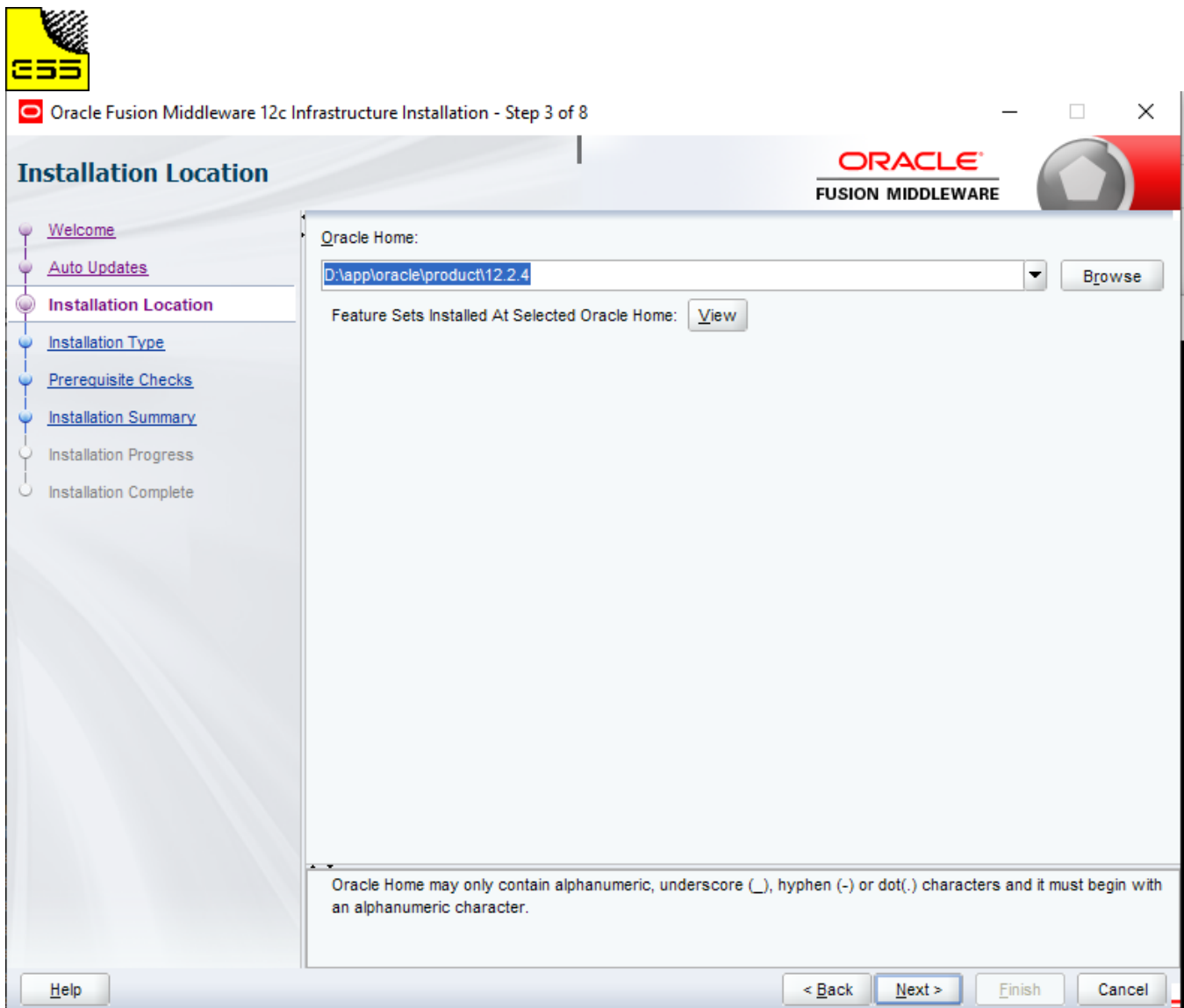
Location:

☐ Search My Oracle Support for Updates

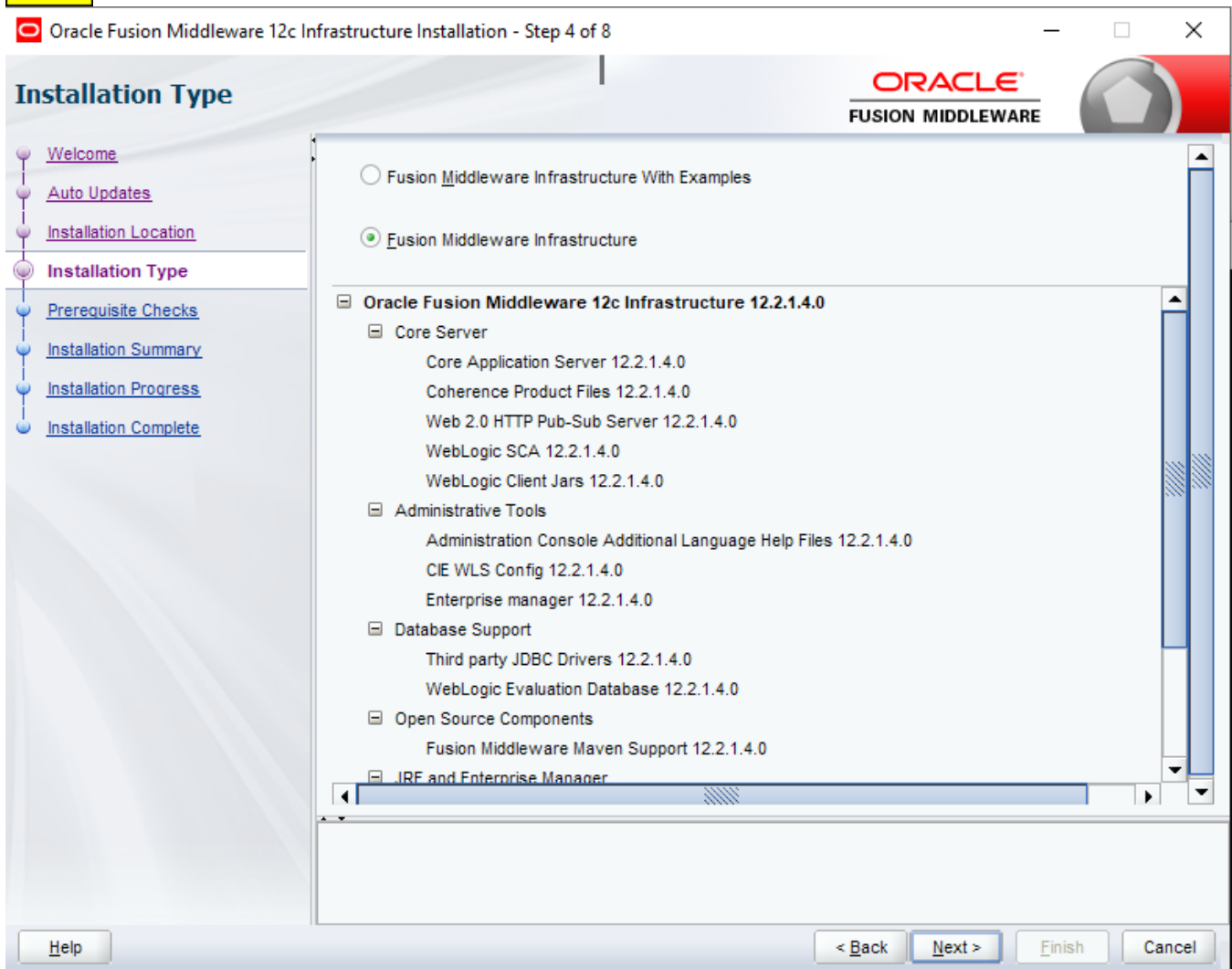
Username:

Password:

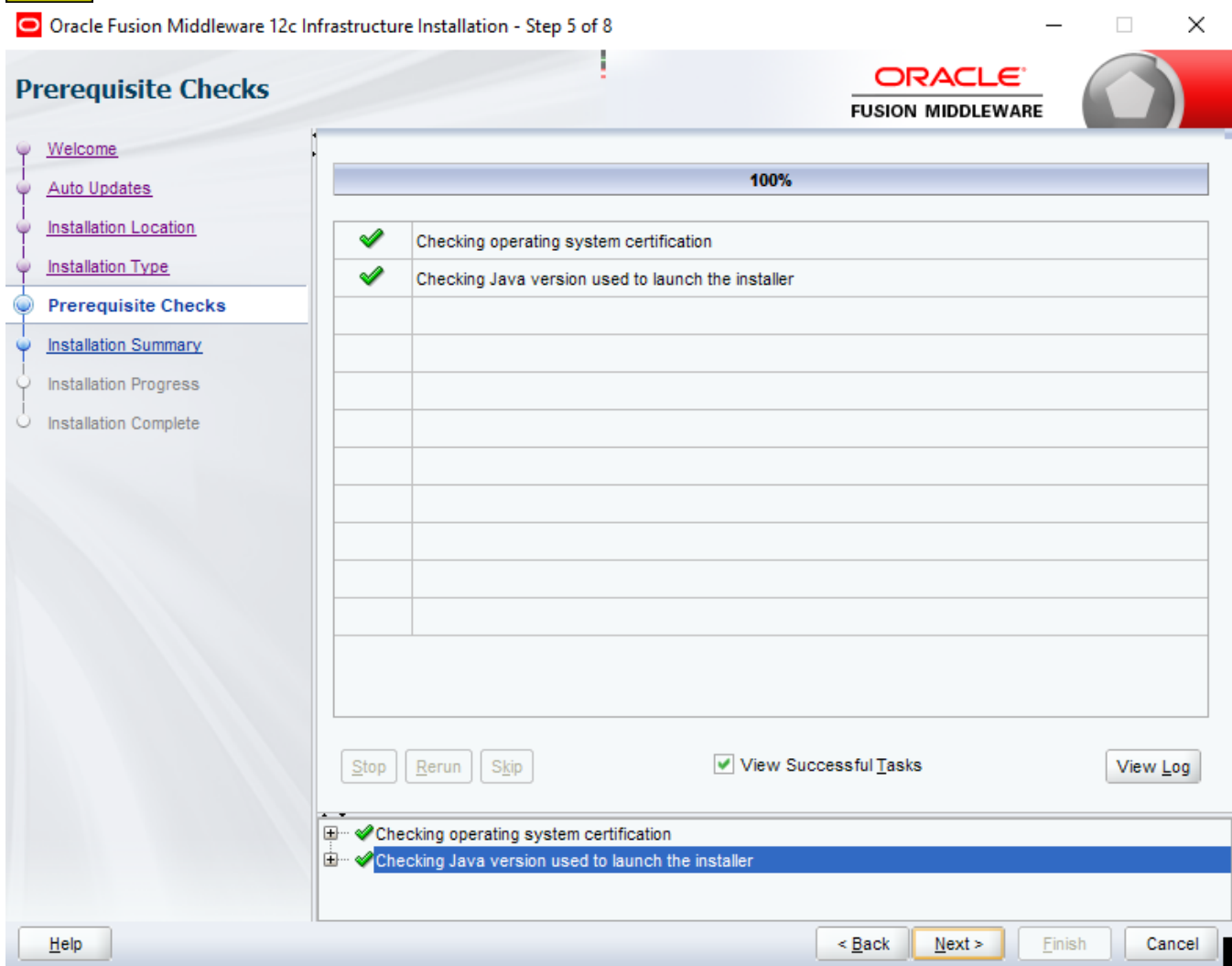
Enter the middleware home ("D:\app\oracle\product\12.2.4") and click the "Next" Button.



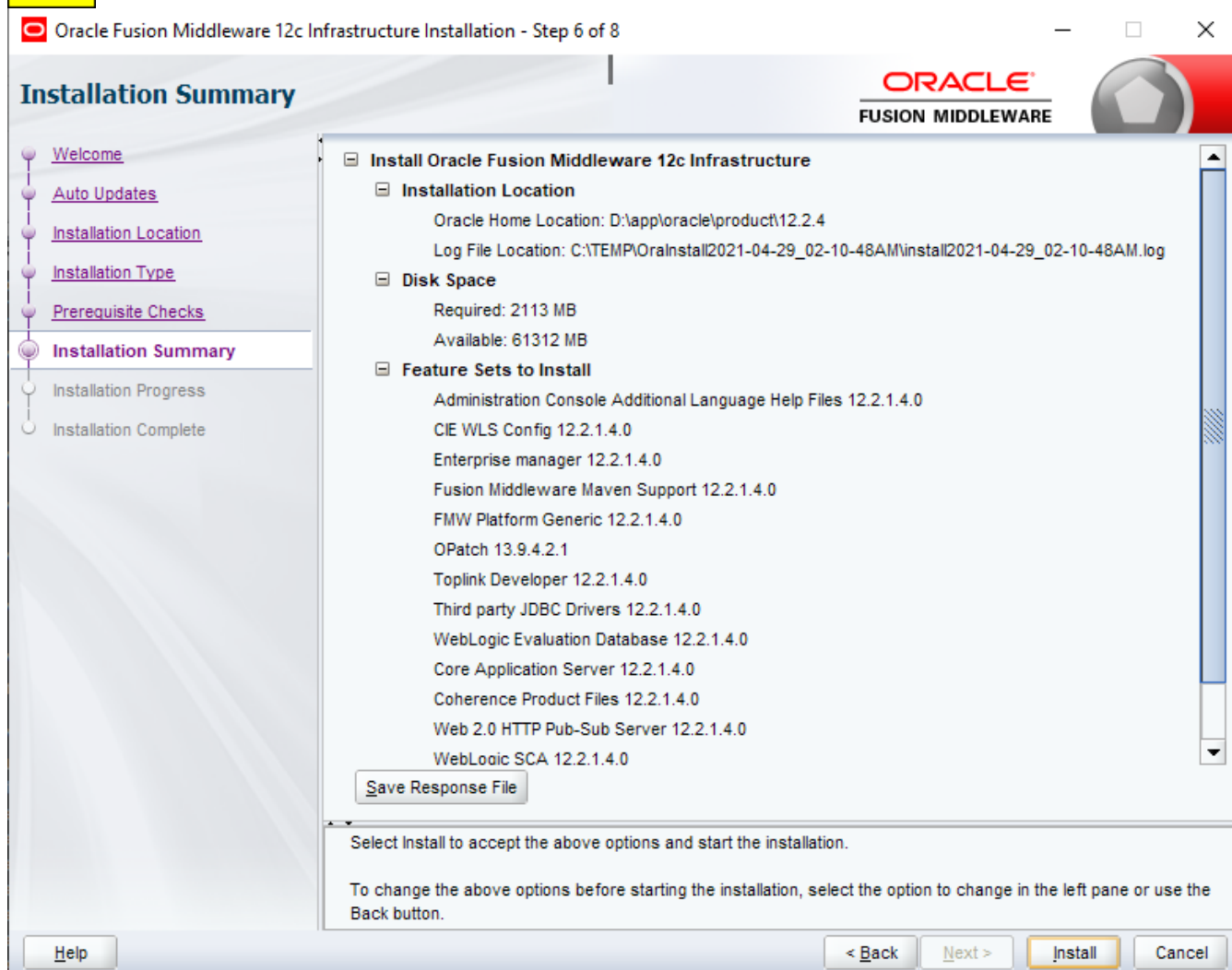
Accept the "Fusion Middleware Infrastructure" option by clicking the "Next" button



Wait for the prerequisite checks to complete. If everything is ok, click the "Next" button.



Click the "Install" button



Wait for the installation to complete, and then click the "Next" button.



Installation Progress

ORACLE
FUSION MIDDLEWARE

- Welcome
- Auto Updates
- Installation Location
- Installation Type
- Prerequisite Checks
- Installation Summary
- Installation Progress**
- Installation Complete

100%

✓	Prepare
✓	Copy
✓	Generating Libraries
✓	Performing String Substitutions
✓	Linking
✓	Setup
✓	Saving the inventory
✓	Post install scripts

View Messages

☒ View Successful Tasks

View Log

**Hardware and Software
Engineered to Work Together**

Help

< Back

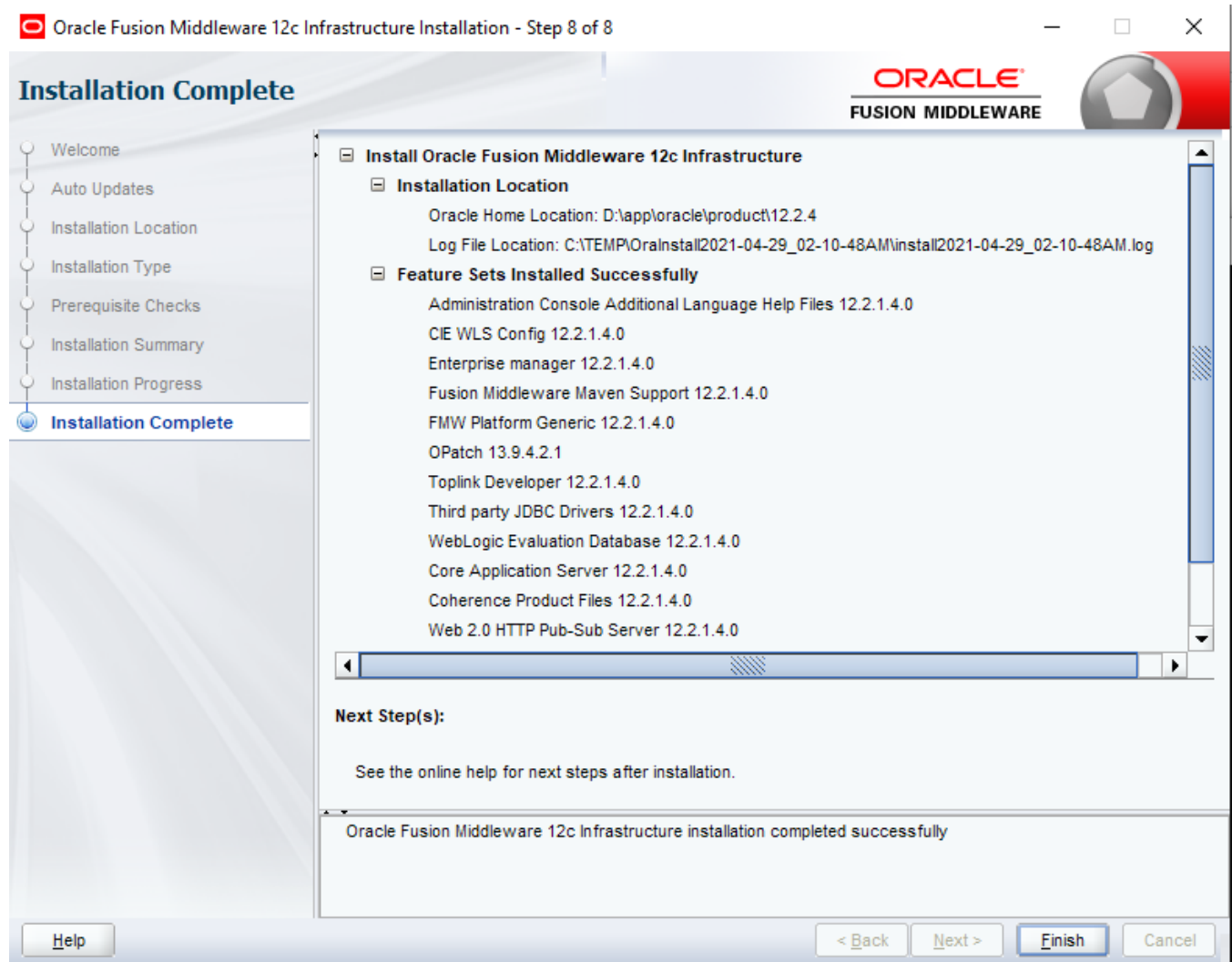
Next >

Finish

Cancel



Once the installation complete, click the "Finish" button. (Do restart to update system variables).





Visual C++ Redistributable for Visual Studio 2012 Update 4 (This may need to change as per Operating System)

Install “vcredist_x64.exe” for C++.Its mandatory to install.

Double click on “vcredist_x64.exe”

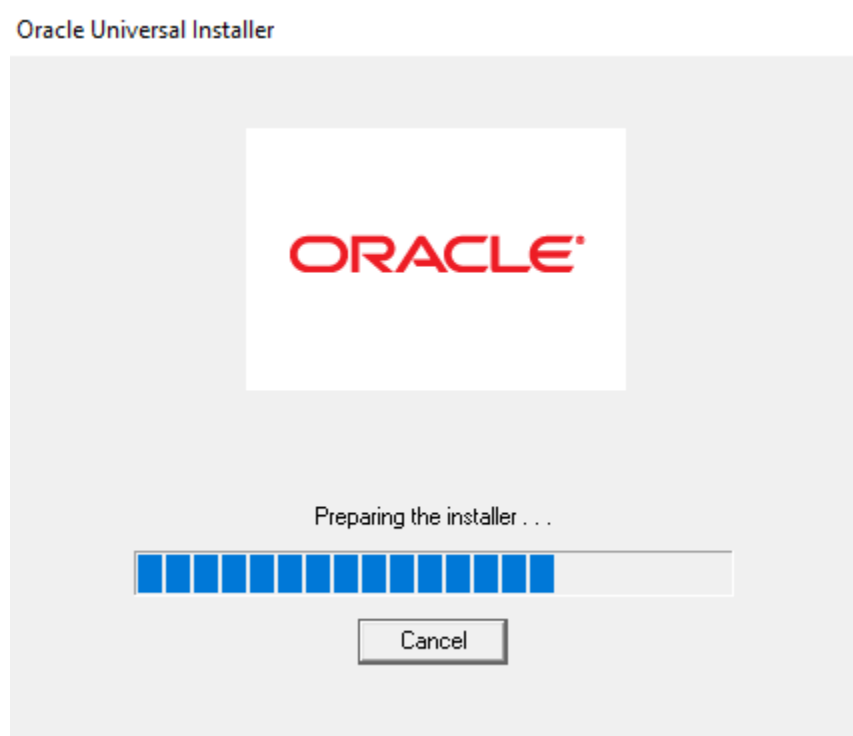


Oracle Forms and Reports Installation

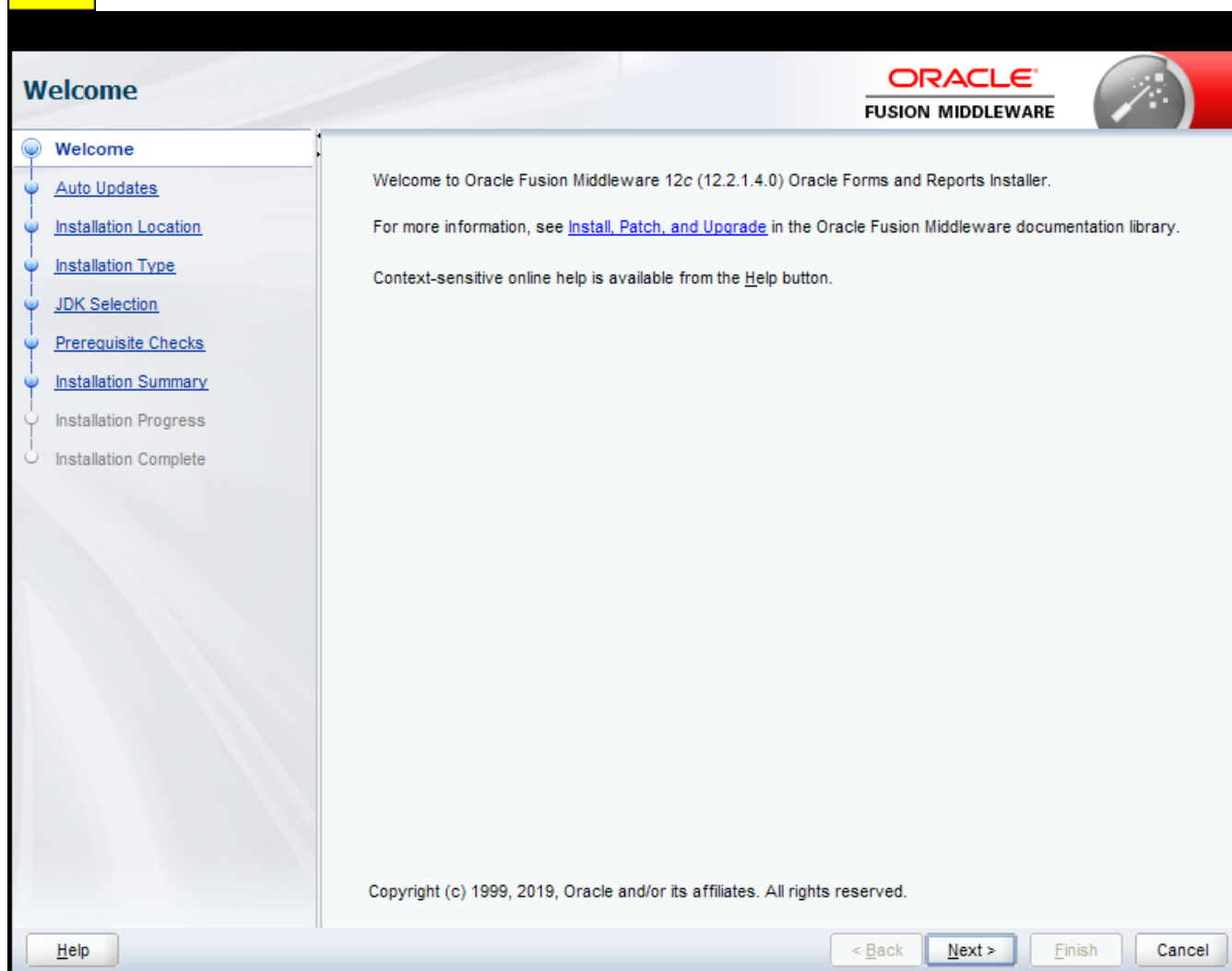
12c (12.2.1.4) installation.

You need:- setup_fmws_12.2.1.4.0_fr_win64.exe

Double click setup_fmws_12.2.1.4.0_fr_win64.exe (Use option run as Administrator)



Click the "Next" button on the Welcome screen.



Select the "Skip Auto Updates" radio group, then click the "Next" button.



Auto Updates

ORACLE®
FUSION MIDDLEWARE



Welcome

Auto Updates

Installation Location

Installation Type

JDK Selection

Prerequisite Checks

Installation Summary

Installation Progress

Installation Complete

☒ Skip Auto Updates

☐ Select patches from directory

☐ Search My Oracle Support for Updates

Location:

Brgwse

Username:

Password:

Proxy Settings

Test Connection

Search

Help

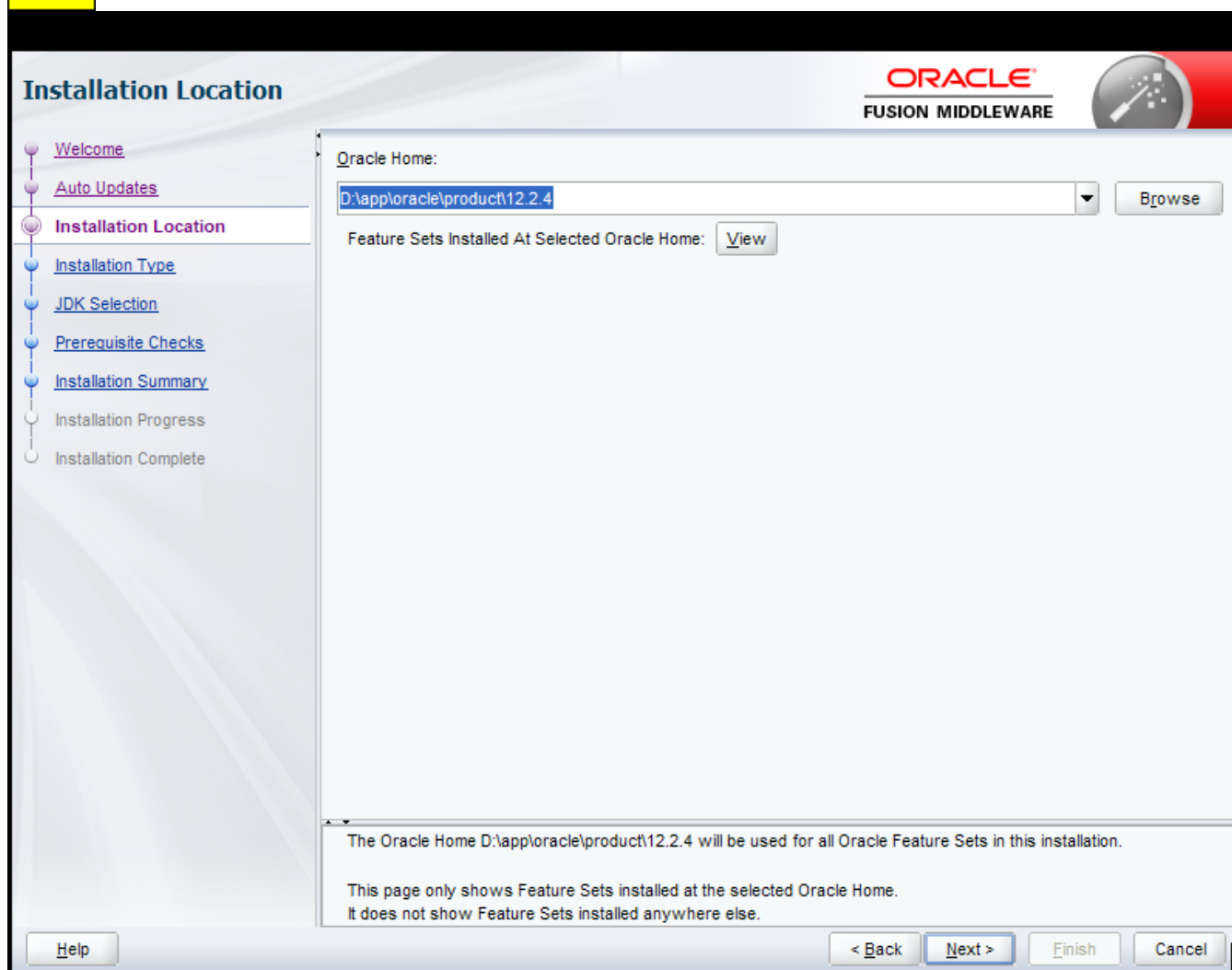
< Back

Next >

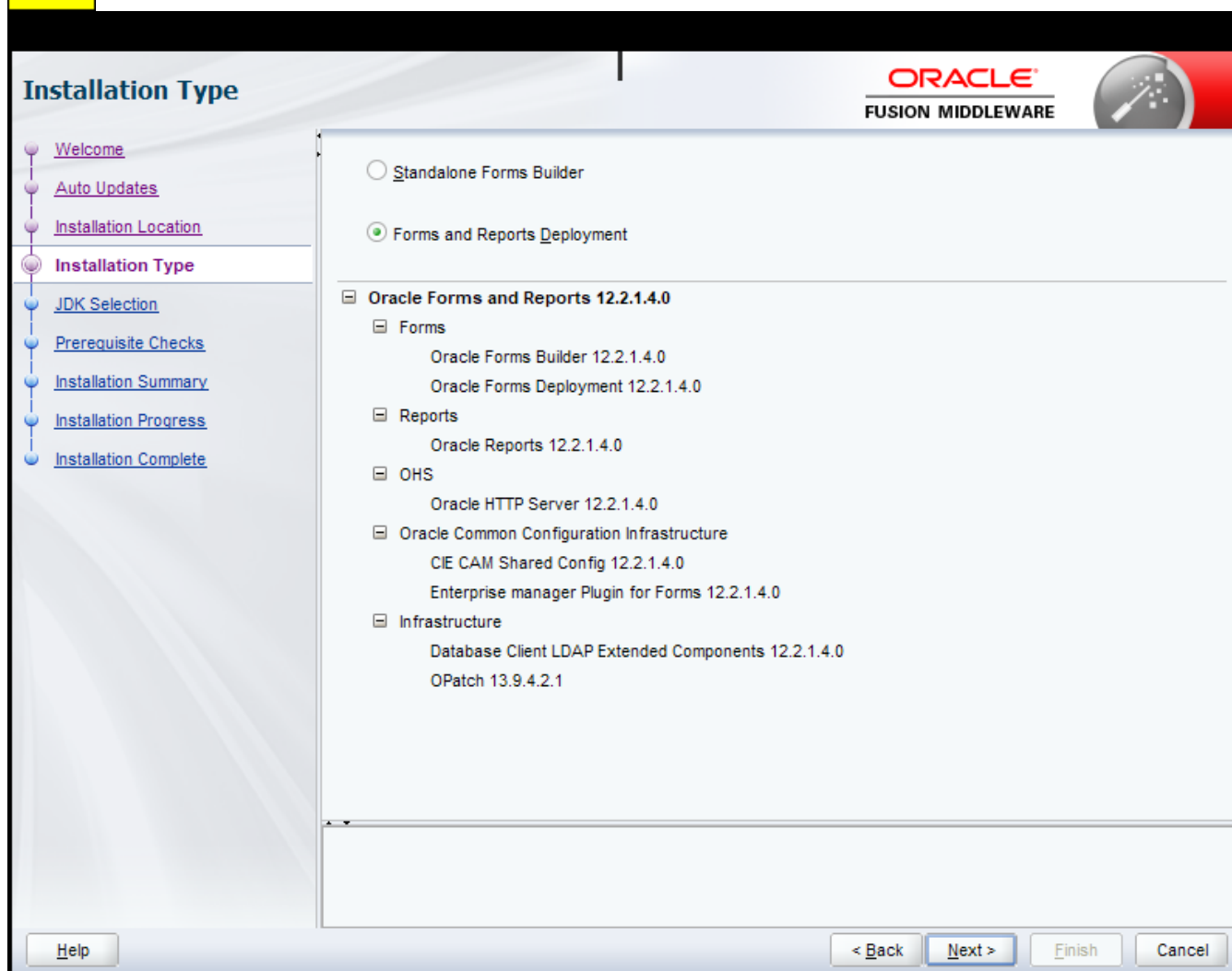
Finish

Cancel

Enter the path 'D:\app\oracle\product\12.2.4' (Or you can chose from dropdown), then click the "Next" button.

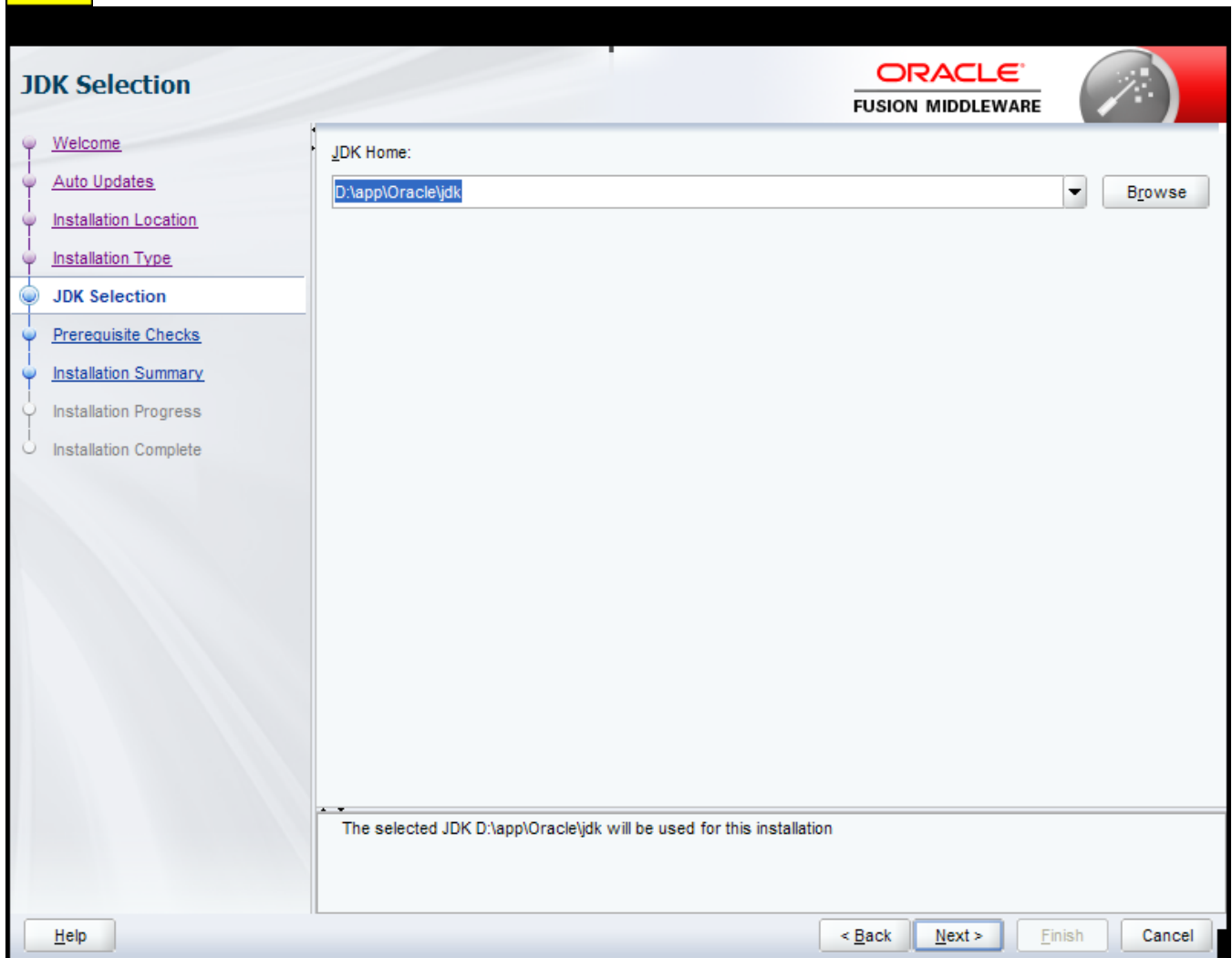


Select the "Forms and Reports Deployment" option, then click the "Next" button.

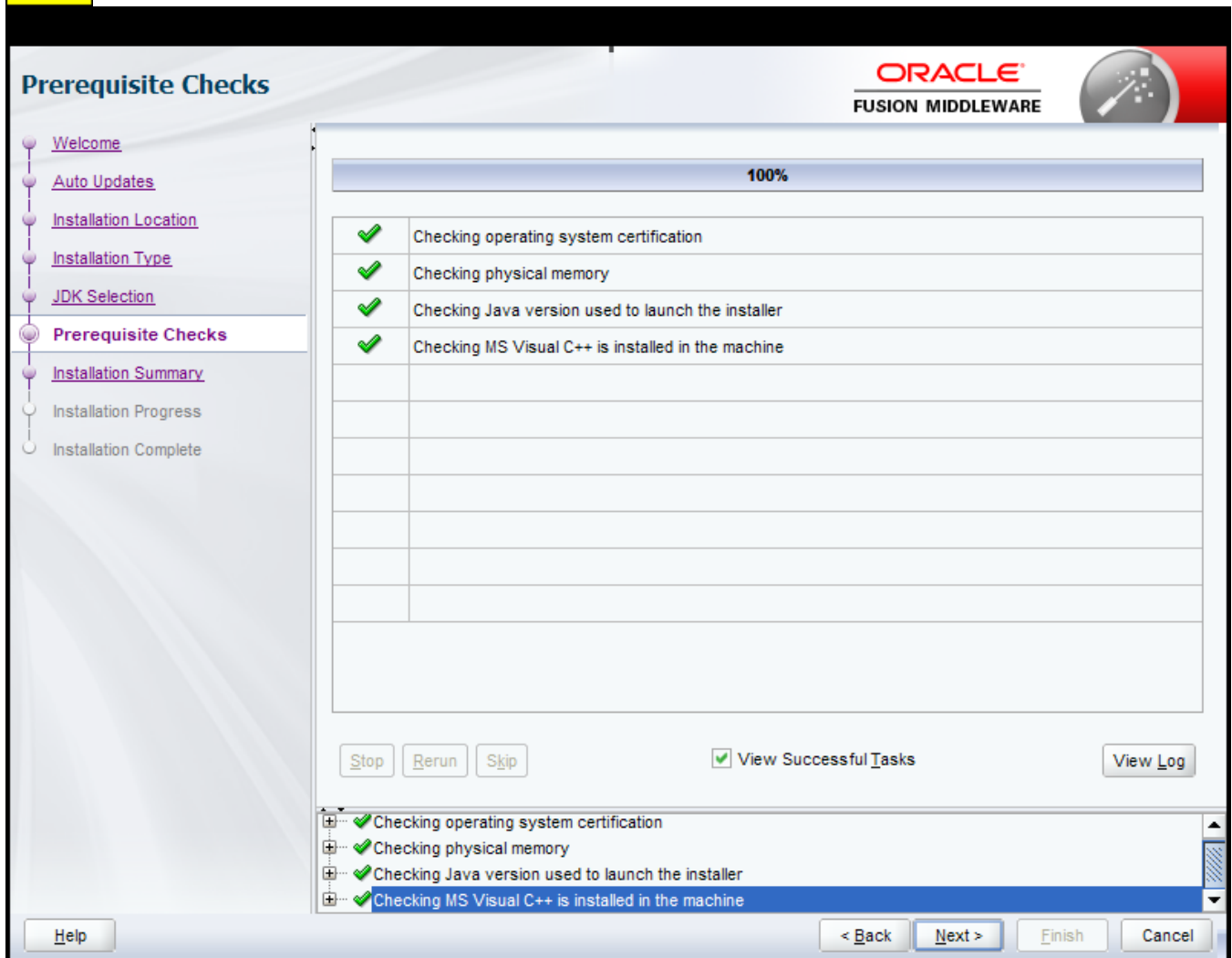


JDK Selection:

Enter D:\app\Oracle\jdk or chose from dropdown.



If the prerequisites were successful, click the "Next" button.



Click the "Install" button



Installation Progress

0%

✓	Prepare
⌚	Copy
	Generating Libraries
	Performing String Substitutions
	Linking
	Setup
	Saving the inventory
	Post install scripts

View Messages

☒ View Successful Tasks

View Log

Hardware and Software
Engineered to Work Together

Help

< Back

Next >

Finish

Cancel

Once the installation complete, click the "Next" button.



Installation Progress

Welcome

Auto Updates

Installation Location

Installation Type

JDK Selection

Prerequisite Checks

Installation Summary

Installation Progress

[Installation Complete](#)

ORACLE®
FUSION MIDDLEWARE

100%

✓	Prepare
✓	Copy
✓	Generating Libraries
✓	Performing String Substitutions
✓	Linking
✓	Setup
✓	Saving the inventory
✓	Post install scripts

View Messages

☒ View Successful Tasks

View Log

**Hardware and Software
Engineered to Work Together**

Help

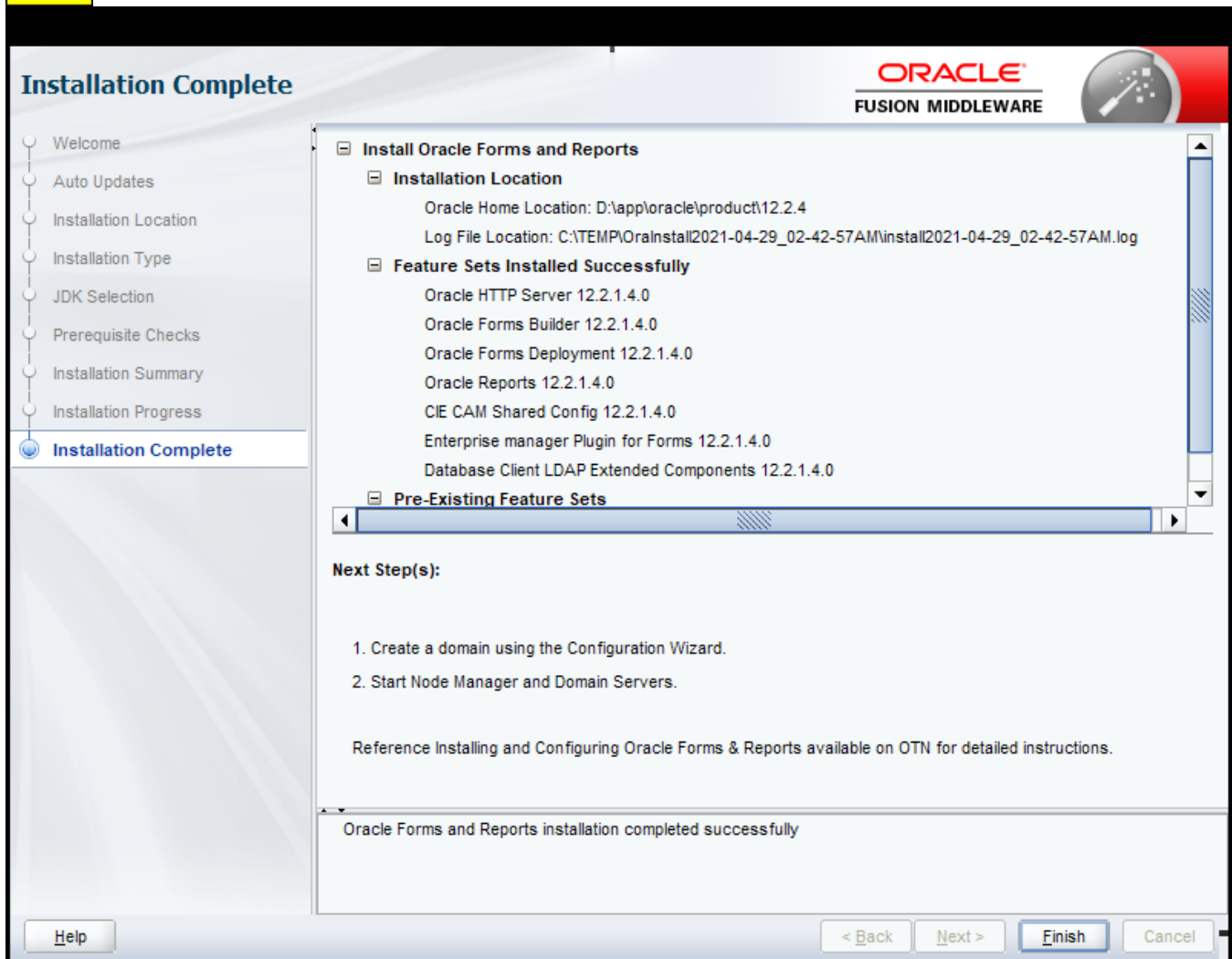
< Back

Next >

Finish

Cancel

Click the "Finish" button.



Your installation is now complete.



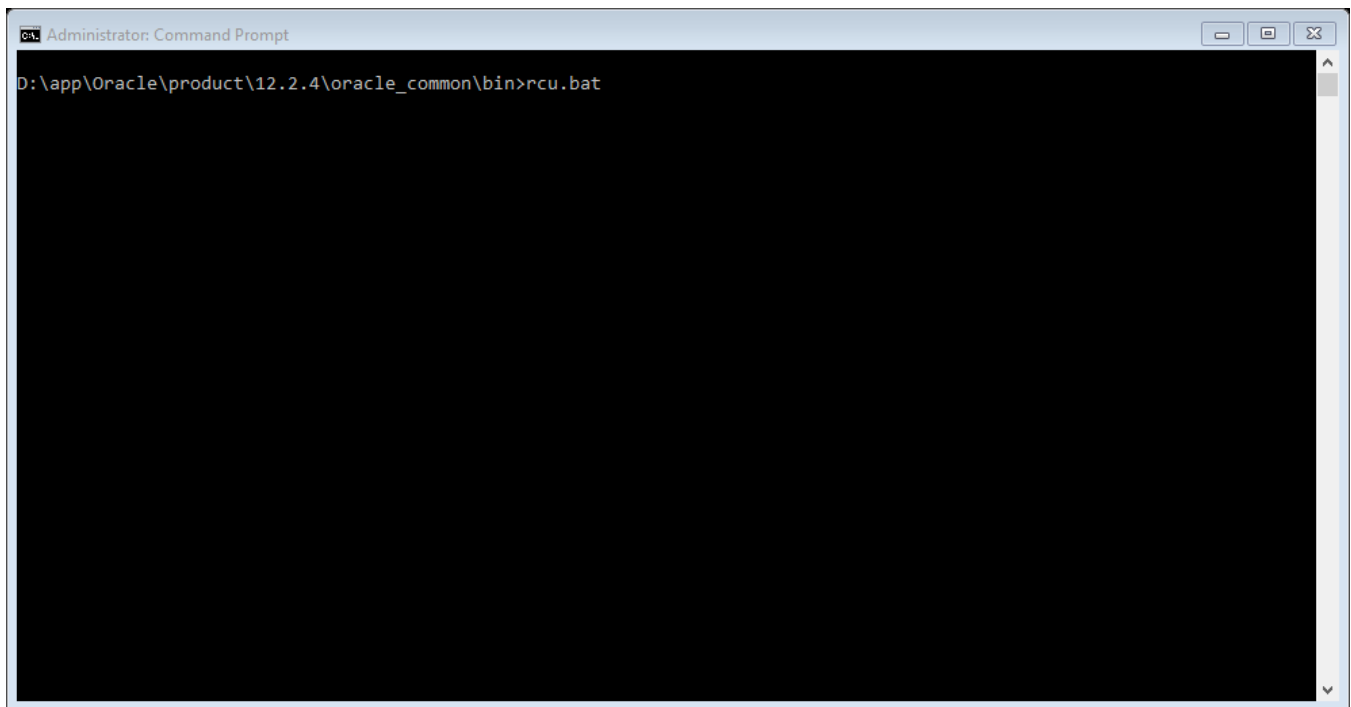
CONFIGURATION

Unlike previous versions, where the Forms and Reports Services installation automatically created the domain, but in 12c you need to create repository and configure domain manually.

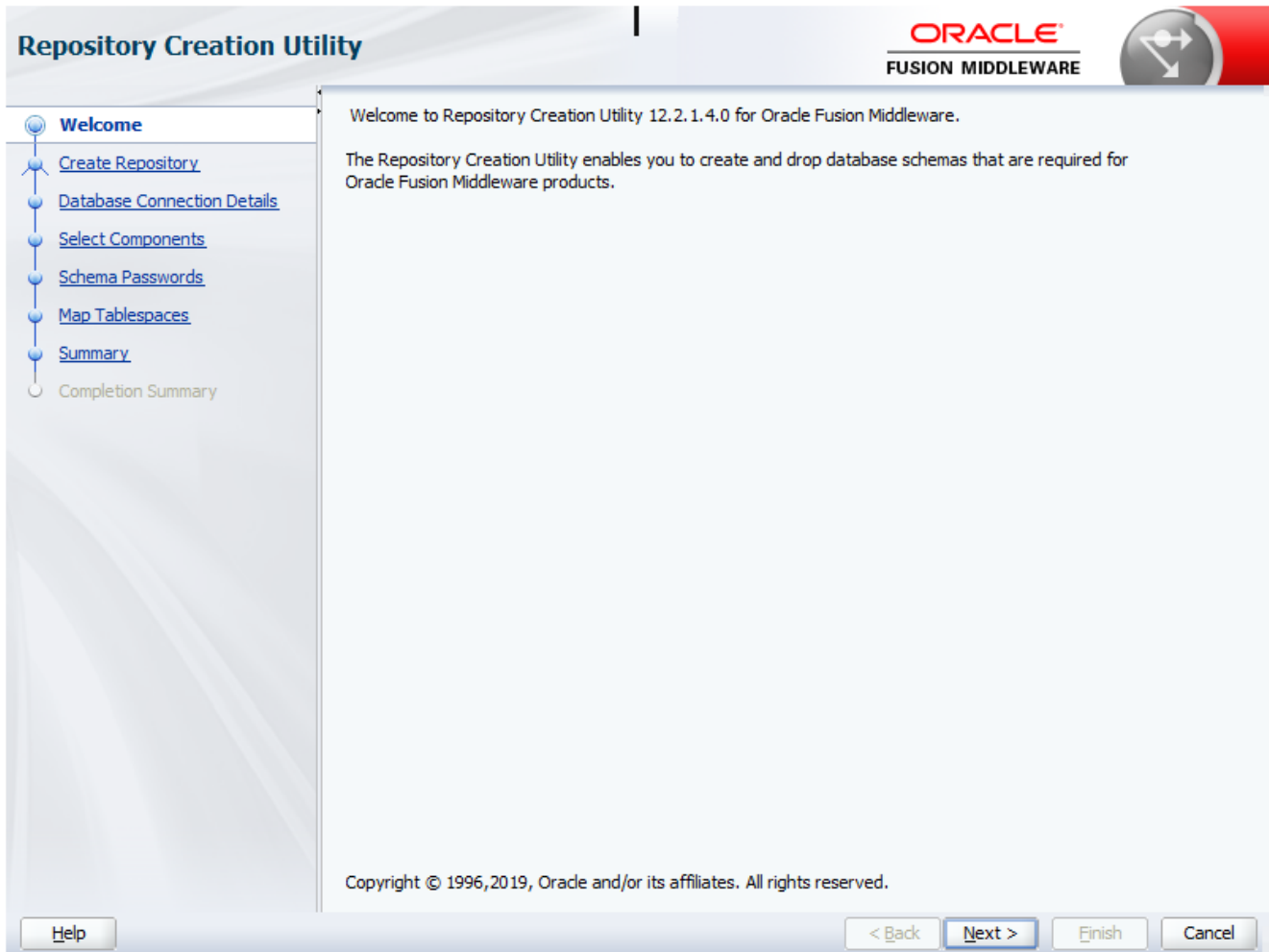
Repository Configuration Utility (RCU)

Start the Repository Configuration Utility (RCU).

Open Command Prompt and type `D:\app\oracle\product\12.2.4\oracle_common\bin\rcu.bat`



Click the "Next" button on the welcome page.



Accept the "Create Repository" and "System Load and Product Load" options by clicking the "Next" button.



Repository Creation Utility

ORACLE
FUSION MIDDLEWARE

[Welcome](#)

Create Repository

[Database Connection Details](#)

[Select Components](#)

[Schema Passwords](#)

[Map Tablespaces](#)

[Summary](#)

[Completion Summary](#)

☒ **Create Repository**
Create and load component schemas into a database.

☒ **System Load and Product Load**
I have DBA privileges.

☐ **Prepare Scripts for System Load**
Create scripts for DBA to run later.

☐ **Perform Product Load**
System Load scripts have been run by DBA.

☐ **Drop Repository**
Remove component schemas from a database.

Help

< Back Next > Finish Cancel

Enter the database connection details and click the "Next" button. Enter Password 'Manager1'



Repository Creation Utility

ORACLE
FUSION MIDDLEWARE

Navigation: Welcome, Create Repository, **Database Connection Details**, Select Components, Schema Passwords, Map Tablespaces, Summary, Completion Summary

Database Type: Oracle Database

Connection String Format: ☒ Connection Parameters ☐ Connection String

Connect String:

Host Name: demo12c

Port: 1521

Service Name: ord

Username: sys

Password:

Role: SYSDBA

Buttons: Help, < Back, Next >, Finish, Cancel

If all the prerequisite checks are successful, click the "OK" button.

Checking Global Prerequisites

✓	Initializing repository configuration metadata	00:01.125(sec)
✓	Obtain properties of the specified database	00:00.125(ms)
✓	Check requirement for specified database	00:01.125(sec)

Operation completed. Click OK to continue to next page.

OK



Enter a schema prefix as ESS and select the components as shown below and click the "Next" Button. For Forms and Reports

Services, you should check the "Oracle Platform Security Services" option, which will automatically select others.

Component	Schema Owner
☐ Oracle AS Repository Components	
☒ AS Common Schemas	
☒ Common Infrastructure Services *	ESS_STB
☒ Oracle Platform Security Services	ESS_OPSS
☐ User Messaging Service	UMS
☒ Audit Services	ESS_IAU
☒ Audit Services Append	ESS_IAU_APPEND
☒ Audit Services Viewer	ESS_IAU_VIEWER
☐ Metadata Services	MDS
☒ Weblogic Services *	ESS_WLS

* Mandatory component. Mandatory components cannot be deselected.

Buttons at the bottom: Help, < Back, Next >, Finish, Cancel.

If all the prerequisite checks are successful, click the "OK" button.



Checking Component Prerequisites

✓	Common Infrastructure Services	00:00.112(ms)
✓	Oracle Platform Security Services	00:00.103(ms)
✓	Audit Services	00:00.103(ms)
✓	Audit Services Append	00:00.131(ms)
✓	Audit Services Viewer	00:00.123(ms)
✓	Weblogic Services	00:00.122(ms)

Operation completed. Click OK to continue to next page.

OK

Enter the credentials Password : 'Manager1' for the schemas and click the "Next" button.



Repository Creation Utility

ORACLE
FUSION MIDDLEWARE

Define passwords for main and auxiliary schema users.

☒ Use same passwords for all schemas

Password:
Alpha numeric only. Cannot start with a number.
No special characters except: \$, #, _.

Confirm Password:

☐ Use main schema passwords for auxiliary schemas

☐ Specify different passwords for all schemas

Help < Back Next > Finish Cancel

Optionally the "Manage Table spaces" button if you want, or accept the defaults by clicking the "Next" button.



Repository Creation Utility

ORACLE

FUSION MIDDLEWARE

Welcome

Create Repository

Database Connection Details

Select Components

Schema Passwords

Map Tablespaces

Summary

Completion Summary

Default and temporary tablespaces for the selected components appear in the table below.
To create new tablespaces or modify existing tablespaces, use the Manage Tablespaces Button

Manage Tablespaces

Component	Schema Owner	Default Tablespace	Temp Tablespace
Common Infrastructur...	ESS_STB	*ESS_STB	*ESS_IAS_TEMP
Oracle Platform Securit...	ESS_OPSS	*ESS_IAS_OPSS	*ESS_IAS_TEMP
Audit Services	ESS_IAU	*ESS_IAU	*ESS_IAS_TEMP
Audit Services Append	ESS_IAU_APPEND	*ESS_IAU	*ESS_IAS_TEMP
Audit Services Viewer	ESS_IAU_VIEWER	*ESS_IAU	*ESS_IAS_TEMP
Weblogic Services	ESS_WLS	*ESS_WLS	*ESS_IAS_TEMP

* Default tablespaces (specified in the configuration files) are to be created upon confirmation.

Help

< Back

Next >

Finish

Cancel

Accept the table space creation confirmation by clicking the "OK" button

Validating and Creating Tablespaces

✓	Check tablespace requirements for selected components	00:00.108(ms)
✓	Create tablespaces in the repository database	00:03.477(sec)

Operation completed. Click OK to continue to next page.

OK



Repository System Load in progress.

✓	Executing pre create operations	00:03.774(sec)
⌚	Common Infrastructure Services	00:02.518(sec)
	Audit Services Append	0
	Audit Services Viewer	0
	Weblogic Services	0
	Audit Services	0
	Oracle Platform Security Services	0
	Executing post create operations	0

Stop

Wait while the repository creation finish and click on close button to exit the RCU.



Repository Creation Utility

ORACLE
FUSION MIDDLEWARE

Welcome

Create Repository

Database Connection Details

Select Components

Schema Passwords

Map Tablespaces

Summary

Completion Summary

Database details:

Host Name
demo12c

Port
1521

Service Name
ORCL

Connected As
sys

Operation
System and Data Load concurrently

Execution Time
1 minute 20 seconds

RCU Logfile
C:\TEMP\RCU2021-04-29_03-02_1488581770\logs\rcu.log

Component Log Directory
C:\TEMP\RCU2021-04-29_03-02_1488581770\logs

View Log
rcu.log

Prefix for (prefixable) Schema ESS
Owners

Component	Status	Time	Logfile(Click to view)
Common Infrastructure Services	Success	00:09.400(sec)	stb.log
Oracle Platform Security Services	Success	00:14.797(sec)	opss.log
Audit Services	Success	00:12.073(sec)	iau.log
Audit Services Append	Success	00:09.193(sec)	iau_append.log
Audit Services Viewer	Success	00:09.203(sec)	iau_viewer.log
Weblogic Services	Success	00:13.220(sec)	wls.log

Help

< Back

Next >

Create

Close

The repository creation is complete



Create Domain

Launch the Configuration Wizard with the following command.

D:\app\oracle\product\12.2.4\oracle_common\common\bin\config.cmd

A screenshot of a Windows Command Prompt window titled "Administrator: Command Prompt". The window has a black background with white text. The command prompt shows the following sequence of commands and output:

```
D:\app\Oracle\product\12.2.4\oracle_common\bin>rcu.bat
      RCU Logfile: C:\TEMP\RCU2021-04-29_03-02_1488581770\logs\rcu.log
D:\app\Oracle\product\12.2.4\oracle_common\bin>cd\
D:\>cd D:\app\oracle\product\12.2.4\oracle_common\common\bin\
D:\app\Oracle\product\12.2.4\oracle_common\common\bin>config.cmd
```

Accept the "Create a new domain" option, enter the domain name at the end of the "Domain Location", and then click the "Next" button. In this case my domain was called "ess", so the path I used was "D:\app\oracle\config\domains\ess".



Configuration Type

ORACLE
FUSION MIDDLEWARE

Create Domain

Templates

Administrator Account

Domain Mode and JDK

Advanced Configuration

Configuration Summary

Configuration Progress

End Of Configuration

What do you want to do?

☒ Create a new domain

☐ Update an existing domain

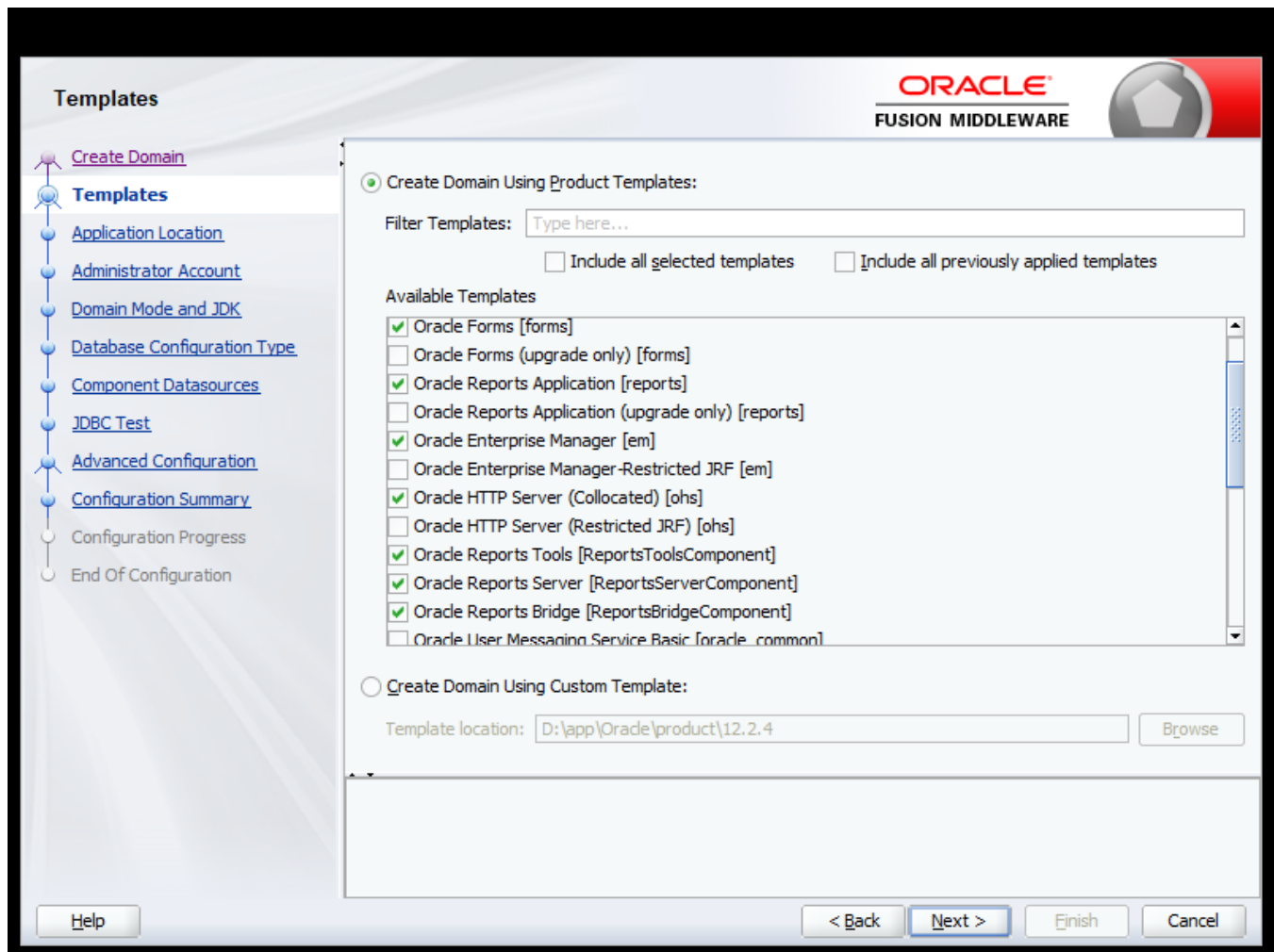
Domain Location: D:\app\oracle\config\domains\ess Browse

Create a new domain.

Help < Back Next > Finish Cancel

Select the required product templates and click the "Next" button. In this case, we need the following, which will bring some dependencies with them.

- Oracle Reports Server - 12.2.1 [ReportsServerComponent]
- Oracle Reports Bridge - 12.2.1 [ReportsBridgeComponent]
- Oracle Reports Application - 12.2.1 [reports]
- Oracle Reports Tools - 12.2.1 [ReportsToolsComponent]
- Oracle HTTP Server (Collocated) - 12.2.1 [ohs]
- Oracle Forms - 12.2.1 [forms]



Enter the application location and click the "Next" button. I used "D:\app\oracle\config\applications\ess".



Application Location

ORACLE
FUSION MIDDLEWARE

Create Domain
Templates
Application Location
Administrator Account
Domain Mode and JDK
Database Configuration Type
Component Datasources
JDBC Test
Advanced Configuration
Configuration Summary
Configuration Progress
End Of Configuration

Domain name: ess

Domain location: D:\app\orade\config\domains

Application location: D:\app\orade\config\applications\ess

Enter the admin name 'weblogic' and password 'weblogic1' then click the "Next" button



Administrator Account

Create Domain

Templates

Application Location

Administrator Account

Domain Mode and JDK

Database Configuration Type

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Configuration Progress

End Of Configuration

ORACLE®
FUSION MIDDLEWARE

Name

weblogic

Password

••••••••

Confirm Password

••••••••

Must be the same as the password. Password must contain at least 8 alphanumeric characters with at least one number or special character.

Help

< Back

Next >

Finish

Cancel

Enter the domain mode as Production and JDK details, then click the "Next" button



The screenshot shows the 'Domain Mode and JDK' configuration window in the Oracle Fusion Middleware Configuration Wizard. The left sidebar contains a navigation tree with the following items: 'Create Domain', 'Templates', 'Application Location', 'Administrator Account', 'Domain Mode and JDK' (selected), 'Database Configuration Type', 'Component Datasources', 'JDBC Test', 'Advanced Configuration', 'Configuration Summary', 'Configuration Progress', and 'End Of Configuration'. The main content area is titled 'Domain Mode and JDK' and features the Oracle Fusion Middleware logo in the top right corner. Under the 'Domain Mode' section, there are two radio buttons: 'Development' (unselected) and 'Production' (selected). The 'Development' option is described as 'Utilize boot.properties for username and password, and poll for applications to deploy.' The 'Production' option is described as 'Require the entry of a username and password, and do not poll for applications to deploy.' Under the 'JDK' section, there are two radio buttons: 'Oracle HotSpot 1.8.0_281 D:\app\Oracle\jdk' (selected) and 'Other JDK Location: [text box]'. A 'Browse' button is located to the right of the 'Other JDK Location' text box. At the bottom of the window, there are four buttons: 'Help', '< Back', 'Next >', and 'Finish'. The 'Next >' button is highlighted with a blue border.

Enter the repository database connection details:-

Host name = "Your computer name", Port as "Your port number", Schema Owner as "ESS_STB", Schema password as "Manager1". click the "Get RCU Configuration" button. If the connection is successful, click the "Next" button

Database Configuration Type

- [Create Domain](#)
- [Templates](#)
- [Application Location](#)
- [Administrator Account](#)
- [Domain Mode and JDK](#)
- [Database Configuration Type](#)**
- [Component Datasources](#)
- [JDBC Test](#)
- [Advanced Configuration](#)
- [Configuration Summary](#)
- [Configuration Progress](#)
- [End Of Configuration](#)



Specify AutoConfiguration Options Using:

☒ RCU Data ☐ Manual Configuration

Enter the database connection details using the schema credentials corresponding to Common Infrastructure Services component in the Repository Creation Utility. The Wizard uses this connection to automatically configure the datasources required for components in this domain.

Vendor: Oracle Driver: *Oracle's Driver (Thin) for Service connections; Version...

☒ Connection Parameters ☐ Connection URL String

Host Name: DEMO12C

DBMS/Service: ORCL Port: 1521

Schema Owner: ESS_STB Schema Password: *****

Connection Result Log

Successfully Done.

Click "Next" button to continue.

If you used the same password for all schemas, click the "Next" button. If not, fill in the correct passwords first.

If all the connection tests were successful, click the "Next" button.

JDBC Component Schema Test

- [Create Domain](#)
- [Templates](#)
- [Application Location](#)
- [Administrator Account](#)
- [Domain Mode and JDK](#)
- [Database Configuration Type](#)
- [Component Datasources](#)
- [JDBC Test](#)**
- [Advanced Configuration](#)
- [Configuration Summary](#)
- [Configuration Progress](#)
- [End Of Configuration](#)



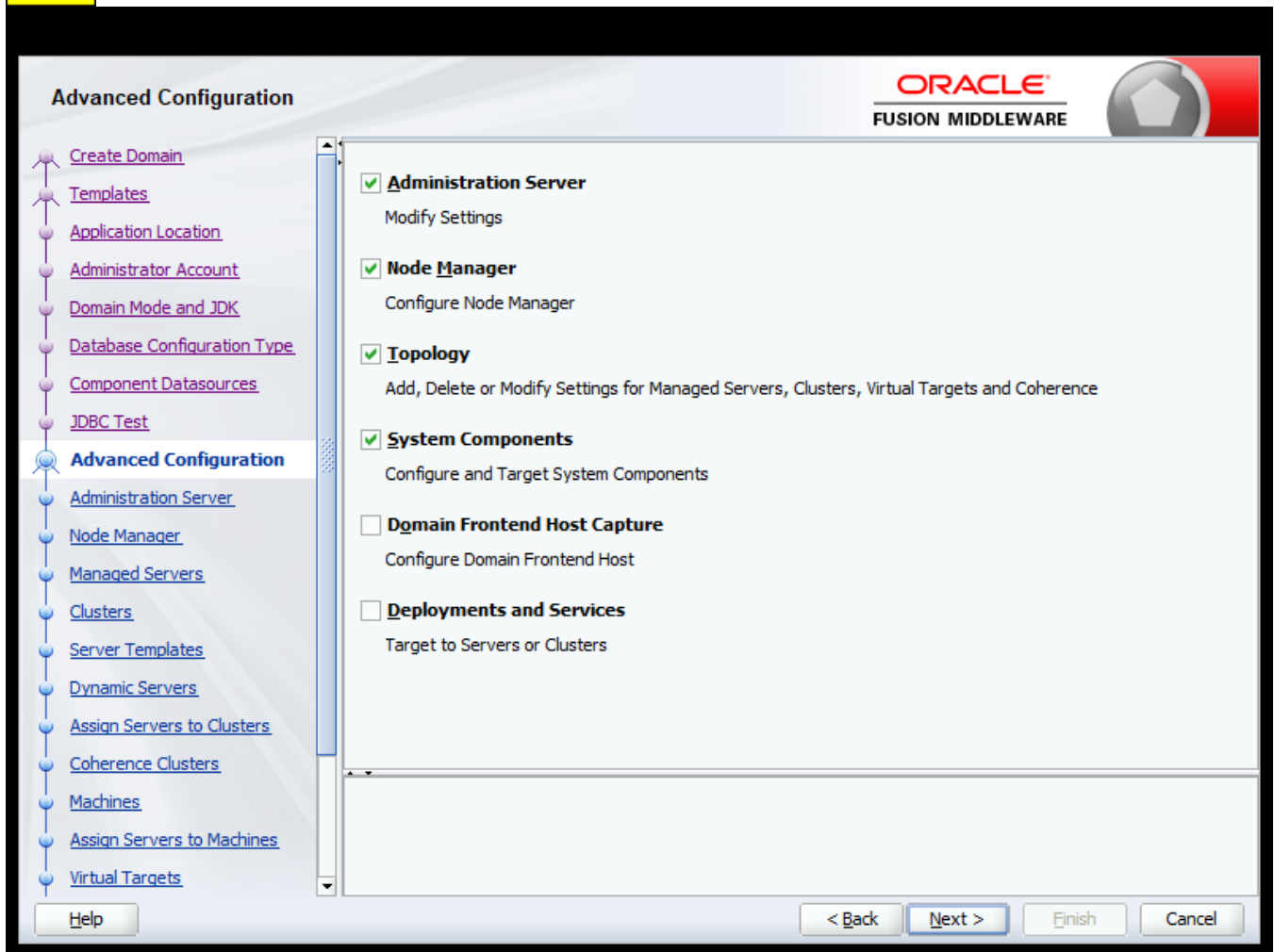
✓	Status	Component Schema	JDBC Connection URL
✓	✓	LocalSvcTbl Schema	jdbc:oracle:thin:@//demo12c:1521/ORCL
✓	✓	WLS Schema	jdbc:oracle:thin:@//demo12c:1521/ORCL
✓	✓	OPSS Audit Schema	jdbc:oracle:thin:@//demo12c:1521/ORCL
✓	✓	OPSS Audit Viewer Schema	jdbc:oracle:thin:@//demo12c:1521/ORCL
✓	✓	OPSS Schema	jdbc:oracle:thin:@//demo12c:1521/ORCL

Connection Result Log

Component Schema=LocalSvcTbl Schema
 Driver=oracle.jdbc.OracleDriver
 URL=jdbc:oracle:thin:@//demo12c:1521/ORCL
 User=ESS_STB
 Password=*****
 SQL Test=SELECT 1 FROM DUAL

CFGFWK-64213: Test Successful!
 CFGFWK-64213: JDBC connection test was successful

Select Administration Server, Node Manager, Topology and System components. And Click Next.



Put system name in Listener address and click Next



Administration Server

ORACLE
FUSION MIDDLEWARE

Create Domain

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Dynamic Servers

Assign Servers to Clusters

Coherence Clusters

Machines

Assign Servers to Machines

Virtual Targets

Server NameAdminServer

Listen Addressdemo12c

Listen Port7001

Enable SSL☐

SSL Listen Port

Server GroupsUnspecified

The name must not be null or empty and may not contain any : , = * ? % / _doned.

Help

< Back

Next >

Finish

Cancel

Enter Node Manager Credentials as weblogic/weblogic1 and click Next

Node Manager

- [Create Domain](#)
- [Templates](#)
- [Application Location](#)
- [Administrator Account](#)
- [Domain Mode and JDK](#)
- [Database Configuration Type](#)
- [Component Datasources](#)
- [JDBC Test](#)
- [Advanced Configuration](#)
- [Administration Server](#)
- Node Manager**
- [Managed Servers](#)
- [Clusters](#)
- [Server Templates](#)
- [Dynamic Servers](#)
- [Assign Servers to Clusters](#)
- [Coherence Clusters](#)
- [Machines](#)
- [Assign Servers to Machines](#)
- [Virtual Targets](#)




Node Manager Type

☒ Per Domain Default Location
☐ Per Domain Custom Location
☐ Manual Node Manager Setup

Node Manager Home:

Node Manager Credentials

Username:
 Password:
 Confirm Password:

Must be the same as the password. Password must contain at least 8 alphanumeric characters with at least one number or special character.

Replace Listen Address with your system name and click next



Managed Servers

ORACLE
FUSION MIDDLEWARE

Create Domain

Templates

Application Location

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Coherence Clusters

Machines

Assign Servers to Machines

Virtual Targets

+ Add

Clone

X Delete

Discard Changes

Server Name	Listen Address	Listen Port	Enable SSL	SSL Listen Port	Server Groups
WLS_FORMS	demo12c	9001	☐	Disabled	FORMS-MA...
WLS_REPORTS	demo12c	9002	☐	Disabled	REPORTS-...

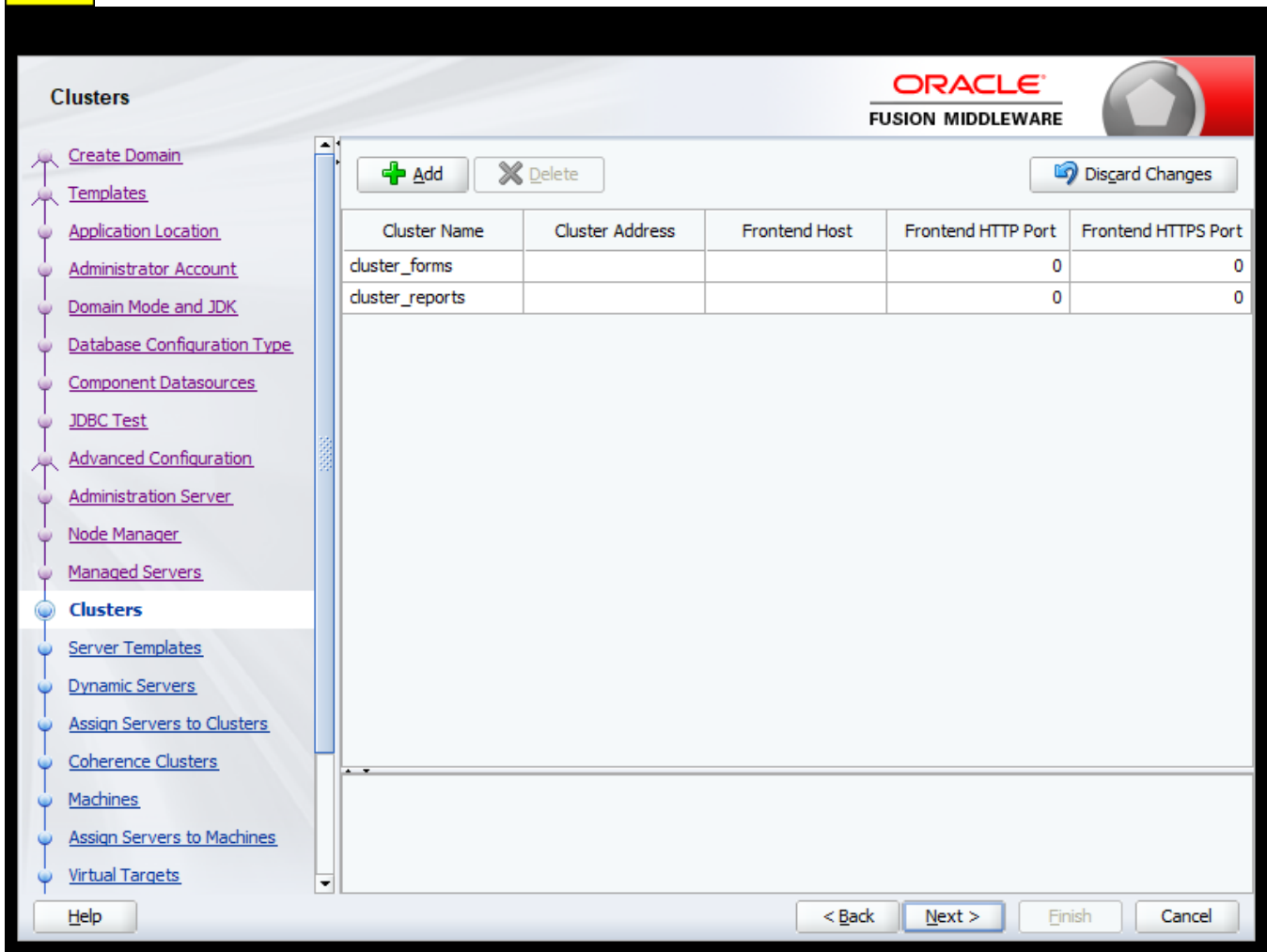
< Back

Next >

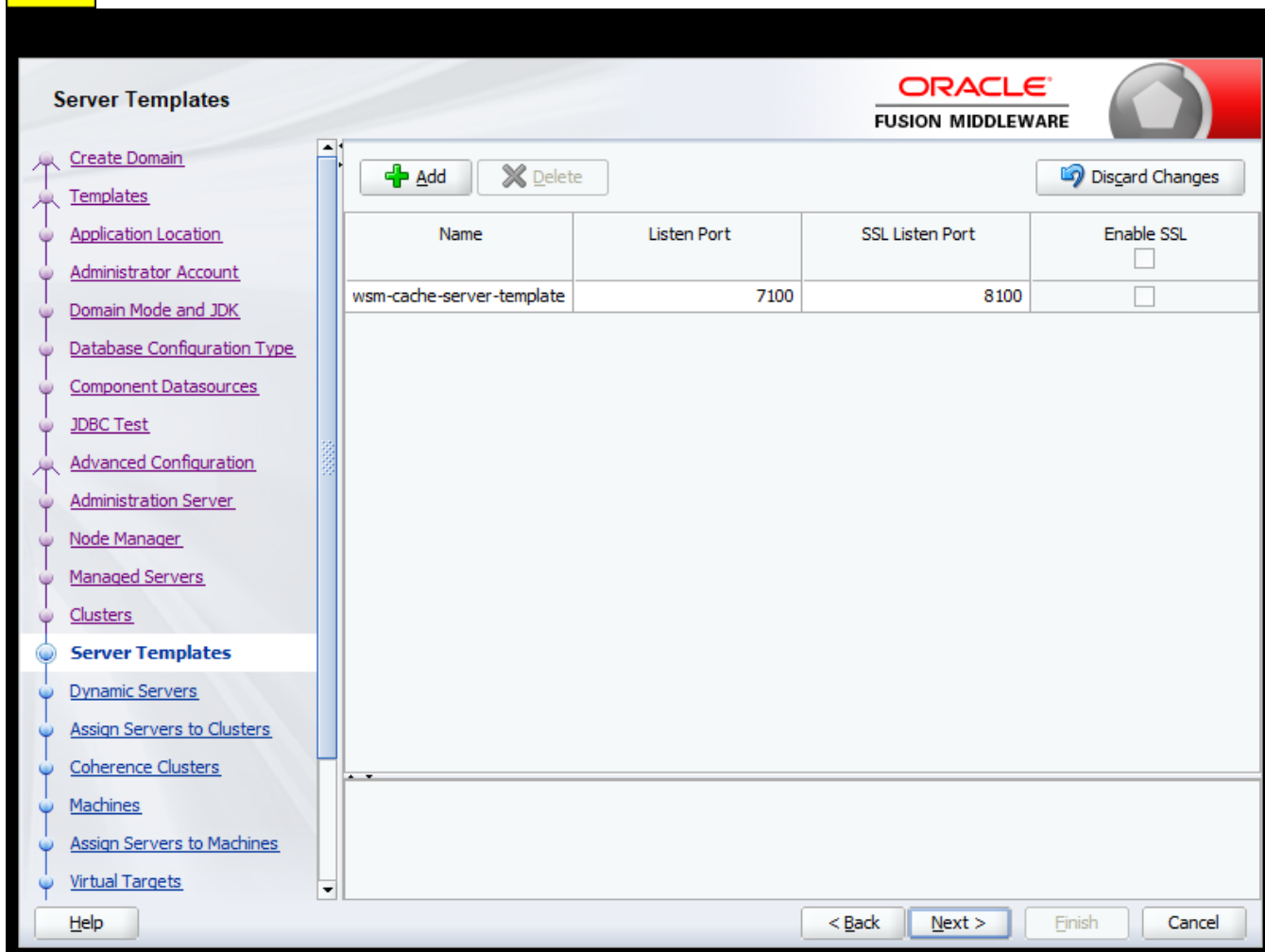
Finish

Cancel

Click Next



ADMINISTRATOR



ADMINISTRATOR



Dynamic Servers

ORACLE

FUSION MIDDLEWARE

Disard Changes

Create Domain

Templates

Application Location

Administrator Account

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Assign Servers to Clusters

Coherence Clusters

Machines

Assign Servers to Machines

Virtual Targets

Cluster Name

Server Name Prefix

Server Template

Dynamic Cluster Size

Machine Name Match Expression

Calculated Machine Names

Calculated Listen Ports

Dynamic Server Groups

cluster_forms	Disabled	Unspecif...	Disabled	Disabled	☐	☐	Unspecif...
cluster_repor	Disabled	Unspecif...	Disabled	Disabled	☐	☐	Unspecif...

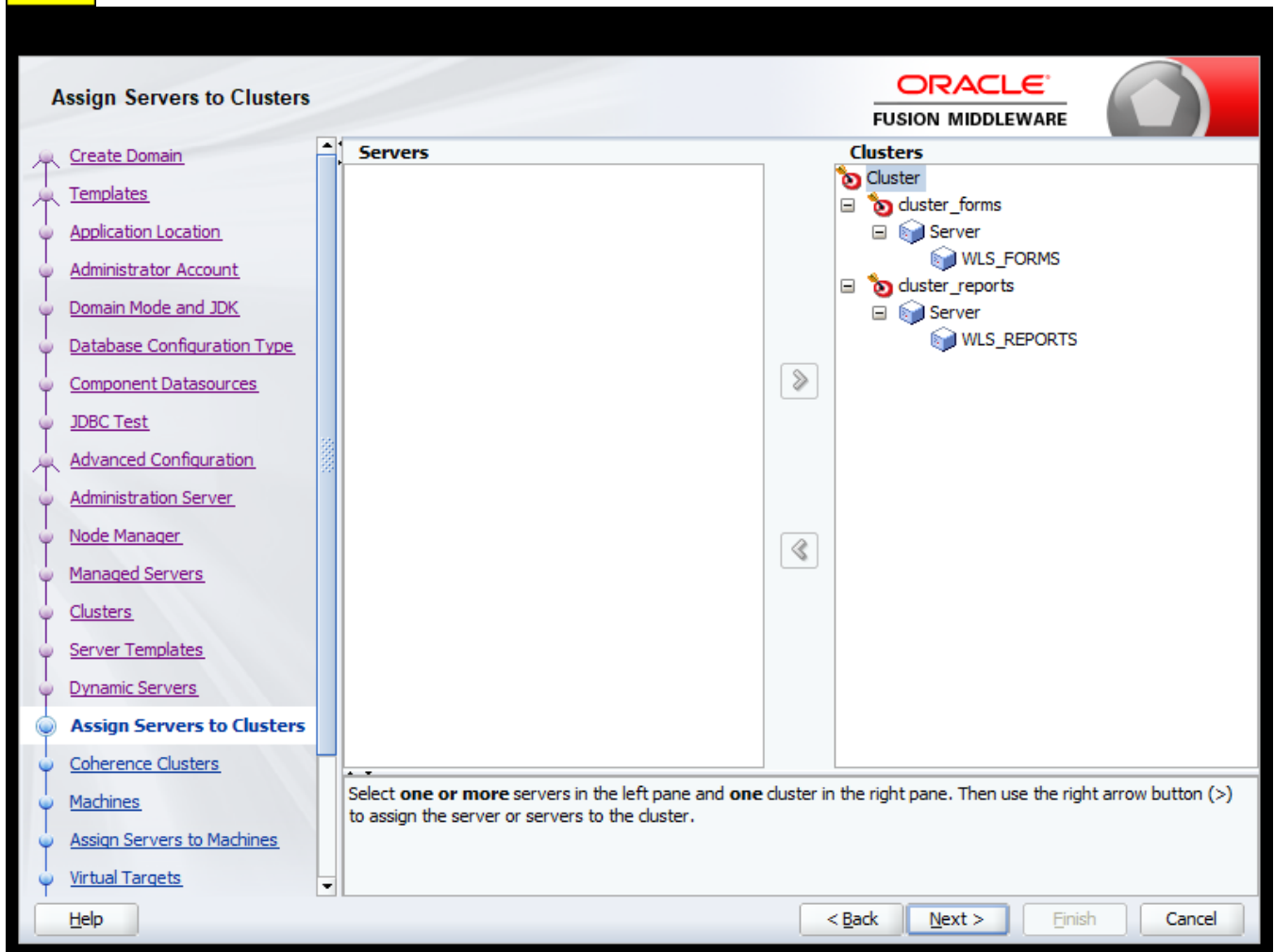
< Back

Next >

Finish

Cancel

Accept the defaults by clicking the "Next" button.



Accept the defaults by clicking the "Next" button



Coherence Clusters

ORACLE

FUSION MIDDLEWARE

Disard Changes

Cluster Name	Cluster Listen Port
defaultCoherenceCluster	7574

Help

< Back

Next >

Finish

Cancel

Create Domain

Templates

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Administrator Account

Domain Mode and JDK

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Dynamic Servers

Assign Servers to Clusters

Coherence Clusters

Machines

Assign Servers to Machines

Virtual Targets

Map AdminServer to AdminSeverMachine. Enter Node Manager listen Address as "Name of Machine" .then click "Next" button.



Machines

ORACLE
FUSION MIDDLEWARE

Create Domain

Templates

Application Location

Administrator Account

Domain Mode and JDK

Database Configuration Type

Component Datasources

JDBC Test

Advanced Configuration

Administration Server

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Dynamic Servers

Assign Servers to Clusters

Coherence Clusters

Machines

Assign Servers to Machines

Virtual Targets

Machine

Unix Machine

Add

Delete

Discard Changes

Name	Node Manager Listen Address	Node Manager Listen Port
AdminServerMachine	demo12c	5556

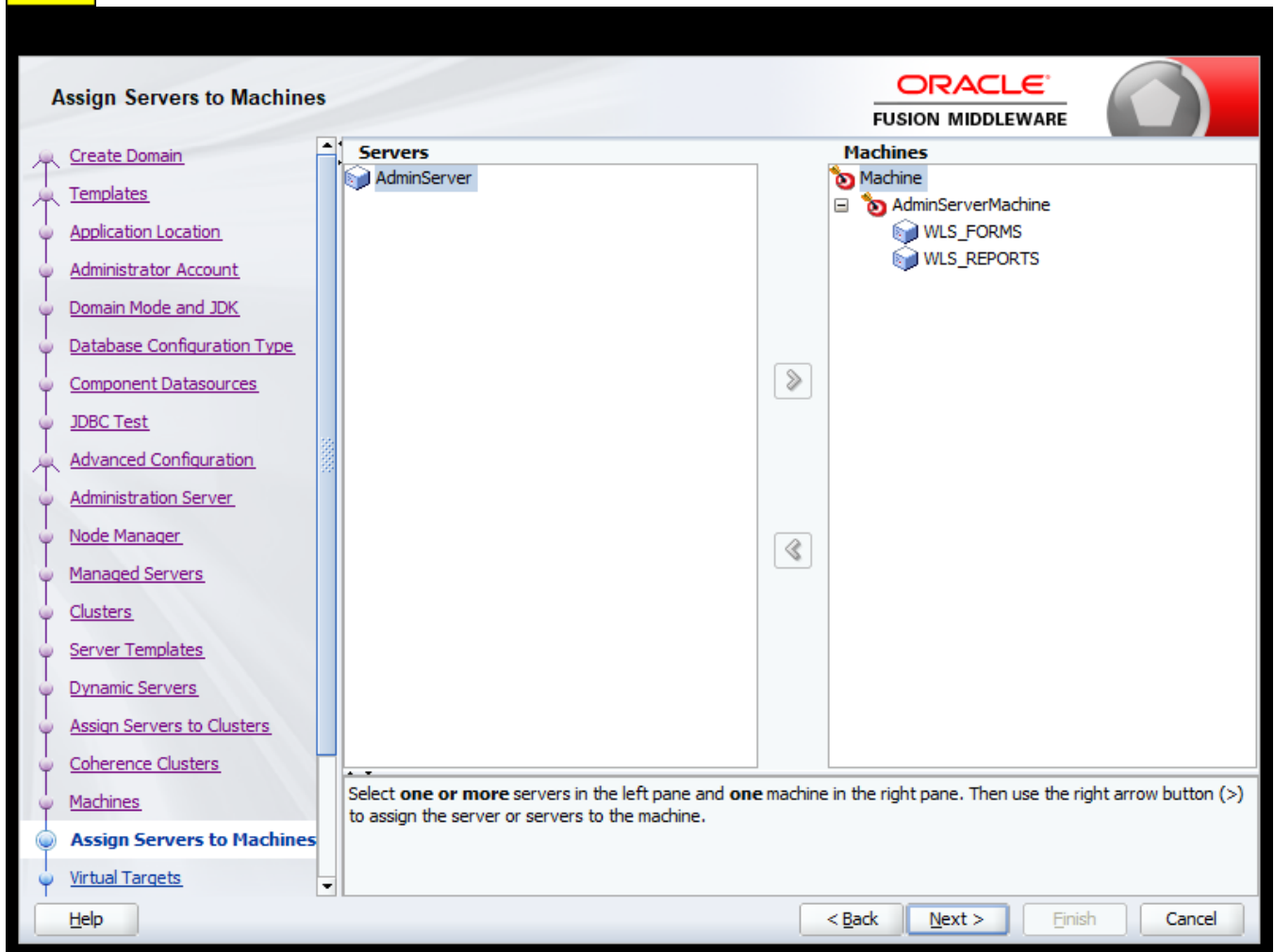
< Back

Next >

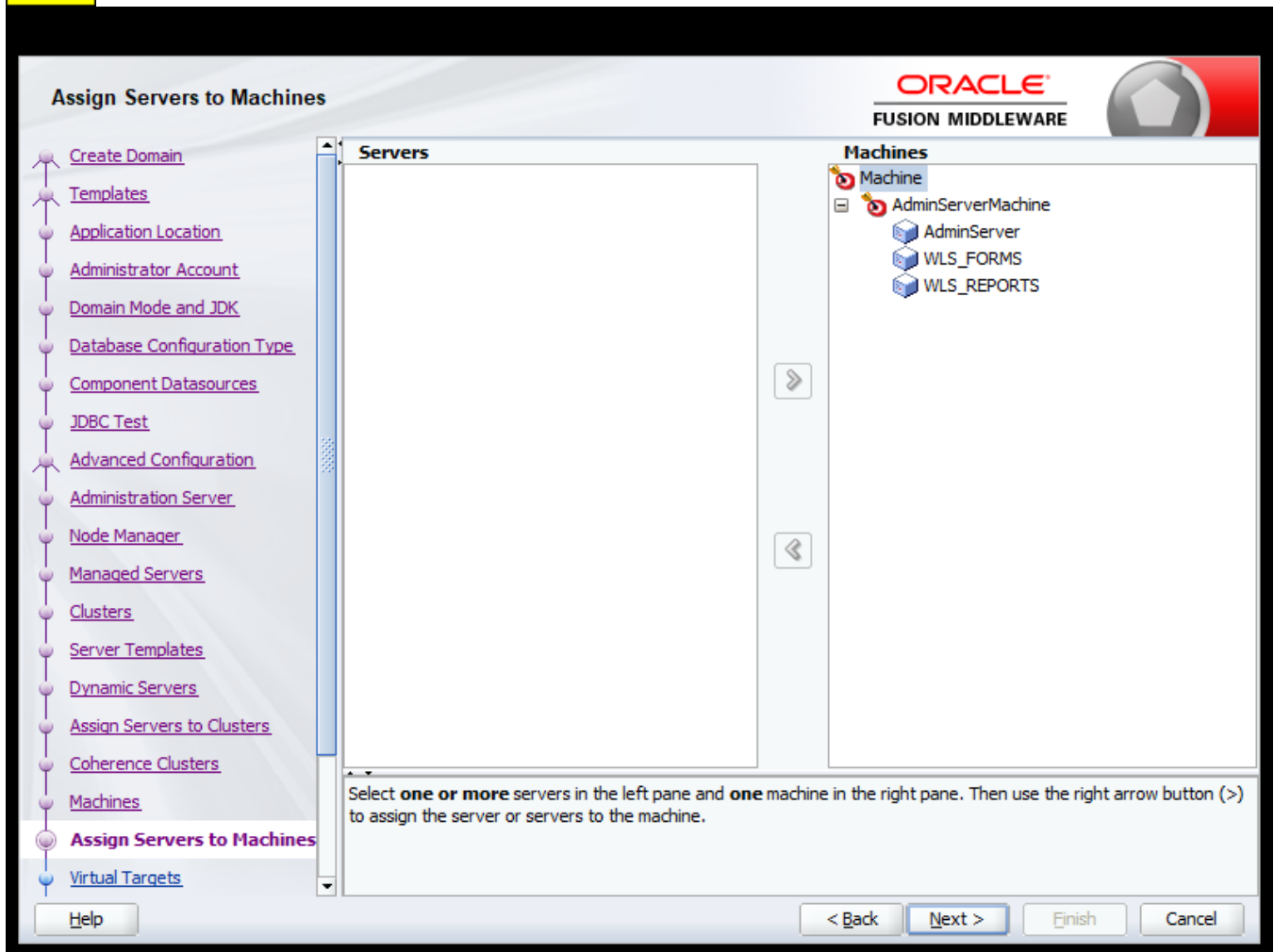
Finish

Cancel

Accept the defaults by clicking the "Next" button



Push the admin server to the right side.



Accept the defaults by clicking the "Next" button

Virtual Targets

- [Administrator Account](#)
- [Domain Mode and JDK](#)
- [Database Configuration Type](#)
- [Component Datasources](#)
- [JDBC Test](#)
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- [Administration Server](#)
- [Node Manager](#)
- [Managed Servers](#)
- [Clusters](#)
- [Server Templates](#)
- [Dynamic Servers](#)
- [Assign Servers to Clusters](#)
- [Coherence Clusters](#)
- [Machines](#)
- [Assign Servers to Machines](#)
- Virtual Targets**
- [Partitions](#)
- [Configuration Summary](#)
- [Configuration Progress](#)

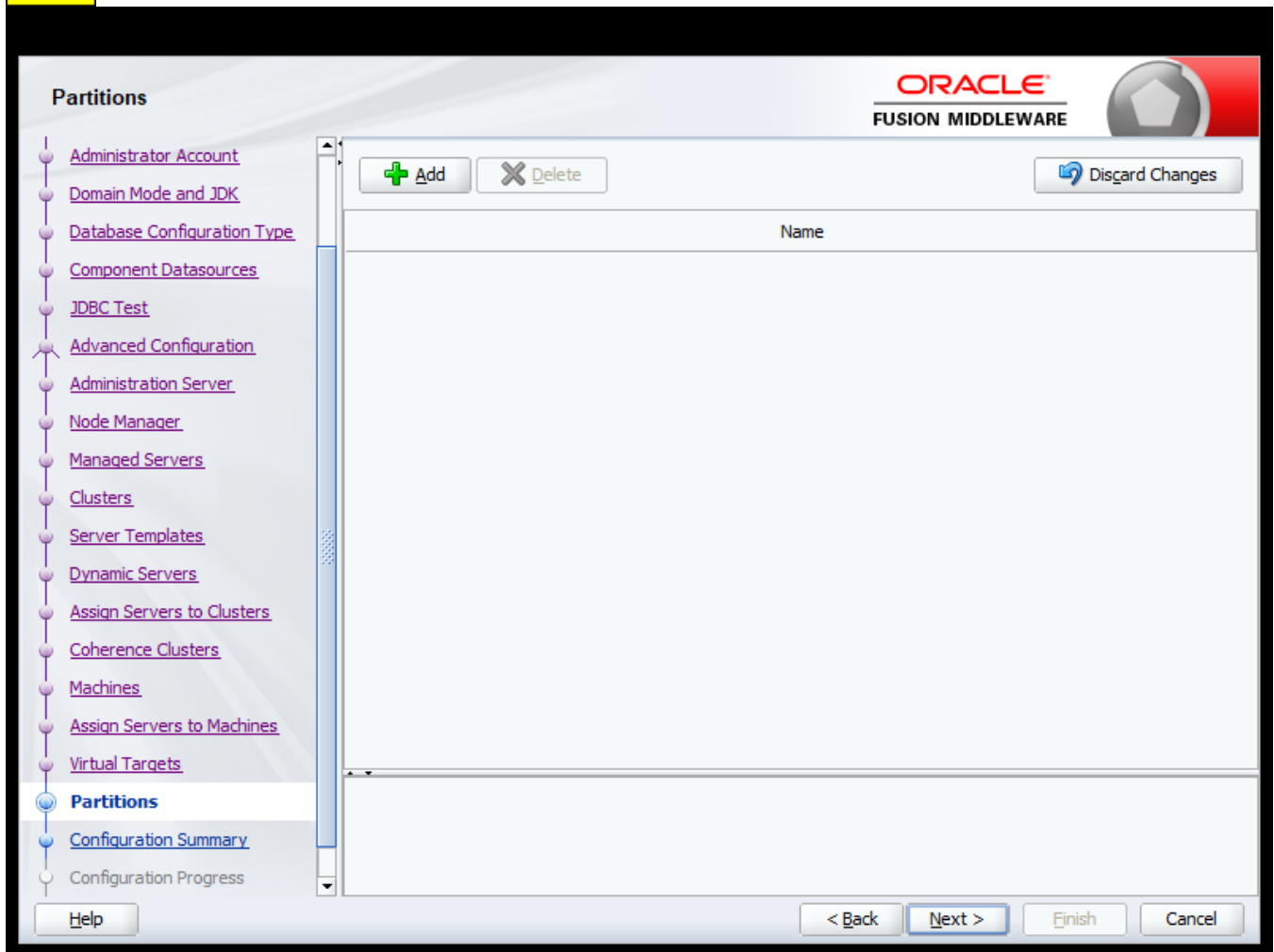
+ Add
 X Delete

 Discard Changes

Name	Target	Host Names	URI Prefix	Explicit Port	Port Offset

< Back
Next >
Finish
Cancel

Accept the defaults by clicking the "Next" button



Click the "+ Add" button to add the OHS component (ohs1), then click the "Next" button.



System Components

Administrator Account

Domain Mode and JDK

Database Configuration Type

Component Datasources

JDBC Test

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Server Templates

Dynamic Servers

Assign Servers to Clusters

Coherence Clusters

Machines

Assign Servers to Machines

Virtual Targets

Partitions

System Components

Assign System Components

Help

ORACLE
FUSION MIDDLEWARE

+ Add

✖ Delete

Discard Changes

System Component	Component Type	Restart Interval Seconds	Restart Delay Seconds
forms1	FORMS	3600	0
ohs1	OHS	3600	0

< Back

Next >

Finish

Cancel

Map the system components to the managed server that controls them. Enter listen Address as "Name of Machine". Click the "Next" button.



OHS Server

Administrator Account

Domain Mode and JDK

Database Configuration Type

Component Datasources

JDBC Test

Advanced Configuration

Administration Server

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Assign Servers to Clusters

Coherence Clusters

Machines

Assign Servers to Machines

Virtual Targets

Partitions

System Components

OHS Server

Help

< Back

Next >

Finish

Cancel

ORACLE
FUSION MIDDLEWARE

System Component

ohs1

Admin Host

127.0.0.1

Admin Port

7779

Listen Address

demo12c

Listen Port

7777

SSL Listen Port

4443

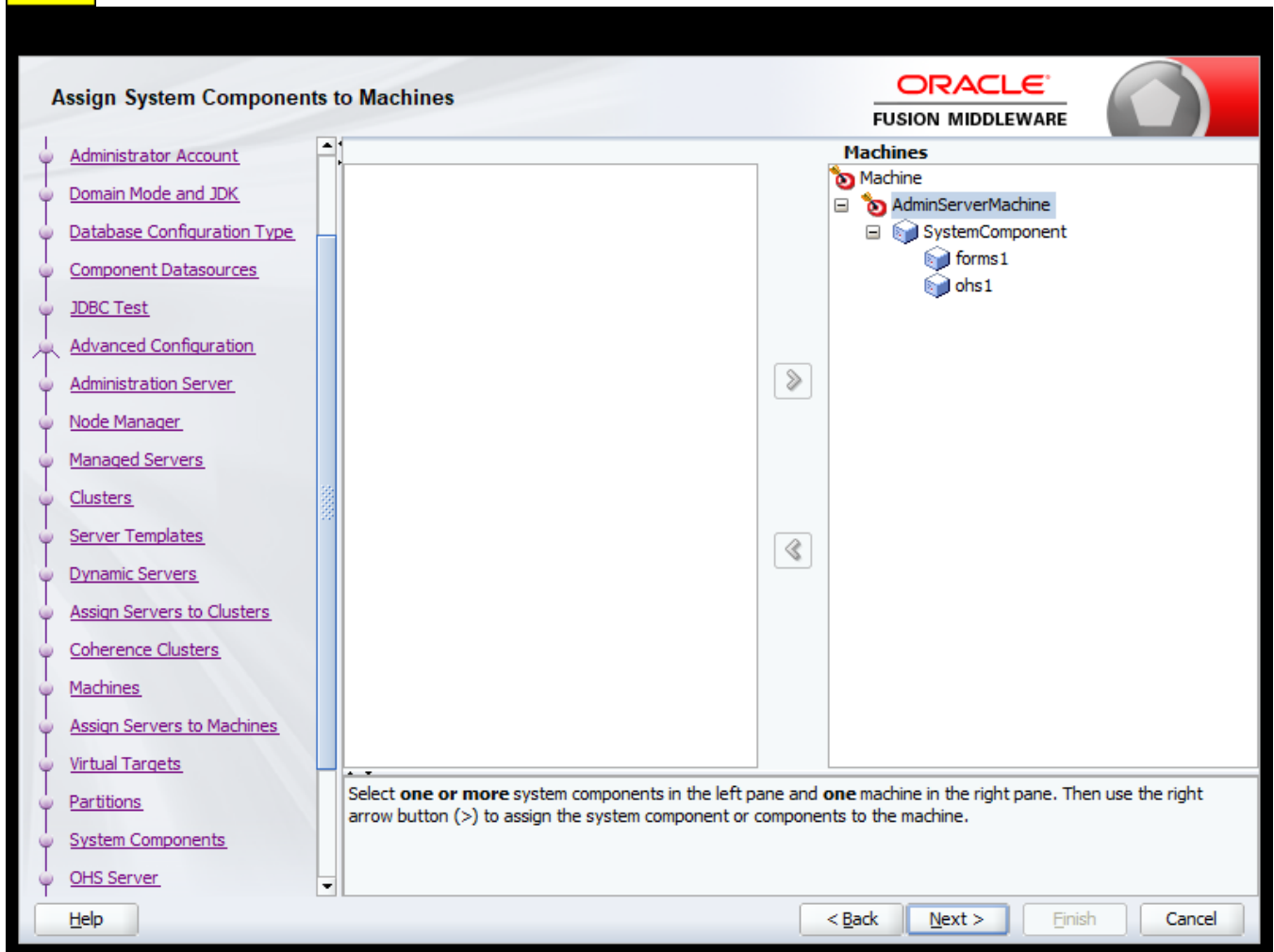
Server Name

http://demo12c:7777

Push ohs1 to right side. Click on Next .

ADMINISTRATOR

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Click the "Create" button.



Configuration Summary

Administrator Account

Domain Mode and JDK

Database Configuration Type

Component Datasources

JDBC Test

Advanced Configuration

Administration Server

Node Manager

Managed Servers

Clusters

Server Templates

Dynamic Servers

Assign Servers to Clusters

Coherence Clusters

Machines

Assign Servers to Machines

Virtual Targets

Partitions

System Components

OHS Server

Help

View: Deployment

ess (D:\app\orade\config\domains\ess)

Server

WLS_FORMS

WLS_REPORTS

AdminServer

AdminServer

AppDeployment

opss-rest

DMS Application #12.2.1.1.0

em

state-management-provider-memory-r

coherence-transaction-rar

Library

orade.bi.adf.model.slib#1.0@12.2.1.3

webtier.em

emcoreintsdk_jar#11.2.0.1.0@12.1.0.

orade.sdp.messaging#2.0@12.2.1.3.0

emcoreconsole_jar

adf.orade.domain.webapp.guava#1.0

orade.wsm.seedpolicies#2.0@12.2.1.3

orade.adf.desktopintegration.model#1

Name

Description

Author

Location

Basic WebLogic Server Domain

Create a basic WebLogic Server dom

Oracle Corporation

D:\app\Oracle\product\12.2.4\wlserv

Name

Description

Author

Location

Orade Reports Server

Orade Reports Server Component

Oracle Corporation

D:\app\Oracle\product\12.2.4\Report

Name

Description

Author

Location

Orade Reports Bridge

Orade Reports Bridge Component

Oracle Corporation

D:\app\Oracle\product\12.2.4\Report

Name

Description

Author

Location

Orade Reports Application

Orade Reports Application Template

Oracle Corporation

D:\app\Oracle\product\12.2.4\report

Name

Description

Author

Location

Orade Reports Tools

Orade Reports Tools Component

Oracle Corporation

D:\app\Oracle\product\12.2.4\Report

Select **Create** to accept the above options and start creating and configuring a new domain. To change the above configuration before starting Domain Creation, go back to the relevant page by selecting its name in the left pane, or by using the **Back** button.

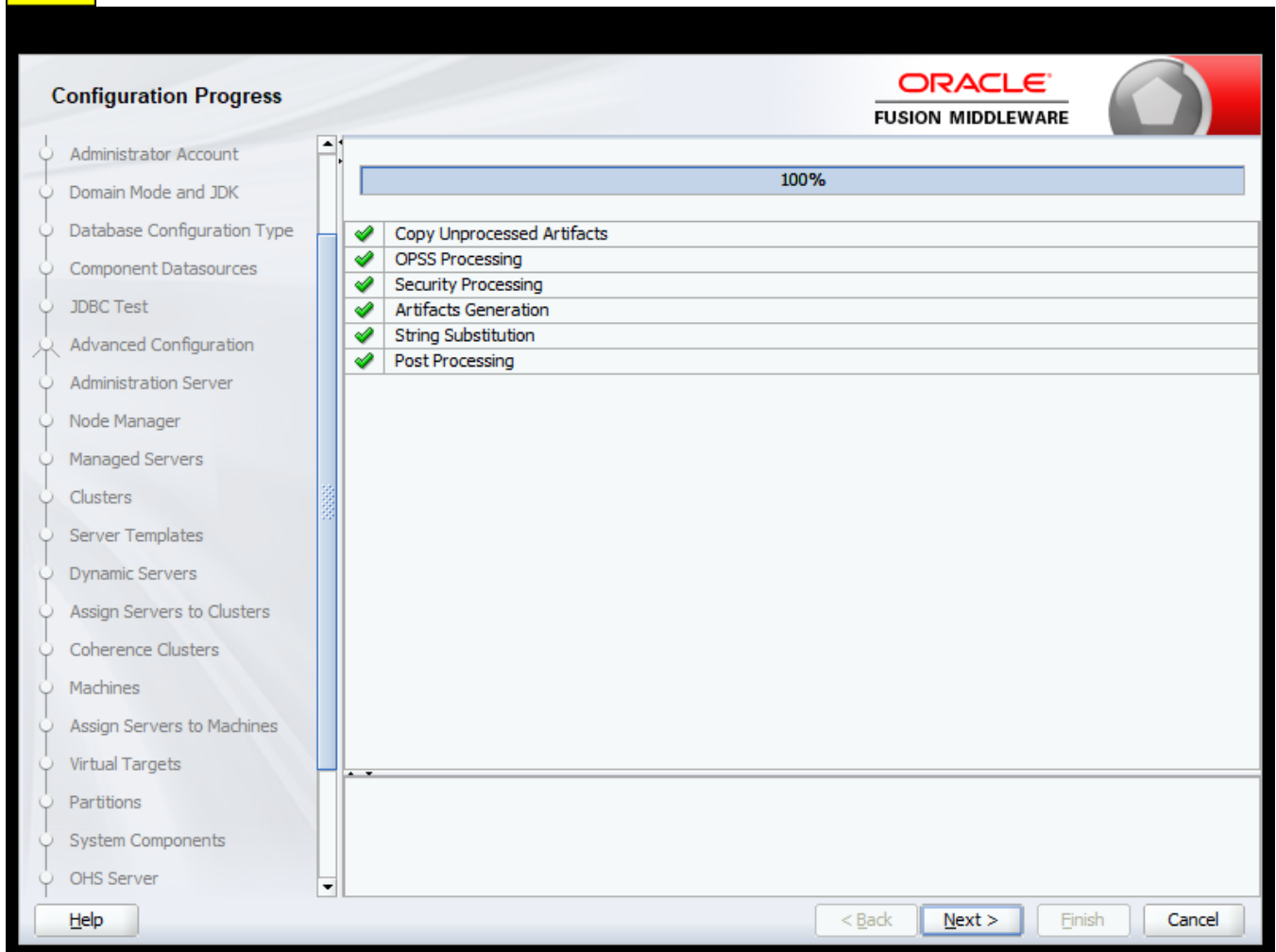
< Back

Next >

Create

Cancel

Once the domain is created, click the "Next" button.



Make a note of the Admin Server URL and click the "Finish" button.



End Of Configuration

Administrator Account

Domain Mode and JDK

Database Configuration Type

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OHS Server

Help

ORACLE
FUSION MIDDLEWARE

✓

Oracle Weblogic Server Configuration Succeeded

New Domain ess Creation Succeeded

Domain Location
<D:/app/oracle/config/domains/ess>

Admin Server URL
<http://demo12c:7001/console>

After successful domain creation, you must run the WLST command `ohs_updateInstances()` to complete all the required steps. Please refer to OHS product documentation for more details.

< Back

Next >

Finish

Cancel

ADMINISTRATOR

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Additional Configuration

(If you chose the "Production Mode" options for the domain, you will need to create a "boot. Properties" file for the scripts referred to later to work without credentials. Adjust the DOMAIN_HOME and credentials appropriately) **

Open Command prompt and copy these:-

```
mkdir D:\app\oracle\config\domains\ess\servers\AdminServer\security
```

```
echo username=weblogic> D:\app\oracle\config\domains\ess\servers\AdminServer\security\boot.properties
```

```
echo password=weblogic1>>D:\app\oracle\config\domains\ess\servers\AdminServer\security\boot.properties
```

```
mkdir D:\app\oracle\config\domains\ess\servers\WLS_FORMS\security
```

```
echo username=weblogic> D:\app\oracle\config\domains\ess\servers\WLS_FORMS\security\boot.properties
```

```
echo password=weblogic1>> D:\app\oracle\config\domains\ess\servers\WLS_FORMS\security\boot.properties
```

```
mkdir D:\app\oracle\config\domains\ess\servers\WLS_REPORTS\security
```

```
echo username=weblogic> D:\app\oracle\config\domains\ess\servers\WLS_REPORTS\security\boot.properties
```

```
echo password=weblogic1>>
```

```
D:\app\oracle\config\domains\ess\servers\WLS_REPORTS\security\boot.properties
```



```
Administrator: Command Prompt - D:\app\oracle\config\domains\ess\startWebLogic.cmd
Microsoft Windows [Version 10.0.19042.906]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>cd\

C:\>mkdir D:\app\oracle\config\domains\ess\servers\AdminServer\security

C:\>echo username=weblogic> D:\app\oracle\config\domains\ess\servers\AdminServer\security\boot.properties

C:\>echo password=weblogic1>>D:\app\oracle\config\domains\ess\servers\AdminServer\security\boot.properties

C:\>mkdir D:\app\oracle\config\domains\ess\servers\WLS_FORMS\security

C:\>echo username=weblogic> D:\app\oracle\config\domains\ess\servers\WLS_FORMS\security\boot.properties

C:\>echo password=weblogic1>> D:\app\oracle\config\domains\ess\servers\WLS_FORMS\security\boot.properties

C:\>mkdir D:\app\oracle\config\domains\ess\servers\WLS_REPORTS\security

C:\>echo username=weblogic> D:\app\oracle\config\domains\ess\servers\WLS_REPORTS\security\boot.properties

C:\>echo password=weblogic1>D:\app\oracle\config\domains\ess\servers\WLS_REPORTS\security\boot.properties

C:\>D:\app\oracle\config\domains\ess\startWebLogic.cmd

.
.
JAVA Memory arguments: -Xms512m -Xmx1024m
.
CLASSPATH=D:\app\Oracle\product\12.2.4\oracle_common\modules\features\com.oracle.db.jdbc7-dms.jar;D:\app\Oracle\jdk\lib\
tools.jar;D:\app\Oracle\product\12.2.4\wlserver\server\lib\weblogic.jar;D:\app\Oracle\product\12.2.4\oracle_common\modul
```

Once the server is started you can access the administrator console using the "http://hostname:port/console" URL . Log in using (User name : Weblogic and Password : weblogic1)



Start/Stop WebLogic

Use following in command prompt to start/stop WebLogic:-

Start Node Manager

```
D:\app\oracle\config\domains\ess\bin\startNodeManager.cmd
```

Start WebLogic Domain

```
D:\app\oracle\config\domains\ess\bin\startWebLogic.cmd
```

Start the managed Servers

```
D:\app\oracle\config\domains\ess\bin\startManagedWebLogic.cmd WLS_FORMS  
D:\app\oracle\config\domains\ess\bin\startManagedWebLogic.cmd WLS_REPORTS
```

Stop the managed Servers

```
D:\app\oracle\config\domains\ess\bin\stopManagedWebLogic.cmd WLS_FORMS  
D:\app\oracle\config\domains\ess\bin\stopManagedWebLogic.cmd WLS_REPORTS
```

Stop WebLogic Domain D:\app\oracle\config\domains\ess\bin\stopWebLogic.cmd

HTTP Setup

Since OPMN is no longer used, starting the HTTP Server must be done using the Enterprise Manager console as below.

Start WebLogic Domain

```
D:\app\oracle\config\domains\ess\bin\startWebLogic.cmd
```



Open chrome and open **http://hostname:port/em** (<http://demo12c:7001/em>)

User name : weblogic. Password : weblogic1

Log into the Enterprise Manager console.

Click on the "Target Navigation" icon.

Expand the "HTTP Server" node.

Click on the "ohs1" instance.

The "ohs1" page has "Start Up" and "Shut Down" links at the top.

****** (If you have any problems starting the OHS, start node manager and try again. If HTTP Server fails to start even after starting node manager then try installing the following windows patches:

<http://www.microsoft.com/enus/download/confirmation.aspx?id=30679>

<http://www.microsoft.com/enus/download/details.aspx?id=16771>

<http://www.microsoft.com/en-us/download/details.aspx?id=2092>)

****** (The HTTP Server must be started once, for the directories associated with "ohs1" instance to be created. The web tier is not configured by default, so you will need to do the following.)

Edit the forms.conf file present in path:-"D:\app\oracle\product\12.2.4\forms\templates\config\forms.conf"
And Replace "System name" with your system name and "port" with the forms port number as below:-

<Location /forms>

SetHandler

weblogic-handler WebLogicHost "System name"

WebLogicPort "port"

</Location>

****Sample

<Location /forms>



SetHandler weblogic-handler

WebLogicHost demo12c

WebLogicPort 9001

</Location>

Edit the "D:\app\oracle\product\12.2.4\reports\conf\reports_ohs.conf" file to contain the following.

Replace "System name" with your system name and "port" with the Report port number.

<Location /reports>

SetHandler weblogic-handler

WebLogicHost "System name"

WebLogicPort "port"

</Location>

*****Sample

<Location /reports>

SetHandler weblogic-handler

WebLogicHost Demo12c

WebLogicPort 9002

</Location>



Copy “forms.conf” from path:-

D:\app\oracle\product\12.2.4\forms\templates\config

And paste in “moduleconf” folder present in path:-

D:\app\oracle\config\domains\ess\config\fmwconfig\components\OHS\instances\ohs1\moduleconf

Similarly,

Copy “reports_ohs.conf” from path:-

D:\app\oracle\product\12.2.4\reports\conf

And paste in “moduleconf” folder present in path:-

D:\app\oracle\config\domains\ess\config\fmwconfig\components\OHS\instances\ohs1\moduleconf

Restart the HTTP Server to take effect



Opening Forms builder

D:\app\oracle\product\12.2.4\bin\frmbld.exe

Opening Reports builder

You need to perform the following steps one time after installation in order to start reports builder

1- Execute wlst.cmd from

D:\app\oracle\product\12.2.4\oracle_common\common\bin

2-Connect to AdminServer in Command:

```
connect("weblogic","weblogic_password","hostname:7001")
```

Run the following wlst command.

```
createReportsToolsInstance(instanceName='reptools1',machine='AdminServerMachine')
```

```
Administrator: Command Prompt - wlst.cmd
C:\>d:

D:\app\Oracle\product\12.2.4\oracle_common\common\bin>wlst.cmd

Initializing WebLogic Scripting Tool (WLST) ...

Welcome to WebLogic Server Administration Scripting Shell

Type help() for help on available commands

wls:/offline> connect("weblogic","weblogic1","demo12c:7001")
Connecting to t3://demo12c:7001 with userid weblogic ...
Successfully connected to Admin Server "AdminServer" that belongs to domain "ess".

Warning: An insecure protocol was used to connect to the server.
To ensure on-the-wire security, the SSL port or Admin port should be used instead.

wls:/ess/serverConfig/> createReportsToolsInstance(instanceName='reptools1',machine='AdminServerMachine')
Location changed to edit custom tree. This is a writable tree with No root.
For more help, use help('editCustom')

Starting an edit session ...
Started edit session, be sure to save and activate your changes once you are done.
Saving all your changes ...
Saved all your changes successfully.
Activating all your changes, this may take a while ...
The edit lock associated with this edit session is released once the activation is completed.
Activation completed
Reports Tools instance "reptools1" was successfully created.
wls:/ess/serverConfig/>
```

Restart WLS_REPORTS

At this moment reports builder can be used:

D:\app\oracle\config\domains\ess\reports\bin\rwbuilder.bat



Create standalone reports server:

Execute `wlst.cmd` from

`D:\app\oracle\product\12.2.4\oracle_common\common\bin`

Connect to the AdminServer.

```
connect("weblogic","weblogic_password","hostname:7001")
```

Run the following WLST command.

```
createReportsServerInstance(instanceName='rep_serverdemo12c',machine='AdminServerMachine')
```

At this point-in-time, the standalone reports server can be used.

Start/Stop reports Server:

Standalone:

```
cd "Domain home"\bin
```

```
cd D:\app\oracle\config\domains\ess\bin startComponent.cmd rep_serverdemo12c
```

or

```
stopComponent.cmd rep_serverdemo12c
```

in-process:

Execute any of the following URLs:

<http://hostname:9002/reports/rwservlet/startserver> or

<http://hostname:9002/reports/rwservlet/stopserver>



Additional Changes for Report Server

Oracle Reports Integration Requires New Environment Variable:

For applications that use the Oracle Forms built-in RUN_REPORT_OBJECT to integrate with Oracle Reports, it will be necessary to add a new environment variable to the Forms environment configuration (that is, default.env).

The new variable name is COMPONENT_CONFIG_PATH and its value should be the fully qualified path to the Reports Tools Component. For example:

DOMAIN_HOME/config/fmwconfig/components/ReportsToolsComponent/<reports_to_ols_component_name>

Add environment variable end of the default.env files

Example

COMPONENT_CONFIG_PATH=D:\app\Oracle\config\domains\ess\config\fmwconfig\components\ReportsToolsComponent\reptools1

Check report server status

Open internet explorer and run showjobs command Example:-

<http://localhost:9002/reports/rwservlet/showjobs?>



Remove Security Setting

In case if it is asking for user name and password, do the following settings:

Step 1: In oracle middleware home search for rwserver.conf, edit and comment the line given below

Before

```
<security id="rwJaznSec" class="oracle.reports.server.RWJAZNSecurity"/>
```

After

```
<!--security id="rwJaznSec" class="oracle.reports.server.RWJAZNSecurity"/--> Do same settings in all  
rwserver.conf files.
```

Step 2: Search for rwservlet.properties and edit line as given below

Before

```
<!--webcommandaccess>L1</webcommandaccess-->
```

After

```
<webcommandaccess>L2</webcommandaccess>
```

Increase no. of report server

Change rwserver.conf at two folder

```
initEngine="1" maxEngine="10" minEngine="1" engLife="50" maxConnect="500"
```

Restart report server WLS_REPORTS



Basic URL

The basic URLs of importance are shown below. Ports may vary between installations.

Console:

<http://localhost:7001/console>

Enterprise Manager :

<http://localhost:7001/em>

Forms:

<http://localhost:9001/forms/frmservlet>

Or

<http://localhost:7777/forms/frmservlet>

Reports

<http://localhost:9002/reports/rwservlet>

Or

<http://localhost:7777/reports>



Web Start (OPTIONAL)

Oracle Forms can easily be run using Java Web Start, freeing you from issues with browser compatibility. For this to work, you need a Java 8 JRE installed on the PC. To run using Web Start for the first time, simply issue the URL in the following form.

<http://hostname:9001/forms/frmservlet?config=webstart>

<http://hostname:7777/forms/frmservlet?config=webstart>

Depending on your browser, the Web Start version of the URL will either just run, or will ask you to save a ".jnlp" file on your computer. Once saved, you can double click on the file and it will launch Forms directly from the PC as a Web Start application, without using a browser.



Ebizframe configuration in 12c

Configuration of TNSNAMES.ORA

Copy tnsnames.ora file of database

D:\db_home\network\admin\tnsnames.ora

And paste it on following location

%ORACLE_FORMS_HOME%\config\domains\ess\config\fmwconfig\tnsnames.ora.

Configuration of formsweb.cfg

NOTE - .

Shut down the report server.

Please take back up of each file before making any modification.

Open D:\app\Oracle and search formsweb.cfg.

There will be three files with the name of formsweb.cfg but we need to make changes only in one file present in the location

D:\app\Oracle\config\domains\ess\config\fmwconfig\servers\WLS_FORMS\applications\formsapp_12.2.4\config\FORMSWEB.CFG

Modify this file as follow. **(Copy and paste not work WITH formsweb.cfg/default.env. Suggested to type these configuration).**

Edit these following parameters

WorkingDirectory=d:\ebiz\makeess\forms

FORM=MKLOGIN.FMX

USERID=MAKESS/MAKESS

HOST=<HOST NAME>

PORT=9001

CODEBASE=/FORMS/JAVA

IMAGEBASE=CODEBASE

WIDTH=100%

HEIGHT=100%

archive=jacob.jar,frmall.jar,frmwebutil.jar,mak_icons.jar,laf_11112.jar

(Hint → Enter your Icons JAR file name followed by the “,”

Copy and paste not work formsweb you may need to type the same)

NETWORKRETRIES=10

pageTitle=Ebizframe Neo

splashScreen=ebiz.gif

background= bg.jpg



Copy your mak_icons.jar file into

D:\app\ORACLE\product\12.2.4\forms\java

Configuration of default.env

Open D:\app\Oracle\ and search default.env.

There will be three files but we need to make changes only in one file present at

a) Open

D:\app\Oracle\config\domains\ess\config\fmwconfig\servers\WLS_FORMS\applications\formsapp_12.2.4\config\default.env

Modify these following parameters

FORMS_PATH parameter with existing value C:/ebiz1/icons;C:/ebiz1/lib;d:;c:; C:\ebiz\makess\forms

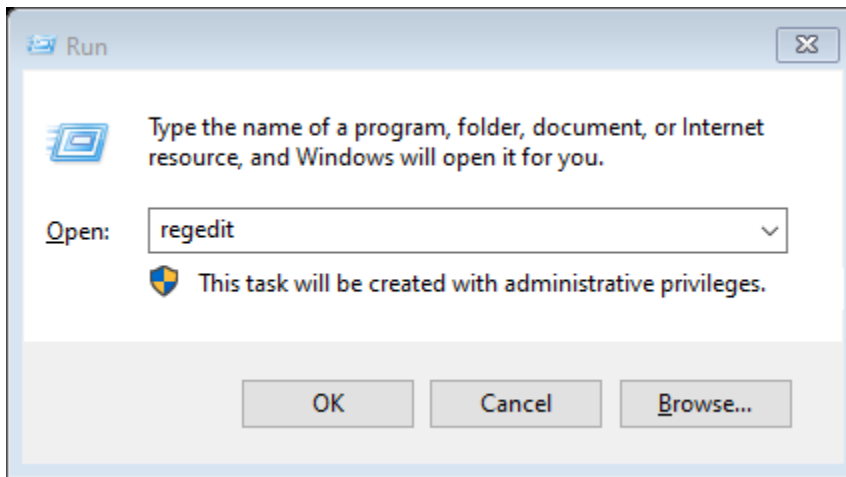
Add forms_mmap parameter with value false (FORMS_MMAP=FALSE)

Add forms_reject_go_disable_item parameter with value false (FORMS_REJECT_GO_DISABLE_ITEM=FALSE)

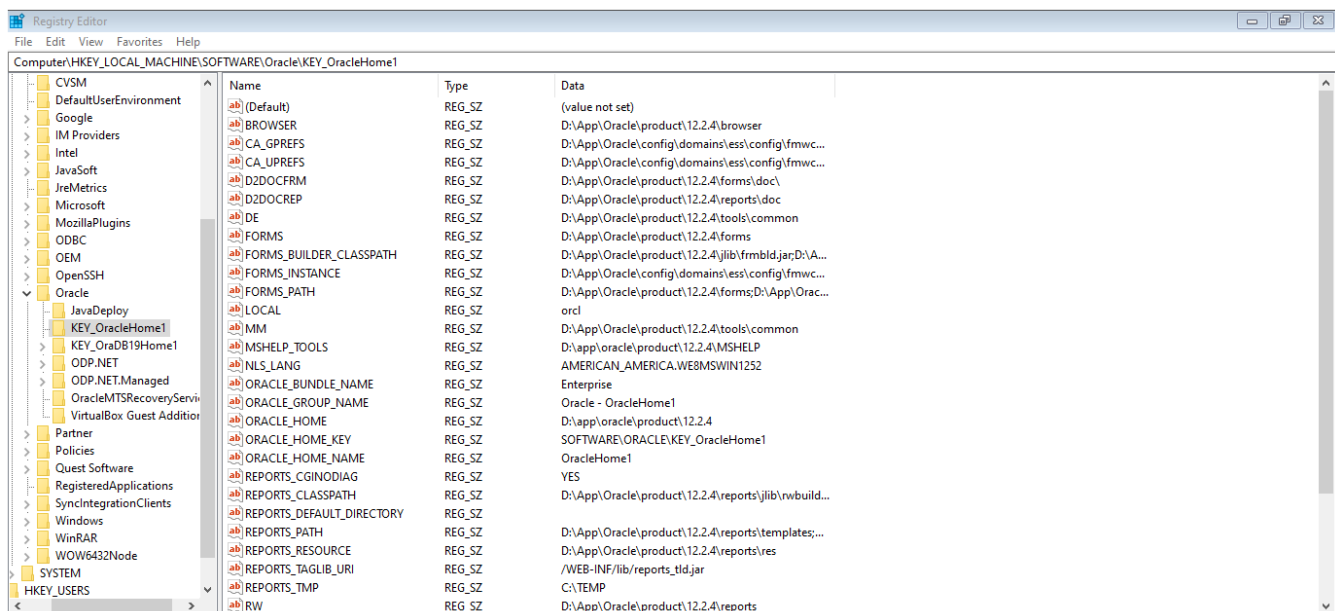


Change in the Windows Registry.

In the run enter regedit



Go to Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Oracle\KEY_OracleHome1



Edit FORMS_PATH add d:\ebiz\icons;d:\ebiz\lib

Edit REPORTS_PATH add d:\ebiz\icons;d:\ebiz\lib

Create 3 new STRING_VALUE as STRING

LOCAL with VALUE of ORCL.



UI_ICONS = d:\ebiz\icons need to specify

NLS_DATE_FORMAT=DD-MM-YYYY

Folder Access

The REVISED Reports 12c folderAccess wildcard (*) Feature is expected to function as follows :

- a. Reports will allow all the sub-directories below a directory like /dir/

```
<folderAccess>
<read>/dir/*</read>
<write>/dir/*</write>
</folderAccess>
```

Example 1 (single location):

```
<folderAccess>
<read>/tmp/*</read>
<write>/tmp/*</write>
</folderAccess>
```

desname=/tmp/a/b/output.pdf ==> tested OK

desname=/tmp/a/output.pdf ==> tested OK

desname=/tmp/output.pdf ==> tested OK

All sub-directories under /tmp were tested to work.

Example 2 (multiple locations):

```
<folderAccess>
<read>/tmp/*;/home/oracle/*</read>
<write>/tmp/*;/home/oracle/*</write>
</folderAccess>
```

Standalone Ebiz

**(Do when necessary)

NOTE: To run ebiz as standalone you need to first configure ebiz normally by using above steps.

TO OPEN EBIZ AS STANDALONE APP (Only if required):

Download frmsal.jar: open browser and type: <http://machine-name:9001/forms/html/fsal.htm>



A new page will open which will provide the link to download fsal.jar.

Download fsal from the new page.

Copy and paste the downloaded jar file into the root directory i.e. c:\

Note: If you place the jar file in other directory then you need to specify that path in the cmd command below.

Open notepad and type the following command:

```
cd\ java -jar frmsal.jar -url "http://machine-name:9001/forms/frmservlet?config=standaloneapp"
```

Save it as .bat file.

Now this file can be used to run Ebizframe as standalone application.

Webutil Configuration

The Webutil utility is largely configured on 12c Forms installation. However, there are some additional steps you will need to accomplish in order to make it work properly. (IN CASE REQUIRED YOU CAN USER ATTACHED WEBUTIL.PLL)

In 12c we can configure Webutil in 2 ways:

1. with OLE.
2. Without OLE (JACOBI.JAR file not required, do only step 5 and step 6)

1. Download the (jacob-1.18-M2) file from

<http://sourceforge.net/projects/jacob-project/files/jacob-project/>

2. Configure your DB for webutil by running script called "create_webutil_db.sql" located in your "\$FR_HOME/forms" directory.
3. Unzip the contents of the file.
4. Copy the "jacob.jar" to the "\$FR_HOME/forms/java" directory.
5. Copy the "jacob-1.18-M2-x86.dll" to the "\$FR_HOME/forms/webutil/win32" directory.
6. Copy the "jacob-1.18-M2-x64.dll" to the "\$FR_HOME/forms/webutil/win64" directory.

Note: Do not change the name of the Jacob file and paste as it is in the respective directory

7. Sign the "jacob.jar" file. (See below) =====
Open sign_webutil.jar from



```
"D:\App\Oracle\config\domains\ess\config\fmwconfig\components\FORMS\instances\forms1\bin"  
SET SET KEYSTORE_PASSWORD=Manager1  
SET JAR_KEY_PASSWORD=Manager1
```

Change path of

```
D:\App\Oracle\product\12.2.4\jdk\bin\keytool  
D:\App\Oracle\product\12.2.4\jdk\bin\jarsigner
```

Open Command prompt and run

```
sign_webutil.bat D:\App\Oracle\product\12.2.4\forms\java\jacob.jar
```

```
Administrator: C:\Windows\System32\cmd.exe  
D:\App\Oracle\product\12.2.4\forms\java\jacob.jar  
WARNING:  
Self-signing Jar files is not supported.  
Jar files should be signed using a certificate obtained  
from a known and trusted certificate authority.  
Do you want to continue anyway [Y,N]?Y  
Generating a self signing certificate for key=webutil2...  
Warning:  
The JKS keystore uses a proprietary format. It is recommended to migrate to PKCS12 which is an industry standard format  
using "keytool -importkeystore -srckeystore C:\Users\Administrator\.keystore -destkeystore C:\Users\Administrator\.keyst  
ore -deststoretype pkcs12".  
...successfully done.  
.  
Backing up D:\App\Oracle\product\12.2.4\forms\java\jacob.jar as D:\App\Oracle\product\12.2.4\forms\java\jacob.jar.old...  
1 file(s) copied.  
Signing D:\App\Oracle\product\12.2.4\forms\java\jacob.jar using key=webutil2...  
jar signed.  
Warning:  
The signer's certificate is self-signed.  
...successfully done.  
D:\App\Oracle\config\domains\ess\config\fmwconfig\components\FORMS\instances\forms1\bin>
```

8. Add "\$FR_HOME/forms/java/frmall.jar" to the CLASSPATH entry in the "default.env" file. This should be done with Enterprise Manager, but it works fine if you manually edit the file. If you use custom environment files, you should add it to those also.
9. Add "frmwebutil.jar" and "jacob.jar" to the "Archive" and "WebutilArchive" entry in the "formsweb.cfg" file for any configurations that require it, for example.

```
WebutilArchive= jacob.jar,frmwebutil.jar,laf_11112.jar,mak_icons.jar
```

You must re-generate webutil.plx from webutil.pll. This file can be found in the Middleware Home \forms directory.

10. Open webutil.cfg from

```
"$fr_home\config\domains\ess\config\fmwconfig\components\FORMS\instances\forms1\server\webutil.cfg"
```



and make the changes in the following parameter.

```
transfer.database.enabled=TRUE
transfer.appsrv.enabled=TRUE
transfer.appsrv.workAreaRoot=
transfer.appsrv.accessControl=TRUE
#List transfer.appsrv.read.<n> directories
transfer.appsrv.read.1=C:\
transfer.appsrv.read.2=D:\
#List transfer.appsrv.write.<n> directories
transfer.appsrv.write.1=c:\
transfer.appsrv.write.2=D:\
```

11. Now run the webutil_demo.fmb and check the webutil functionality is working.

Direct Printing (OPTIONAL)

For Direct Printing we need to configure webutil apart from this we need to create 3 procedures on the form which are mentioned below:

1- Download_as:

PROCEDURE DOWNLOAD_AS (fname VARCHAR2)

IS

```
l_success    BOOLEAN;
l_bare_filename VARCHAR2 (50);
BEGIN
l_success :=
webutil_file_transfer.AS_to_Client_with_progress (
clientFile    => 'C:\Ebiz1\Home\Pivot\' || fname || '.rrpa',
serverFile    => 'C:\ebiz1\Home\Pivot\' || fname || '.rrpa',
progressTitle => 'Download from Application Server    Inprogress',
progressSubTitle    => 'Please wait');
EXCEPTION
    WHEN OTHERS
    THEN
        MESSAGE ('File download failed: ' || SQLERRM);
```

END;

2- Files_Wbp

PROCEDURE FILES_WBP (l_copy_name VARCHAR2) IS

```
l_file_list webutil_file.file_list;
l_temp_text VARCHAR2 (8000);
l_filename VARCHAR2 (80);
l_success    BOOLEAN;
BEGIN
```



```
l_file_list.delete;
l_success := webutil_file.delete_file (l_copy_name);
IF l_success
THEN
NULL;
ELSE
MESSAGE ('Delete failed');
END IF;
END;
```

3- Print report

```
PROCEDURE proc_call1 IS
BEGIN
DECLARE
al_id    alert;
al_report    NUMBER;
store_var    VARCHAR2 (150);
pi_id    paramlist;
compcode    VARCHAR2 (2);
temp    VARCHAR2 (30);
vc_reportserverjob    VARCHAR2 (100);
report_job_id    VARCHAR2 (100);
repid    report_object;
v_rep_status    VARCHAR2 (100);
makrep    VARCHAR2 (150);
BEGIN
repid := FIND_REPORT_OBJECT ('RP2RRO');
PROC_PATH ('FINANCE', 'reports', store_var);
proc_path ('MAKREP', 'REGISTRY', makrep);
store_var := store_var || 'comp_lst.rdf';
SET_REPORT_OBJECT_PROPERTY (repid, report_filename, store_var);
SET_REPORT_OBJECT_PROPERTY (repid, report_server, :global.r_server);
SET_REPORT_OBJECT_PROPERTY (repid, report_execution_mode, RUNTIME);
SET_REPORT_OBJECT_PROPERTY (repid, report_comm_mode, SYNCHRONOUS);
SET_REPORT_OBJECT_PROPERTY (repid, report_destype, cache);
SET_REPORT_OBJECT_PROPERTY (repid, report_desformat, :global.dsfmt);
-----
pi_id := GET_PARAMETER_LIST ('cust_param');
IF NOT ID_NULL (pi_id)
THEN
DESTROY_PARAMETER_LIST (pi_id);
END IF;
pi_id := CREATE_PARAMETER_LIST ('cust_param');
SELECT makess.nstring ( :global.vc_comp_code, 'R') INTO temp FROM DUAL;
ADD_PARAMETER (pi_id,
'comp_code',
TEXT_PARAMETER,
```



```
:global.vc_comp_code);  
ADD_PARAMETER (pi_id,  
    'format',  
TEXT_PARAMETER,  
temp);  
ADD_PARAMETER (pi_id,  
    'PARAMFORM',  
TEXT_PARAMETER,  
'NO');  
vc_reportserverjob := RUN_REPORT_OBJECT (repid, pi_id); report_job_id :=  
SUBSTR (vc_reportserverjob,  
LENGTH ( :global.r_server) + 2,  
LENGTH (vc_reportserverjob));  
v_rep_status := REPORT_OBJECT_STATUS (vc_reportserverjob);  
IF v_rep_status = 'FINISHED'  
THEN  
web.show_document ( " || :global.HOST || ':' || :global.port  
|| '/reports/rwservlet/getjobid' || report_job_id || '?server=' || :global.r_server, '_blank');  
ELSE  
MESSAGE ('error when running report' || v_rep_status);  
END IF;  
END;  
END;
```

Creating Additional Report Server (OPTIONAL)

PLEASE CHANGE THE YELLOW MARKED AS PER INSTALLATION NEEDS.

STEP 1:

Start Admin Server

D:\app\oracle\product\12.2.4\oracle_common\common\bin\wlst

Start Admin Server

STEP 2:

connect('weblogic','weblogic1','<MachineName>7001')

createReportsServerInstance(instanceName='rep_serversls',machine='AdminServerMachine')

exit()



STEP 3:

D:\app\Oracle\config\domains\ess\config\fmwconfig\components\ReportsServerComponent\rep_serversls

open rwserver.conf

change as follow

In case you don't need any kind of security for your Oracle Reports align the following section as follows:

```
<job jobType="report" engineId="rwEng" securityId="rwJaznSec"/>
```

To, remove the part securityId="rwJaznSec":

```
<job jobType="report" engineId="rwEng"/>
```

STEP 4: [OPTIONAL]

D:\app\Oracle\config\domains\ess\config\fmwconfig\servers\WLS_REPORTS\applications\reports_12.2.4\configuration

OPEN rwservlet.properties

Add the line <webcommandaccess>L2</webcommandaccess>

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<rwservlet xmlns="http://xmlns.oracle.com/reports/rwservlet"
```

```
xmlns:xsd="http://www.w3.org/2001/XMLSchema">
```

```
  <server>rep_wls_reports_greenfield</server>
```

```
  <signon>no</signon>
```

```
  <inprocess>yes</inprocess>
```

```
  <webcommandaccess>L2</webcommandaccess>
```

```
</rwservlet>
```

STEP 5:

START WLS_REPORTS

Start Node manager

D:\app\Oracle\config\domains\ess\bin

startComponent rep_serversls storeUserConfig

http://<MachineName>:9002/reports/rwservlet/showjobs?server=rep_serversls



Secure Showjobs Web Command In A Non Secured Reports Server

There are 2 things that must be configured to protect SHOWJOBS Reports Server page in a non-secured Reports Server for Reports 12c:

1. <webcommandaccess> element in rwservlet.properties
2. <identifier> element in rwserver.conf

WEBCOMMANDACCESS Element

The webcommandaccess element that is used to control access is in the rwservlet.properties file, which is located under:

`$DOMAIN_HOME\config\fmwconfig\servers\<WLS_SERVER_NAME>\applications\reports_\configuration\rwservlet.properties`

`D:\App\Oracle\config\domains\ess\config\fmwconfig\servers\WLS_REPORTS\applications\reports_12.2.1\configuration`

Since Reports 12c is no longer integrated in EM, you need to modify webcommandaccess either manually or via System Mbean Browser.

Valid settings are:

L0: no Web commands allowed.

L1: only end user Web commands allowed (GETJOBID, KILLJOBID, SHOWAUTH, SHOWJOBID).

L2: administrator Web commands (DELAUTH, GETSERVERINFO, KILLENGINE, PARSEQUERY, SHOWENV, SHOWJOBS, SHOWMAP, SHOWMYJOBS) are also allowed.

When enabling L0 or L1 settings you will have to enable AUTHID for the Reports Server administrator to be able to execute all the above administrator Web commands included in L2 setting.

Manual Instructions for Setting webcommandaccess:

Back up rwservlet.properties (under `$DOMAIN_HOME\config\fmwconfig\servers\<WLS_SERVER_NAME>\applications\reports_\configuration`).

`D:\App\Oracle\config\domains\ess\config\fmwconfig\servers\WLS_REPORTS\applications\reports_12.2.1\configuration`

Stop WLS_REPORTS managed server.

Modify rwservlet.properties by setting webcommandaccess to L1 (since you want to secure SHOWJOBS command), like the following example.



```
<?xml version="1.0" encoding="UTF-8"?>
<rwervlet xmlns="http://xmlns.oracle.com/reports/rwervlet"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">
<server>your_inprocess_repserver</server>
<signon>no</signon>
<inprocess>yes</inprocess>
<webcommandaccess>L1</webcommandaccess>
</rwervlet>
```

Note: It matters where the webcommandaccess is placed in file.

Save rwervlet.properties.

Restart WLS_REPORTS managed server for changes to take effect.

'System Mbean Browser' Instructions for Setting webcommandaccess:

1- Go to Enterprise Manager console:

`http://<hostname>:<port>/em`

2- From the WebLogic Domain menu, select "System MBean Browser".

3- Go to:

- > "Application Defined MBeans"
- > oracle.reportsApp.config
- > Server: WLS_REPORTS
- > Application: reports
- > ReportsApp
- > rwervlet
- > Attributes tab
- > WebcommandAccess
 - Select value: L1
- > "Apply"

4- Restart WLS_REPORTS managed server for changes to take effect.

IDENTIFIER Element

The admin user that will be able to access the SHOWJOBS web command is configured using the identifier element, which is set in the rwserver.conf file.

Steps:

Back up rwserver.conf.

The location of this rwserver.conf depends on whether it is an in-process Reports Server or Standalone Reports



Server. The file locations are detailed in:

D:\App\Oracle\config\domains\ess\config\fmwconfig\servers\WLS_REPORTS\applications\reports_12.2.1\configuration

Modify rwserver.conf.

Include identifier element in between <queue> and <pluginParam> elements, like this:

Example

```
<queue maxQueueSize="1000"/>
<identifier encrypted="no"><admin_user></password></identifier>
<pluginParam name="mailServer" value="%MAILSERVER_NAME%"/>
```

*Be sure to replace the above <admin_user> and <password> with the actual user and password that you want to use as your admin user. This information will be encrypted once the Reports Server is re-started. Remember that the order of elements in rwserver.conf file matter. Adding an element in a wrong Reports Server configuration file location will not allow Reports Server to start.

Then restart WLS_REPORTS (in-process Reports Server) and/or Standalone Reports Server

After restart, the <admin_user></password> in rwserver.conf should be encrypted.

Now you can do a SHOWJOBS and pass in admin user using AUTHID parameter, like this:

```
http://<hostname>:<port>/reports/rwservlet/showjobs?server=<ReportsServerName>&authid=<admin_user>/<password>
```

Also, the web commands like SHOWJOBS should not be accessible by everyone.

Create a batch file to open show jobs and exit so that showjob will not be required by user.

Put the below command in batch file.

```
@echo off
```

```
start iexplore "
http://<hostname>:<port>/reports/rwservlet/showjobs?server=<ReportsServerName>&authid=<admin_user>/<password>"
timeout /t 3
taskkill /IM iexplore.exe /F
:loop
exit
goto loop
```



Using JVM Controllers (JVMControllers)

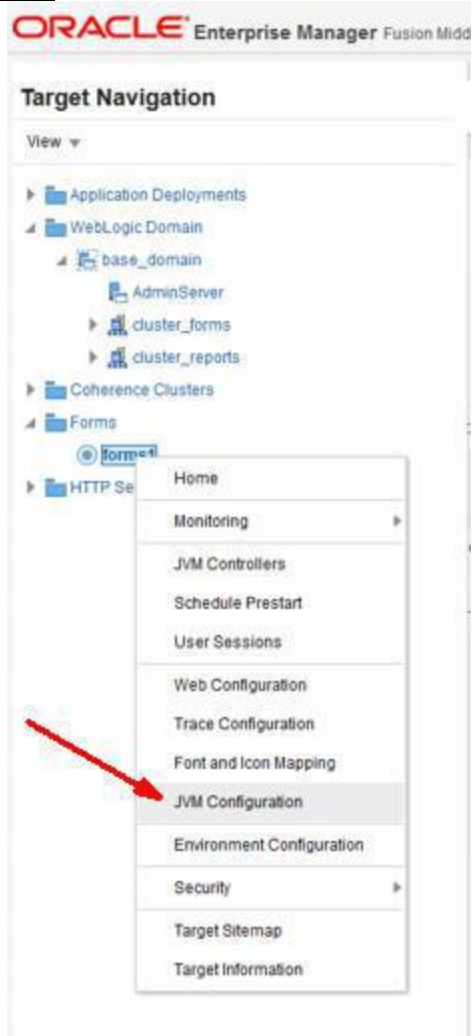
How to Decrease Memory Usage in Forms 12c Version 12.2.1.0.0 and later Using JVM Controllers (JVMControllers)

Oracle Forms - Version 12.2.1.0.0 and later

When making Java calls from Forms when using `RUN_REPORT_OBJECT()` (RRO) - either to call Oracle Reports or Oracle BI Publisher Reports (BIP) - or imported java, the size of the Forms runtime processes (frmweb.exe) can be significantly reduced by configuring JVM Controllers. This note demonstrates how to set up basic JVM Controller configuration for 12c Forms. For the purpose of this document, we will use Oracle Reports as the external Java call target. The steps are not significantly different from those for 11g. However, there are new features in 12c that can improve performance

Solution

1. First, create a new RRO specific jvmcontroller. Bring up Oracle Enterprise Manager Fusion Middleware Control 12c (EM) and open the JVM Configuration option as indicated below:



2. Using the "Create Like" option, duplicate the "example" configuration and name it "repjvmc".

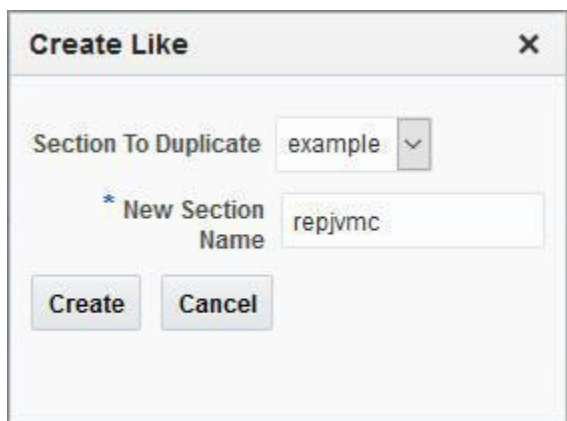


Figure 2



3. Notice that the entries for the new section were exactly copied from "example." The values derived from the "example" config are just examples. They are not necessarily recommended values or parameters. In this example, the following parameters were used in jvmcontroller "repjvmc" Parameter values such as those specified for "jvmoptions", "maxsessions", "loadbalance", "autoremoval" are only notional. See JVM Configuration Parameters and Default Values for more specifics.

jvmoptions = -Xms512m -Xmx1024m #Stands for initial heap and max heap.

classpath = ORACLE_HOME\jlib\zrclient.jar;ORACLE_HOME\reports\jlib\rwrun.jar
#Replace ORACLE_HOME with the physical ORACLE_HOME directory.

These .jar files are needed for RRO.

maxsessions = 50 #Specifies number of frmweb processes allowed to attach to a JVM before a new child JVM is created.

logdir = d:\temp #jvmcontroller log file directory. The file will be called repjvmc.log after the jvmcontroller name.

Logging = info #Recommend leaving as "info" until verification that all is working okay.

loadbalance= random #Determines how the JVM controller will send new requests for processing.

autoremoval=MODERATE #Allows the JVM Controller to monitor and terminate unneeded child JVM processes.

4. In the default section of the formsweb.cfg, add "jvmcontroller=%jvmcontroller%" to the value of "otherparams". For example, "otherparams" should have this value ->

obr=%obr% record=%record% tracegroup=%tracegroup% log=%log% term=%term%
ssoProxyConnect=%ssoProxyConnect% jvmcontroller=%jvmcontroller%

5. One can directly reference a particular jvmcontroller in a Forms url by specifying a parameter like ".....&otherparams=jvmcontroller=repjvmc" for our case here. However, in this example we will configure a specific jvmcontroller to be associated with a particular formsweb.cfg configuration section called "reportsjvm". If you already have a specific configuration section for your Forms that call reports then just modify it similarly as done here in the "reportsjvm" configuration section.

Find the otherparams parameter in the "reportsjvm" configuration section and set its value as the following:

obr=%obr% record=%record% tracegroup=%tracegroup% log=%log% term=%term%
ssoProxyConnect=%ssoProxyConnect% jvmcontroller=repjvmc

This will cause the "repjvmc" JVM Controller to be used when that config section is used.



6. Now Run a form that calls reports via RRO using the config section that you created. In this case we will assume the config section is called “reportsjvm”.

<http://machine.domain:port/forms/frmservlet?config=reportsjvm>

The JVM Controller starts automatically the first time a config section that references it is used. This is because the default value of “allowJVMControllerAutoStart” is true. If this value is preferred to be set to false, then the “allowJVMControllerAutoStart” parameter should be added to the default section of the formsweb.cfg using FMW Control and set to false

7. Once you accomplish your RRO call to reports in your form, you will see the jvmcontroller process start as indicated by the following

a. From the Forms -> JVM Controllers option in EM, you will see the following:

ORACLE Enterprise Manager Fusion Middleware Control 12c

forms1 Forms

Home > JVM Controllers

JVM Controllers

A JVM Controller manages the JVMs which can be shared between Forms Sessions.

Total Memory Used (KB) 95092

View Start Stop Restart Delete logfile

Name	Status	CPU Usage (%)	Private Memory Usage (KB)	JVMs	Current Sessions	Maximum sessions per JVM	Logging	Logfile
example	Down					50	Off	
repjvmc	Up	0.01	95092	1	1	50	info	

Figure 3

b . You can click on the eyeglasses icon to the right of your jvmcontroller in Figure 3 and see information like the following:

01/10/16 17:31:59 Standard Time::JVM PID 17924: Starting.

01/10/16 17:31:59 Standard Time::JVM PID 17924: JVM Controller version : 12.2.1.2.0

01/10/16 17:31:59 Standard Time::JVM PID 17924: Copyright (c) 1994, 2016, Oracle and/or its affiliates. All rights reserved.

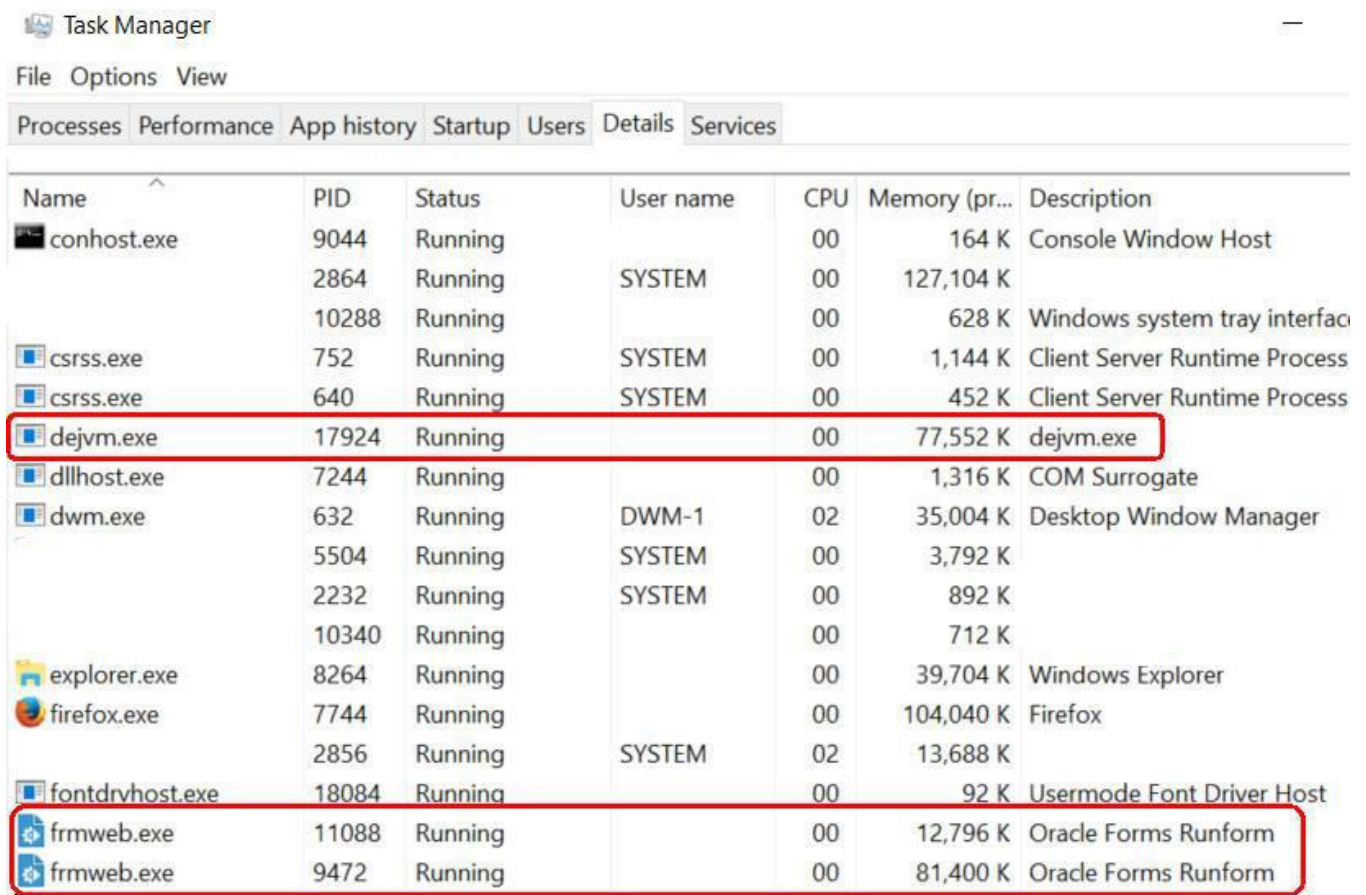
01/10/16 17:31:59 Standard Time::JVM PID 17924: Parameter -- JVM controller name = repjvmc

01/10/16 17:31:59 Standard Time::JVM PID 17924: Parameter -- Maximum sessions = 50



01/10/16 17:31:59 Standard Time::JVM PID 17924: Parameter -- Log Dir = d:\temp
01/10/16 17:31:59 Standard Time::JVM PID 17924: Parameter -- Logging = info
01/10/16 17:31:59 Standard Time::JVM PID 17924: Parameter -- JVM Options = -Xms512m -Xmx1024m
01/10/16 17:31:59 Standard Time::JVM PID 17924: Parameter -- JVM Classpath =
d:\middleware\jlib\zrclient.jar;d:\middleware\reports\jlib\rwrun.jar
01/10/16 17:31:59 Standard Time::JVM PID 17924: Parameter -- JVM Load Balance = random
01/10/16 17:31:59 Standard Time::JVM PID 17924: Parameter -- JVM Auto Removal = moderate
01/10/16 17:31:59 Standard Time::JVM PID 17924: Scheduled for accepting connection requests.
01/10/16 17:32:00 Standard Time::JVM PID 17924: New connection from Forms Runtime Process
(PID:11088)

c. A dejvm process will start as is shown by the windows task manager image below:



Name	PID	Status	User name	CPU	Memory (pr...	Description
conhost.exe	9044	Running		00	164 K	Console Window Host
	2864	Running	SYSTEM	00	127,104 K	
	10288	Running		00	628 K	Windows system tray interfac
csrss.exe	752	Running	SYSTEM	00	1,144 K	Client Server Runtime Process
csrss.exe	640	Running	SYSTEM	00	452 K	Client Server Runtime Process
dejvm.exe	17924	Running		00	77,552 K	dejvm.exe
dllhost.exe	7244	Running		00	1,316 K	COM Surrogate
dwm.exe	632	Running	DWM-1	02	35,004 K	Desktop Window Manager
	5504	Running	SYSTEM	00	3,792 K	
	2232	Running	SYSTEM	00	892 K	
	10340	Running		00	712 K	
explorer.exe	8264	Running		00	39,704 K	Windows Explorer
firefox.exe	7744	Running		00	104,040 K	Firefox
	2856	Running	SYSTEM	02	13,688 K	
fontdrvhost.exe	18084	Running		00	92 K	Usermode Font Driver Host
frmweb.exe	11088	Running		00	12,796 K	Oracle Forms Runform
frmweb.exe	9472	Running		00	81,400 K	Oracle Forms Runform

Figure 4

Note in Figure 4 the frmweb process consuming 81,400 K of memory was a form making a RRO call "without" using the JVM Controller. The frmweb process consuming 12,796 K is from the same form making the exact same RRO call but using the JVM Controller. The JVM Controller process is the dejvm process.

A note about JVM Controller Memory Utilization. There is no easy formula for determining the ideal settings for your JVM Controller parameters. While it is certain that you should configure JVM Controllers when your



forms are making RRO calls the ideal parameter settings must be determined during load testing in Development/QA environments. This is because the amount of memory usage of the JVM Controllers is highly dependent on the frequency of calls to them as well as complexity of the java code involved. The settings listed in this note have generally been found to be adequate for most customers. The parameter values listed here were tested internally using a moderately complex form making RRO calls to a basic Oracle report. For 10 frmweb sessions each frmweb process each initially consumed about 13 MB of memory and the dejvm process grew from 77 MB to about 90 MB. However, this is just an example so extrapolating from this will most likely result in inaccurate conclusions. Realistic load testing before going into Production is the only safe way to determine your application's resource needs.

IMPORTANT NOTE: Before restarting the Oracle WebLogic managed server, all the JVM Controller processes (dejvm) started by that server must be stopped. Otherwise, WLS_FORMS will not restart after a shutdown. Note also that if too much memory is consumed, then the Reports Server can hang. If this is the case, you will see the following in the system event log.

Upgrading to New Password Setup.

```
conn sys/manager1@orcl as sysdba
```

```
grant execute on sys.dbms_crypto to makess with grant option;
```

```
conn makess/makess@orcl;
```

```
CREATE OR REPLACE FUNCTION GET_ENC_VAL(P_IN IN VARCHAR2, P_KEY IN RAW)
RETURN RAW IS
  L_ENC_VAL RAW(2000);
  L_MOD NUMBER := DBMS_CRYPTO.ENCRYPT_AES128 + DBMS_CRYPTO.CHAIN_CBC +
DBMS_CRYPTO.PAD_PKCS5;
BEGIN
  L_ENC_VAL := DBMS_CRYPTO.ENCRYPT(UTL_I18N.STRING_TO_RAW(P_IN, 'AL32UTF8'), L_MOD, P_KEY);
  RETURN L_ENC_VAL;
END;
/
```

```
CREATE OR REPLACE FUNCTION GET_DEC_VAL(P_IN IN RAW, P_KEY IN RAW)
RETURN VARCHAR2 IS
  L_RET VARCHAR2(2000);
  L_DEC_VAL RAW(2000);
  L_MOD NUMBER := DBMS_CRYPTO.ENCRYPT_AES128 + DBMS_CRYPTO.CHAIN_CBC +
DBMS_CRYPTO.PAD_PKCS5;
BEGIN
  L_DEC_VAL := DBMS_CRYPTO.DECRYPT(P_IN, L_MOD, P_KEY);
  L_RET := UTL_I18N.RAW_TO_CHAR(L_DEC_VAL, 'AL32UTF8');
  RETURN L_RET;
END;
```



```
END;  
/
```

```
CREATE OR REPLACE FUNCTION MAKESS.DECRYPT_CRYPT0(ENCRYPTED_TEXT IN VARCHAR2)  
  RETURN VARCHAR2 AS  
BEGIN  
  RETURN GET_DEC_VAL(ENCRYPTED_TEXT, 'A1A2A3A4A1A2A3A4A1A2A3A4A1A2A3A4');  
END;  
/
```

```
CREATE OR REPLACE FUNCTION MAKESS.ENCRYPT_CRYPT0(DECRYPTED_TEXT IN VARCHAR2)  
  RETURN VARCHAR2 AS  
BEGIN  
  RETURN GET_ENC_VAL(DECRYPTED_TEXT, 'A1A2A3A4A1A2A3A4A1A2A3A4A1A2A3A4');  
END;  
/
```

```
GRANT EXECUTE ON MAKESS.DECRYPT_CRYPT0 TO SALES, PURCHASE, INVENT, FINANCE, HRPAY, PRODUCTION,  
MAINTAIN;
```

```
GRANT EXECUTE ON MAKESS.ENCRYPT_CRYPT0 TO SALES, PURCHASE, INVENT, FINANCE, HRPAY, PRODUCTION,  
MAINTAIN;
```

```
CREATE OR REPLACE FUNCTION MAKESS.DECRYPT(ENCRYPTED_TEXT IN VARCHAR2)  
  RETURN VARCHAR2 AS  
BEGIN  
  RETURN GET_DEC_VAL(ENCRYPTED_TEXT, 'A1A2A3A4A1A2A3A4A1A2A3A4A1A2A3A4');  
END;  
/
```

```
CREATE OR REPLACE FUNCTION MAKESS.ENCRYPT(DECRYPTED_TEXT IN VARCHAR2)  
  RETURN VARCHAR2 AS  
BEGIN  
  RETURN GET_ENC_VAL(DECRYPTED_TEXT, 'A1A2A3A4A1A2A3A4A1A2A3A4A1A2A3A4');  
END;  
/
```

```
GRANT EXECUTE ON MAKESS.DECRYPT TO SALES, PURCHASE, INVENT, FINANCE, HRPAY, PRODUCTION,  
MAINTAIN;
```

```
GRANT EXECUTE ON MAKESS.ENCRYPT TO SALES, PURCHASE, INVENT, FINANCE, HRPAY, PRODUCTION,  
MAINTAIN;
```



Running ebizframe on IOS

1. Install jdk-8u201-macosx-x64.dmg.
2. Click on FINDER then go to Application.
3. Create new folder in Application and name it "ERP" then copy frmsal.jar file in the ERP folder.
4. Copy ERP.command file to Desktop
5. Edit ERP.command file to change the erp path like
`http://server-ip:9001/forms/frmservlet?config=webutil_standaloneapp"`
6. Double click on the ERP.command Icon to run ERP.

Apex Installation 19.2

(Both Database and Application are installed on same server)

STEP 1 Pre Configuration

Copy apex_19.2 files or any other version of apex to D folder of our ebiz.

Then make a shortcut of sql plus on desktop. Rename sql to sql_apex.

Right click sql plus>>properties>>start in>>

Replace the location of apex files only.

"D:\apex_19.2_en\apex"

Press APPLY then OK

Open Sql Plus and Connect To Sys/<Password> As SYSDBA

Now run below alter commands:-

```
ALTER PROFILE DEFAULT LIMIT PASSWORD_LIFE_TIME UNLIMITED ;
```

```
ALTER USER ANONYMOUS ACCOUNT UNLOCK;
```

```
CREATE THE USER WHICH YOU ARE USING TO INSTALL
```



```
CREATE TABLESPACE EBIZ DATAFILE 'D:\ORACLE\APP\ADMINISTRATOR\ORADATA\ORCL\EBIZ.DBF' SIZE 1G;
```

STEP 2 Apex Installation

Full development environment –

Run apexins.sql passing the following four arguments in the order shown:

```
@apexins.sql tablespace_apex tablespace_files tablespace_temp images
```

Where:

- *tablespace_apex* is the name of the tablespace for the Application Express application user.
- *tablespace_files* is the name of the tablespace for the Application Express files user.
- *tablespace_temp* is the name of the default temporary tablespace or tablespace group.
- *images* is the virtual directory for Application Express images. To support future Application Express upgrades, define the virtual image directory as /i/.

For example:

```
@apexins.sql EBIZ EBIZ TEMP /i/
```

This command will take sometime after completion sql will popup at end.

STEP 3 Apex Installation

```
@apxchpwd.sql
```

“Enter the administrator's username [ADMIN]” Press enter to create or use if existing.

“Enter ADMIN's email [ADMIN]” Enter your own email

“Enter ADMIN's password []” Password\$1234

```
SQL Plus
http://host:port/apex (Oracle REST Data Services)

timing for: Phase 3 (Switch)
Elapsed: 00:00:07.92
timing for: Complete Installation
Elapsed: 00:06:05.78

PL/SQL procedure successfully completed.

1 row selected.

...null1.sql
SYS> @apxchpwd.sql
...set_appun.sql
=====
This script can be used to change the password of an Application Express
instance administrator. If the user does not yet exist, a user record will be
created.
=====
Enter the administrator's username [ADMIN]
User "ADMIN" does not yet exist and will be created.
Enter ADMIN's email [ADMIN] neeraj.kumar@ebizframe.com
Enter ADMIN's password [ ]
Created instance administrator ADMIN.

SYS>
```



STEP 4 Apex Installation

@apex_epg_config.sql d:\apex_19.2_en

check port apex installed default is 8080

Now this port 8080 must be occupied

```
select dbms_xdb.gethttpport from dual;  
GETHTTPPORT
```

```
-----  
0
```

check if port is zero then set port 8080

execute dbms_xdb.sethttpport(8080);

STEP 5 Apex Installation Verification.

After that run the below URL on Browser.

http://localhost:8080/apex/apex_admin

Enter Details:-

Username :- Admin

Password :- Password\$1234

STEP 6 Apex Configuration.

Chose "Create Workspace"

ENTER "WSTS" and press next.

Reuse exisiting schema?: YES

Schema name:- MAKESS

Schema password: MAKESS

Press Next

Username :- Admin

Password :- Password\$1234

email:- own email

Press Next

Just read out details and press button create workspace



Logout admin at right top corner and again return to sign in page.

Apex Installation 21.1, 21.2

Pre - requisites: -

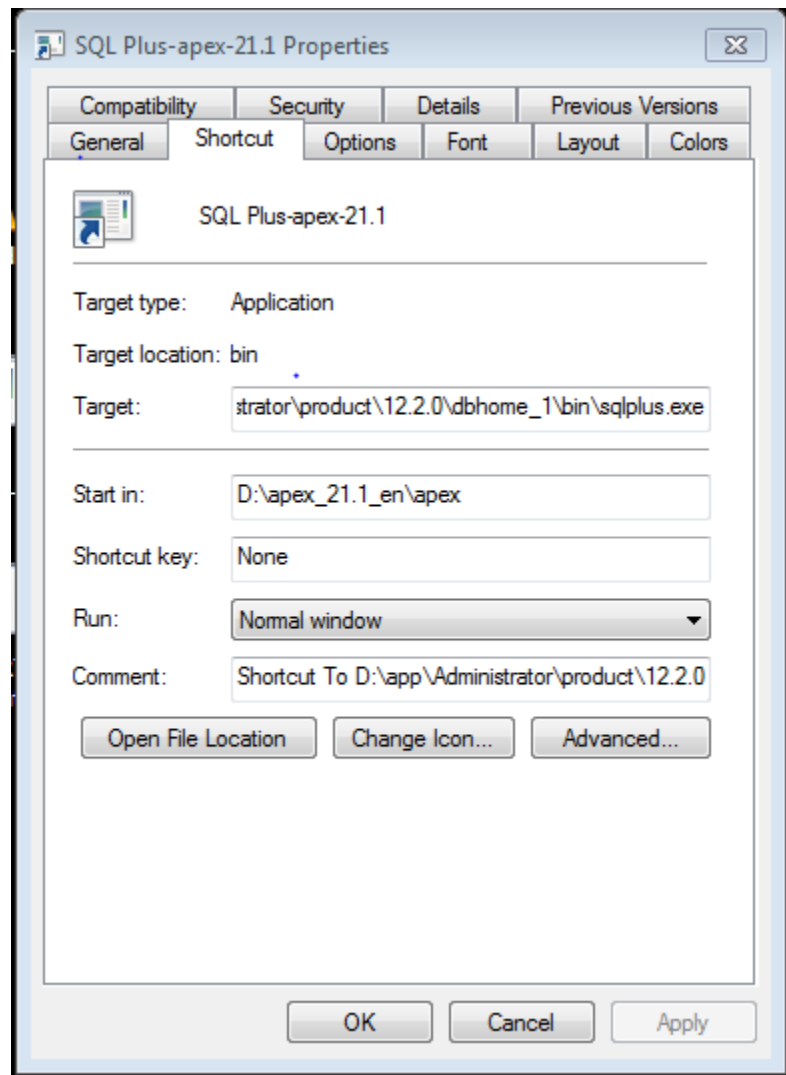
- ORACLE Database 12c version 12.2.0.2.0 or later.
- Requires a JavaScript-enabled browser and supports the current and prior major release of Google Chrome, Mozilla Firefox, Apple Safari, and Microsoft Edge.

Installation Steps: -

1. Create tablespace EBIZ for apex installation (it can be changed according to ERP). Run the below commands in sqlplus:
Create Tablespace ebiz datafile 'D:\app\{user}\oradata\orcl\ebiz.ora' Size 200m autoextend on; (NOTE: Ensure that the tablespace must be created inside app database oradata folder else APEX will not detect it while installation)
2. Now, get the **apex_21.1_en.zip/ apex_21.2_en.zip** file from the apex installations folder and copy to zip file to D drive.
3. Now, to extract the file right click and choose extract to apex_21.2_en folder. Now the working directory would look like: **"D:\apex_21.2_en\apex"**.
4. After extraction create a shortcut of SQLPLUS on desktop by going to D drive app folder of and then bin folder of database. Normally it is: **"D:\app\'user'\product\'db_version'\dbhome_1\bin"** (Here user and db_version is your system user and database version)
5. Now right click on **sqlplus** file and send to desktop as shortcut
6. Now, go to D drive and go to apex_21.2_en directory **"D:\apex_21.2_en\apex"** and copy the path.



- Now, right click to the sqlplus shortcut you created on desktop and go to properties and paste the link in start in:



- Now, click on ok
- Now open sqlplus shortcut we just created and in username use: --- **sys as sysdba** and in password hit enter
- Now run the below commands

```
ALTER PROFILE DEFAULT LIMIT PASSWORD_LIFE_TIME UNLIMITED;  
ALTER USER apex_public_user IDENTIFIED BY password ACCOUNT UNLOCK;  
ALTER USER SYS_AUX IDENTIFIED BY password ACCOUNT UNLOCK;  
ALTER USER ANONYMOUS ACCOUNT UNLOCK;
```

- In the above command if sysaux user is not found then leave it as it is and continue further



12. Now, run

Spool on;

@apexins.sql EBIZ EBIZ TEMP /i/

(Note: Here first EBIZ is the ebiz tablespace name we just created in step 1, second EBIZ is ebiz tablespace filename and TEMP is the database temporary tablespace, you can also use ERP temporary tablespace here, instead of TEMP but remember ERP temporary tablespace must be created inside app database oradata folder else APEX will not detect it while installation)

13. The above command takes 20-30 mins according to the machine's performance

```
sqlplus -xe apex
SQL> alter session set container= <epdb1,
Session altered.

SQL> ALTER PROFILE DEFAULT LIMIT  PASSWORD_LIFE_TIME UNLIMITED ;
Profile altered.

SQL> ALTER USER apex_public_user IDENTIFIED BY password ACCOUNT UNLOCK;
ALTER USER apex_public_user IDENTIFIED BY password ACCOUNT UNLOCK
*
ERROR at line 1:
ORA-01918: user 'APEX_PUBLIC_USER' does not exist

SQL> ALTER USER SYSAUX IDENTIFIED BY password ACCOUNT UNLOCK;
ALTER USER SYSAUX IDENTIFIED BY password ACCOUNT UNLOCK
*
ERROR at line 1:
ORA-01918: user 'SYSAUX' does not exist

SQL> alter user ANONYMOUS account unlock;
User altered.

SQL> @apexins.sql EBIZ EBIZ TEMP /i/
...set_appun.sql
```

14. Now, after installation when sql prompt returns run next command,

@apxchpwd.sql

For username hit enter,

for email id enter the desired email id,

For password enter- "Password\$1234" and hit enter

(Refer the below screenshot)



```
sqlplus - xe apex
http://host:port/apex (Oracle REST Data Services)
timing for: Phase 3 (Switch)
Elapsed: 00:00:09.76
timing for: Complete Installation
Elapsed: 00:06:58.74
PL/SQL procedure successfully completed.

1 row selected.

...null1.sql
SYS> @apxchpwd.sql
...set_appun.sql
-----
This script can be used to change the password of an Application Express
instance administrator. If the user does not yet exist, a user record will be
created.
-----
Enter the administrator's username [ADMIN]
User "ADMIN" does not yet exist and will be created.
Enter ADMIN's email [ADMIN] kunal.kishor@essindia.com
Enter ADMIN's password []
Created instance administrator ADMIN.

SYS>
```

15. Now, run the below command:

@apex_rest_config.sql

And for apex_listener enter password- "Password\$1234",
for apex_rest_public_user- "Password\$1234"

NOTE: This above command is used for ORACLE Rest data Services this must be run while apex installation

```
sqlplus - xe apex

Loading Images directory: d:\apex_18.2_en\apex\Images
timing for: Load Images
Elapsed: 00:01:32.27
SYS> @apex_rest_config.sql
-----
Enter a password for the APEX_LISTENER user []
Enter a password for the APEX_REST_PUBLIC_USER user []
...set_appun.sql
...create APEX_LISTENER and APEX_REST_PUBLIC_USER users
SYS>
```

Note: Apex 21.2 will not run without the installation of ords. After the installation of 21.2 apex, we need to install the ords-21.1.1.116.2032



Installation and configuration of ORDS 21.2 with Oracle APEX 21.2

Overview: -

Oracle REST Data Services (ORDS), formerly known as the APEX Listener, allows APEX applications to be deployed without the use of Oracle HTTP Server (OHS) and mod_plsql or the Embedded PL/SQL Gateway. ORDS version 3.0 onward also includes JSON API support to work in conjunction with the JSON support in the 12c database. ORDS can be deployed on WebLogic, Glassfish or Tomcat.

The Oracle Application Express architecture requires some form of web server to proxy requests between a web browser and the Oracle Application Express engine. Oracle REST Data Services satisfies this need, but its use goes beyond that of Oracle Application Express configurations. Oracle REST Data Services simplifies the deployment process because there is no Oracle home required, as connectivity is provided using an embedded JDBC driver.

Purpose: -

ORDS is a Java EE-based alternative for Oracle HTTP server which allows a great deal of functionality including command line-based deployment and configuration, enhanced security, and most importantly RESTful Web Services, also ORDS must be used for APEX PDF report printing and sometimes with some of the PLUGINS that require ORDS connection with apex applications, ORDS must be used for 3-tier system like where Database and Application servers are on different machines.

NOTE: - If there is 3-tier system then ORDS must be installed on application server, and if the database and application server are on the same machine then ORDS can be installed on the same machine as well.

Version: -

Oracle ORDS- ords-21.4.0.348.1956

Pre-requisites: -

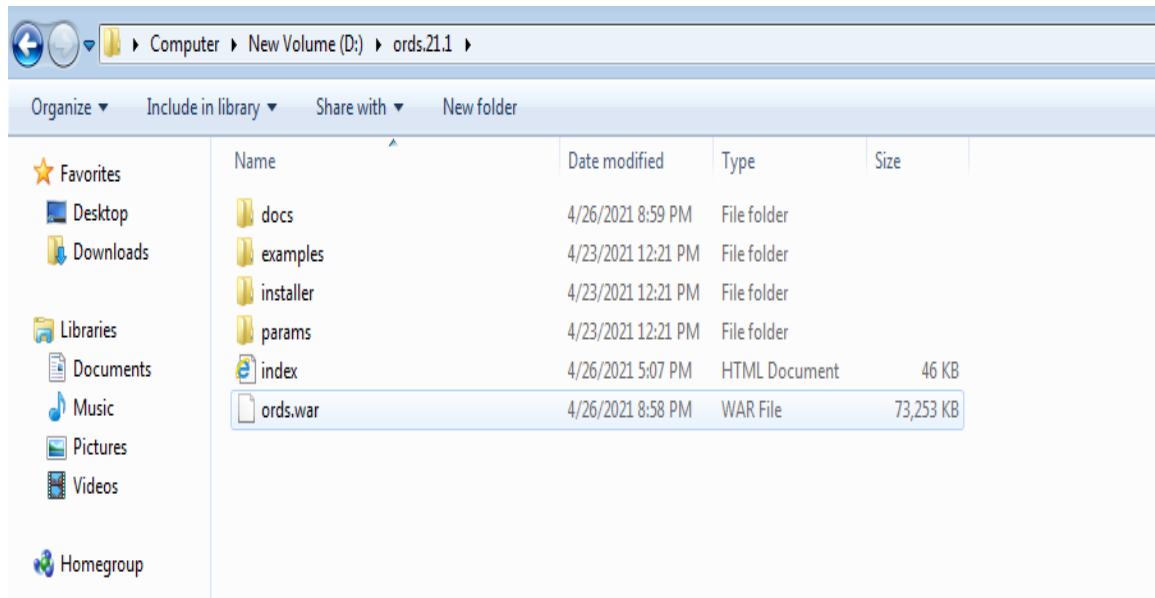
- ORACLE Database 12c version 12.2.0.2.0 or later.
 - Requires a JavaScript-enabled browser and supports the current and prior major Release of Google Chrome, Mozilla Firefox, Apple Safari, and Microsoft Edge.
- Oracle APEX 21.2 installed on the database.

NOTE: - If you have database and application server on the different machines then please copy this "apex_21.2_en" folder in D drive from database server to D drive in the application server and then install ORDS on the application server to connect apex from database server to application server.

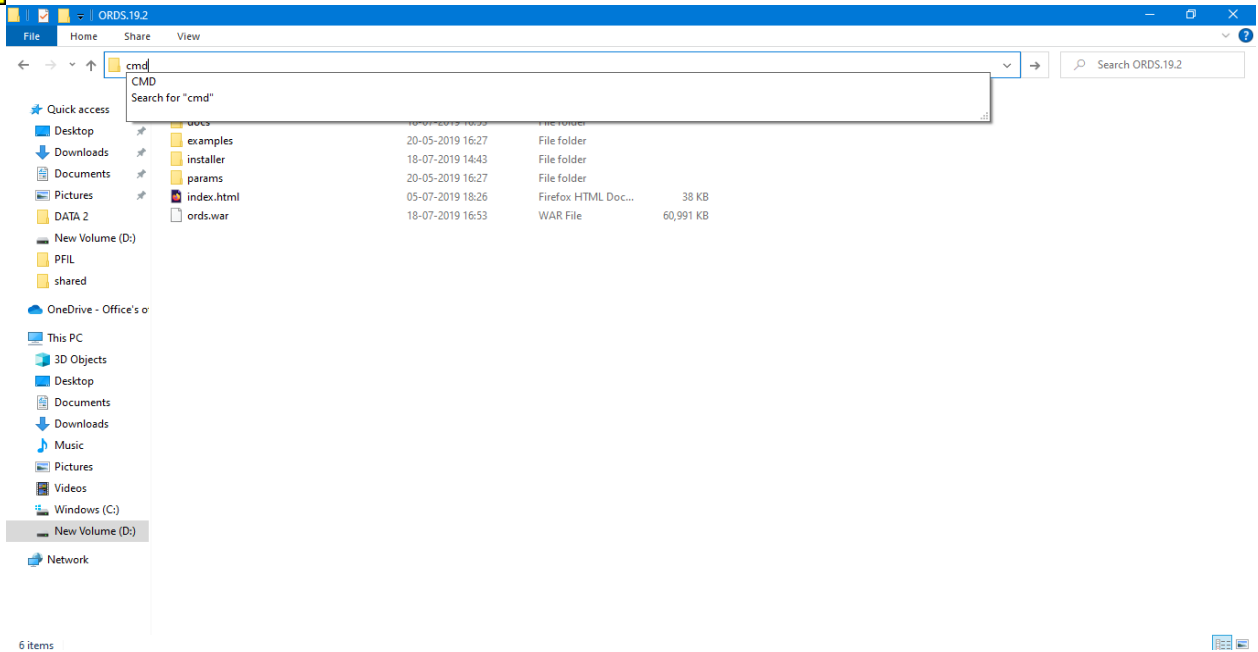


Installation and Configuration: -

1. Now, Get the **ords-21.4.0.348.1956.zip** file from the ORDS installation folder/APEX installation folder.
2. Create a folder in D drive namely ORDS.21.2 and double click the above zip file and click extract to on the top of WINRAR window and extract to the folder ORDS.21.2 in D drive
3. Contents would look like:



4. Now in the same folder from where you got the zip file for ords.21.2 get the file **jre-8u25-windows-i586.exe**
5. Double click to run the java JRE setup and follow the instruction and install
6. Now, after installation of java JRE
7. Go to D drive and go to ORDS.21.2 folder open the folder and in the address bar type cmd and hit enter and command prompt will open like this:



8. Now, in the command prompt window type
java -jar ords.war

NOTE: - For this version of ORDS jre 8 minimum is required, if on running above command JAVA is not detected make sure java is set in environment variable and if then also it does not work, then instead of the above command run:

"JAVA_PATH" -jar ords.war

(in JAVA_PATH enter java bin folder path)

For example :c:\java_1.8.xx\bin\java -jar ords.war

Note: If you get error : Caused by: java.lang.OutOfMemoryError: Java heap space
then use following command:

java -jar -Xmx800M ords.war install advanced



```
Administrator: C:\Windows\System32\cmd.exe - java -jar ords.war
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

D:\ords.21.1>java -jar ords.war
This Oracle REST Data Services instance has not yet been configured.
Please complete the following prompts

Enter the location to store configuration data:
```

9. enter the configuration data location as **"conf"** hit enter
10. then for database machine name it is taken as localhost by default:
now, if database and application server are different then enter the IP address for database machine
and if they both are on the same machine hit enter do not enter anything
11. Now the database port in by default is 1521 for that hit enter
12. Its asking if you want to enter database service name or SID for the database, for this installation we
are using service name and it is also the default selection so hit enter do not enter anything
13. Now, for next it is asking for database service name hit enter and enter it as: **"orcl"** or change according
to your installation and hit enter
14. Now for ORDS_PUBLIC_USER password enters - **"Password\$1234"**
then confirm password as same **"Password\$1234"** and hit enter



```
Administrator: C:\Windows\System32\cmd.exe - java -jar ords.war
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

D:\ords.21.1>
D:\ords.21.1>java -jar ords.war
This Oracle REST Data Services instance has not yet been configured.
Please complete the following prompts

Enter the location to store configuration data: conf
Enter the name of the database server [localhost]:
Enter the database listen port [1521]:
Enter 1 to specify the database service name, or 2 to specify the database SID [1]:
Enter the database service name:orcl
Enter the database password for ORDS_PUBLIC_USER:
Confirm password:
Requires to login with administrator privileges to verify Oracle REST Data Services schema.

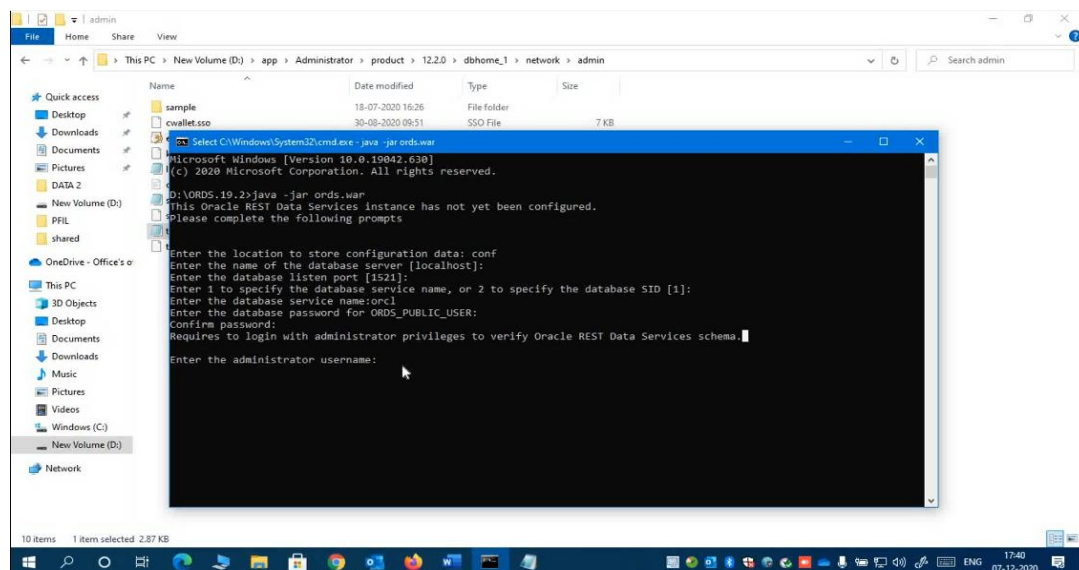
Enter the administrator username:
```

15. Now, it is asking for database administrator username enter as **sys as sysdba**

16. Now, it is asking for database password for SYS AS SYSDBA

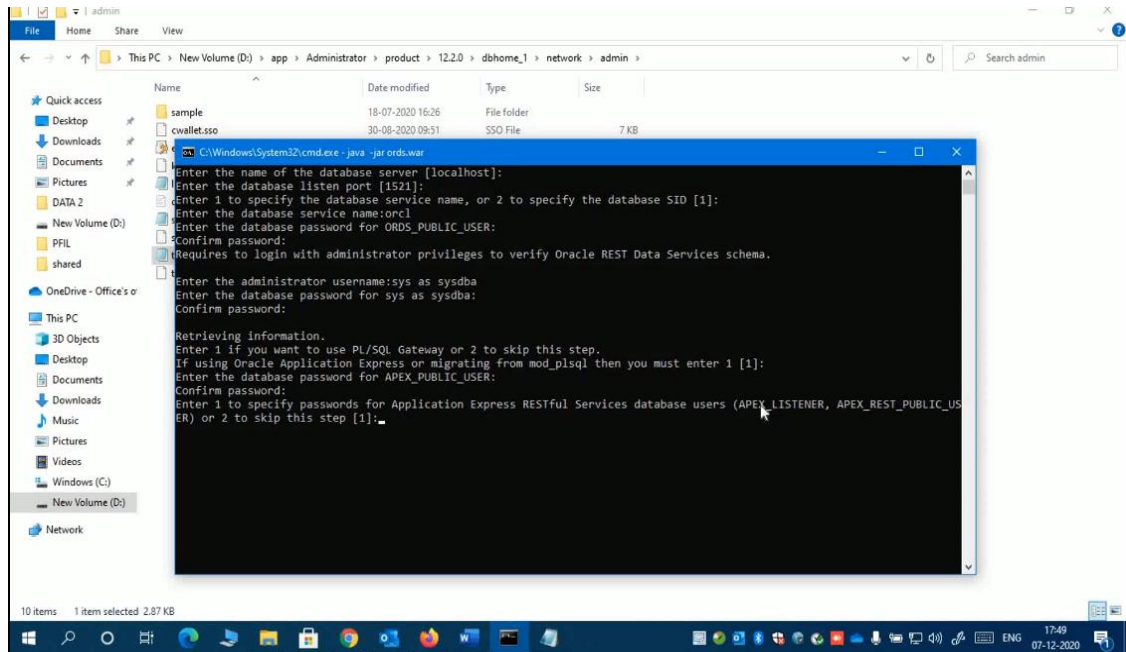
17. Enter it as “manager1” and confirm it as “manager1” hit enter or according to your database configuration

NOTE: - If password is unknown to you then connect sqlplus with sys as sysdba and use command “alter user sys identified by manager1;”



18. Now, it asks for if we want to use pl/sql gateway of Oracle apex then do not choose anything just hit enter

19. Now, enter the APEX_PUBLIC_USER password as –
“Password\$1234” and confirm it as **“Password\$1234”** hit enter



20. Now, hit enter again and now it is asking for password for application express restful services database users (apex listener, apex_rest_public_user), do not choose anything hit enter.

21. Enter the passwords for both apex_listener and apex_rest_public_user
"Password\$1234" and confirm it as **"Password\$1234"**

```
Retrieving information.
Enter 1 if you want to use PL/SQL Gateway or 2 to skip this step.
If using Oracle Application Express or migrating from mod_plsql then you must enter 1 [1]:
Enter the database password for APEX_PUBLIC_USER:
Confirm password:
Enter 1 to specify passwords for Application Express RESTful Services database users (APEX_LISTENER, APEX_REST_PUBLIC_USER) or 2 to skip this step [1]:
Enter the database password for APEX_LISTENER:
Confirm password:
Enter the database password for APEX_REST_PUBLIC_USER:
Confirm password:
```

22. Now, it is asking to select a number to enable features, do not enter anything hit enter

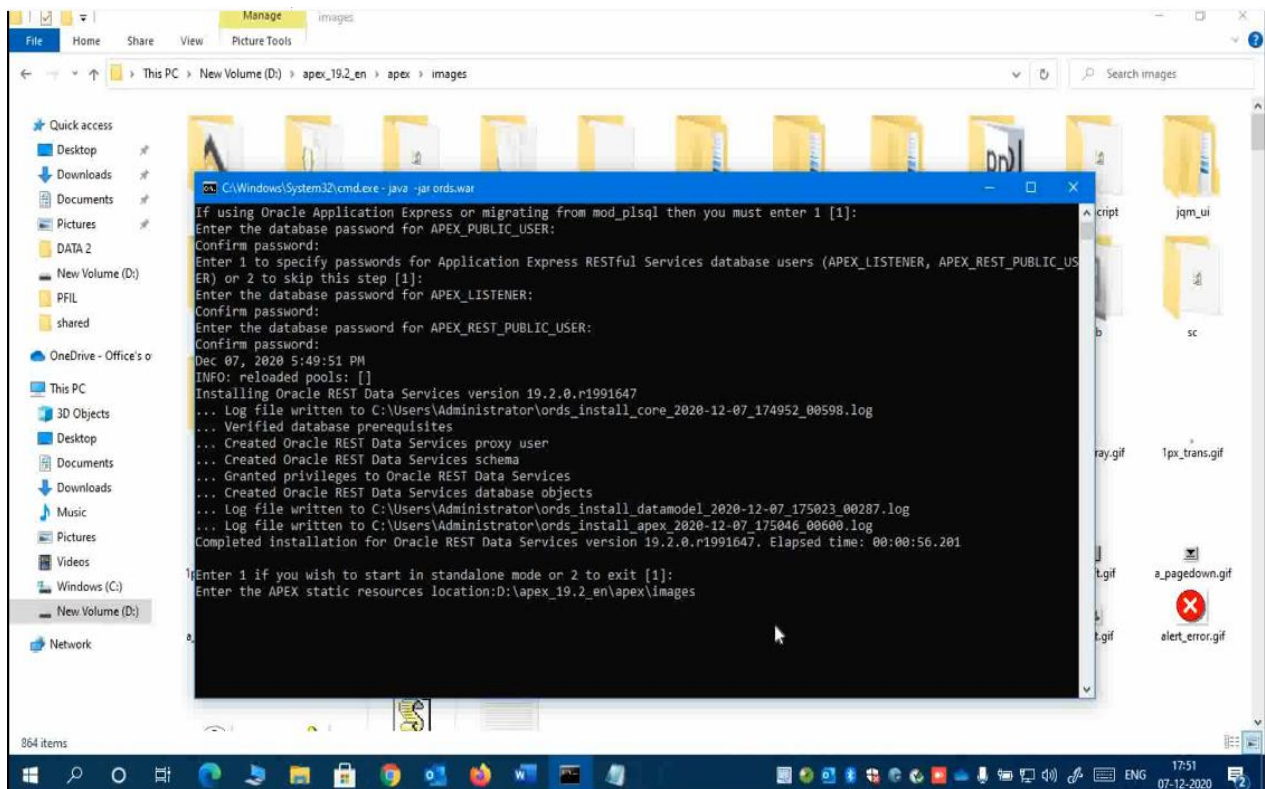


```
... Enter the database password for APEX_LISTENER:
Confirm password:
Enter the database password for APEX_REST_PUBLIC_USER:
Confirm password:
Enter a number to select a feature to enable:
  [1] SQL Developer Web (Enables all features)
  [2] REST Enabled SQL
  [3] Database API
  [4] REST Enabled SQL and Database API
  [5] None
Choose [1]:
2021-07-02T17:03:37.702Z INFO      reloaded pools: []
Installing Oracle REST Data Services version 21.1.1.r1162032
... Log file written to C:\Users\Administrator\ords_install_core_2021-07-02_2233
37_00936.log
... Verified database prerequisites
... Created Oracle REST Data Services proxy user
... Created Oracle REST Data Services schema
... Granted privileges to Oracle REST Data Services
```

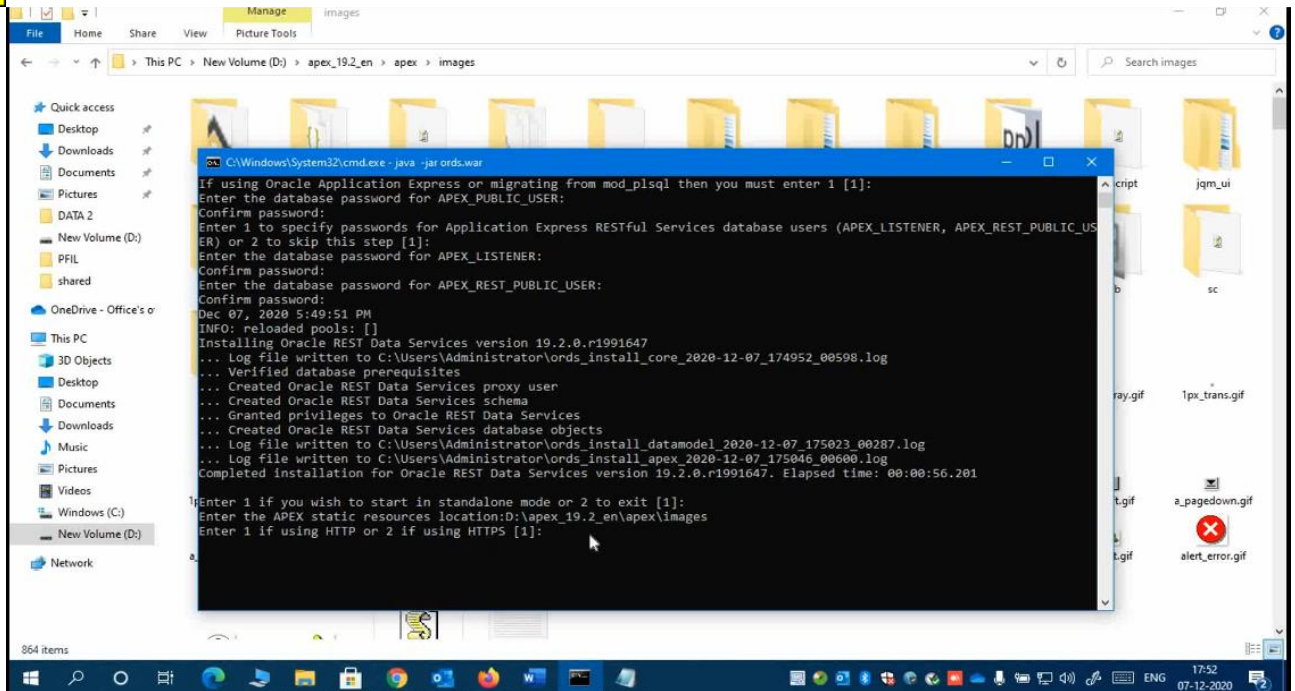
23. Now, let the installation continue, when the prompt return it will as for standalone mode or exit hit enter

24. Now it will ask for apex static resource location file for that navigate to D: drive apex_21.2_en folder apex folder and images folder inside it and copy the path generally it would look like:

D:\apex_21.2_en\apex\images

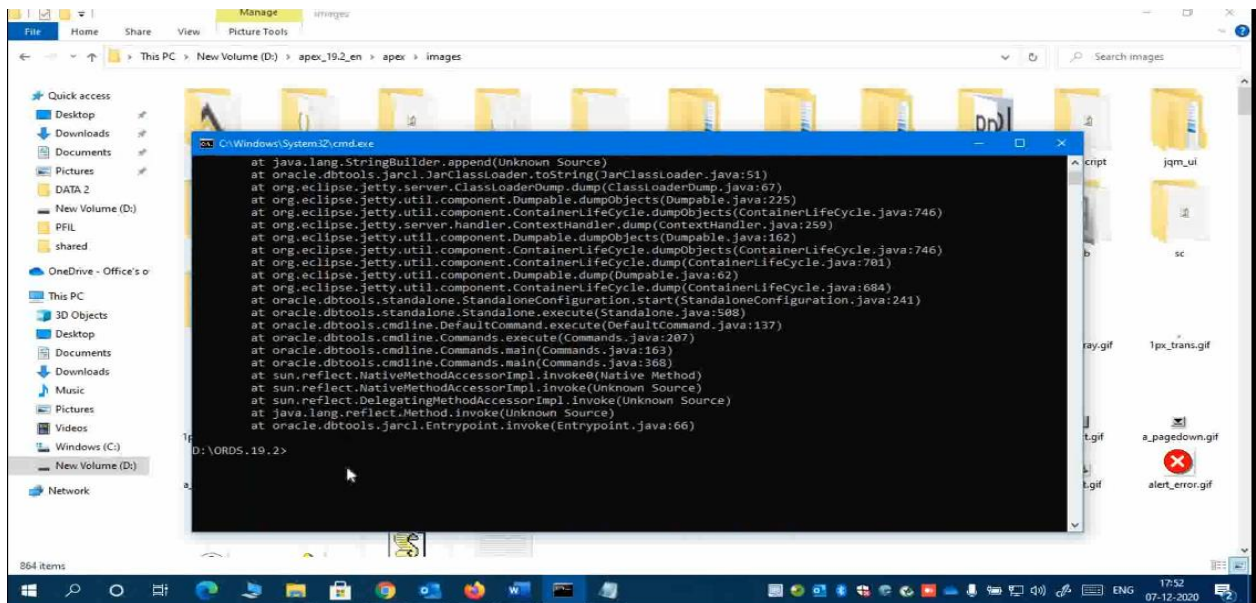


Then for http or https hit enter





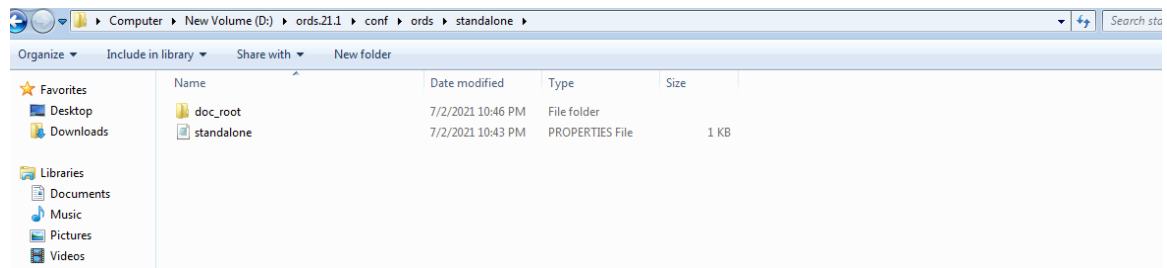
25. Now hit enter when the command running is completed exit the command prompt



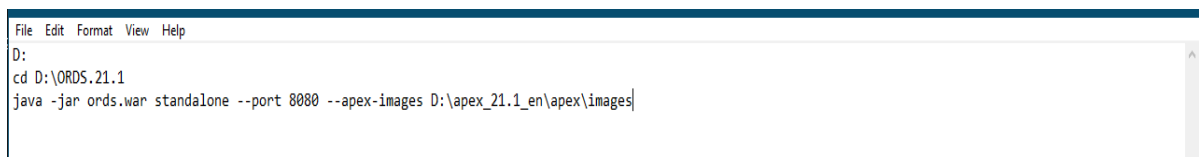
26. Exit cmd on this state

27. Now go to D drive open folder ORDS.21.2 and open conf folder then open ords then open standalone folder

28. Create folder doc_root in the same folder where standalone.properties file is present as shown below



29. Now, in the apex folder find the batch file PMSSTART.BAT and edit with notepad and it should look like below, if it does not look like below change as shown in screenshot below





30. Now open the sqlplus shortcut created on desktop while installing the apex 21.2 and for username – sys as sysdba and password as hit enter

31. Now run the below command as follows:

```
alter user APEX_210100 identified by Password$1234 account unlock;  
alter user APEX_REST_PUBLIC_USER identified by Password$1234 account unlock;  
alter user APEX_INSTANCE_ADMIN_USER identified by Password$1234 account unlock;  
alter user APEX_PUBLIC_USER identified by Password$1234 account unlock;  
alter user APEX_LISTENER identified by Password$1234 account unlock;
```

32. Now, run PMSSTART.BAT file as administrator and the window will look like:

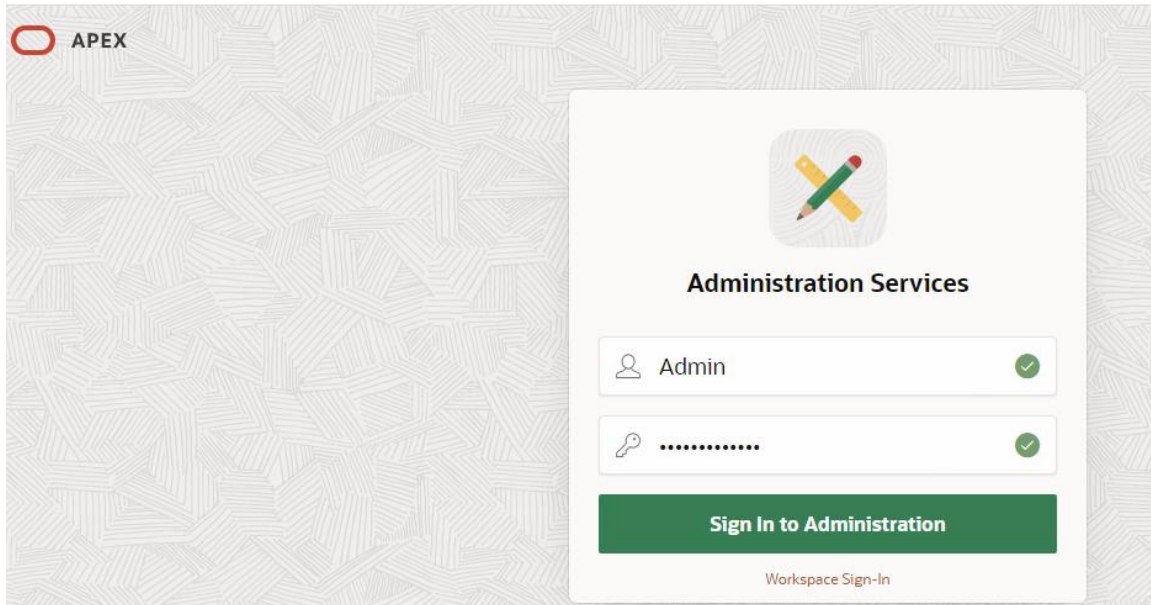
```
D:\>D:  
D:\>cd D:\ORDS.21.1  
D:\ords.21.1>java -jar ords.war standalone --port 9191 --apex-images D:\apex_21.2\en\apex\images  
2021-07-02T17:26:02.522Z INFO HTTP and HTTP/2 cleartext listening on http://localhost:9191  
2021-07-02T17:26:02.663Z INFO The document root is serving static resources located in: D:\ords.21.1\conf\ords\standalone\doc_root  
2021-07-02T17:26:05.913Z INFO Configuration properties for: !apex!!  
database.api.enabled=true  
db.connectionType=basic  
db.hostname=localhost  
db.port=1521  
db.servicename=orcl  
feature.sdw=true  
restEnabledSql.active=true  
security.requestValidationFunction=www_flow_epg_include_modules.authorize  
security.validationFunctionType=plsql  
db.password=*****  
db.username=APEX_PUBLIC_USER  
resource.templates.enabled=true
```

33. Now, open chrome and write <http://localhost:8080/ords> and apex 21.2 will be opened and you have successfully installed ORDS.21.2



37. Now, go to browser and run

Note: Apex 21.2 will not run without the installation of ords http://localhost:8080/ords/apex_admin

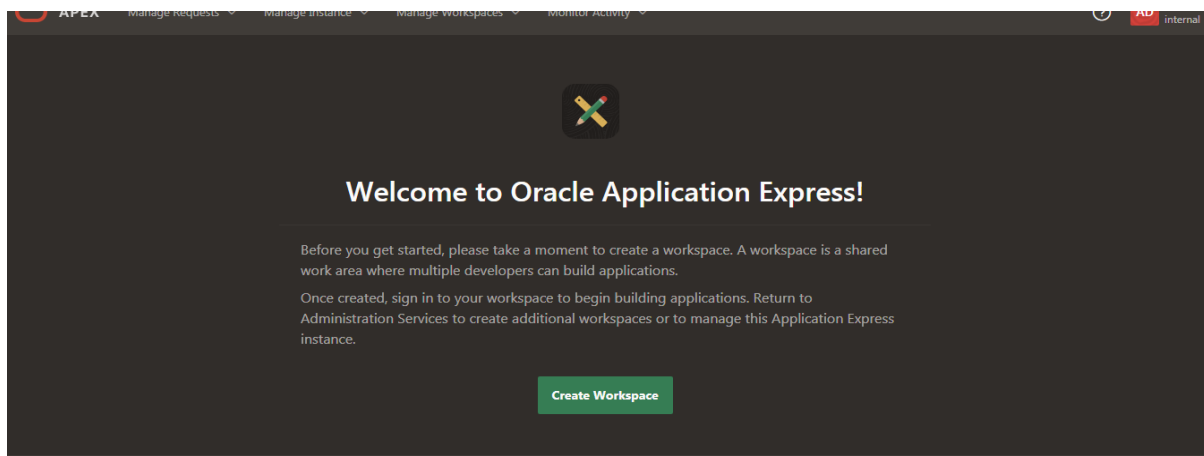


38. Enter below Credentials:

Username: Admin

Password: Password\$1234

After login into admin Account click Create Workspace Button:





39. Enter Workspace Name as “WSTS” and Workspace ID should be blank. Click on next button.

APEX Manage Requests Manage Instance Manage Workspaces Monitor Activity

Manage Workspaces Create Workspace

Create Workspace

Identify Workspace

Workspace Name WSTS

Workspace ID

Workspace Description

Tasks

- Create Multiple Workspaces

Cancel Next >

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40. Select ‘Yes’ for Re-use existing Schema.

Select schema name as ‘Makess’. Click Next.

Create Workspace

Identify Schema

Select whether or not the schema already exists. If the schema exists, select the schema from the list. If the schema does not exist, enter a name and password and choose the size of the associated tablespace to be created.

Re-use existing schema? Yes

Schema Name MAKESS

Schema Password

Space Quota (MB) 100

Information on Schema

Account Status	OPEN
Default Tablespace	MAKESS
Suggested Additional Privileges	CREATE DIMENSION, CREATE JOB, CREATE MATERIALIZED VIEW, CREATE SYNONYM

< Cancel Next >

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41. Enter Username as ADMIN and password as Password\$1234 and email ID as your desired email ID then press Next button.



APEX Manage Requests Manage Instance Manage Workspaces Monitor Activity ? AD admin internal

Manage Workspaces Create Workspace

Create Workspace

Identify Administrator

Administrator Username Admin

Administrator Password

First Name

Last Name

Email kunalkishor@essindia.com

< Cancel Next >

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42. Check all the Workspace Information and Press **Create Workspace** button.

Create Workspace

Confirm Request

Workspace Information

Workspace name WSTS

Workspace ID System Assigned

Description

Administrator Information

User name Admin

E-Mail kunalkishor@essindia.com

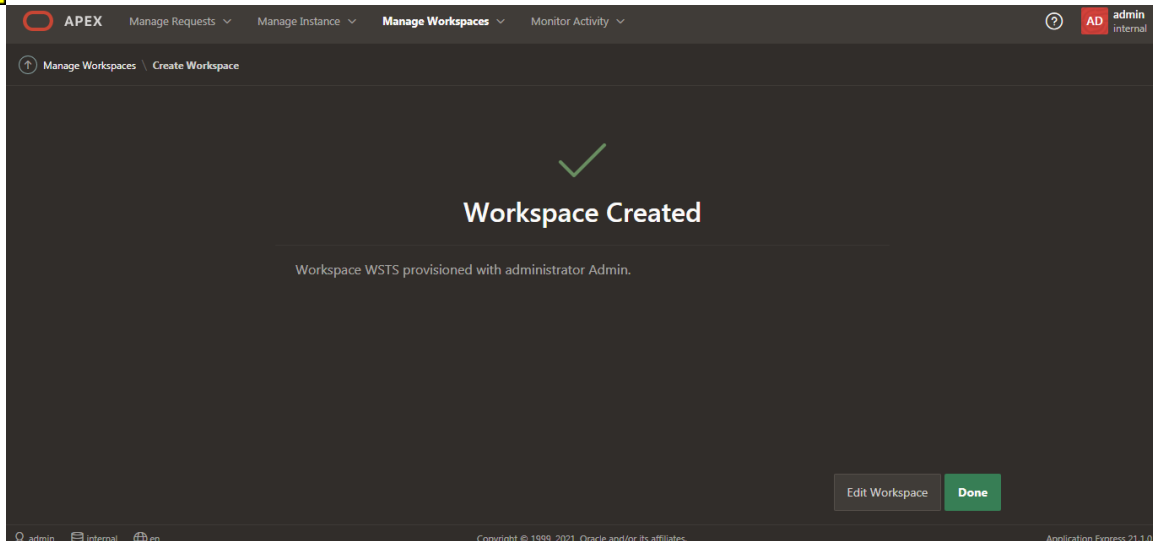
Database Schema Information

Reuse Existing Schema Yes

Schema Name MAKESS

< Cancel Create Workspace

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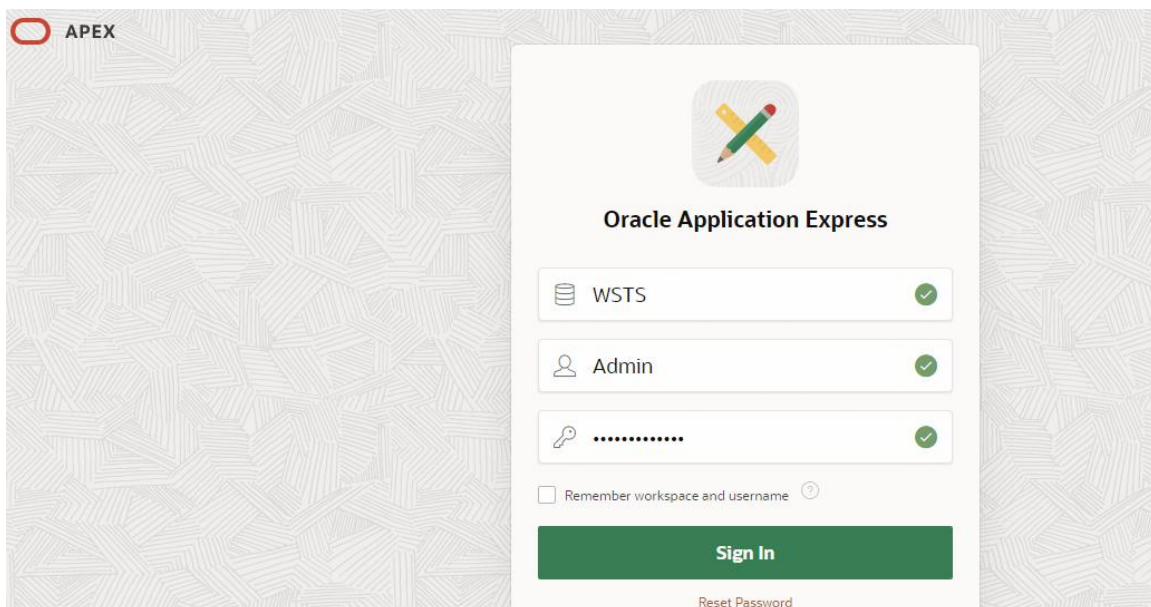


43. Click on **Done** button after successful creation of Workspace.

Now that we have successfully created the Workspace now it is time to login the Workspace so that we can develop applications.

44. Login using URL <http://localhost:8080/ords/> (Change Port according to installation)

45. Enter Workspace name i.e., “WSTS”, Username (Admin) and password (Password\$1234) for login Apex application and change the password for admin user as per your preference.



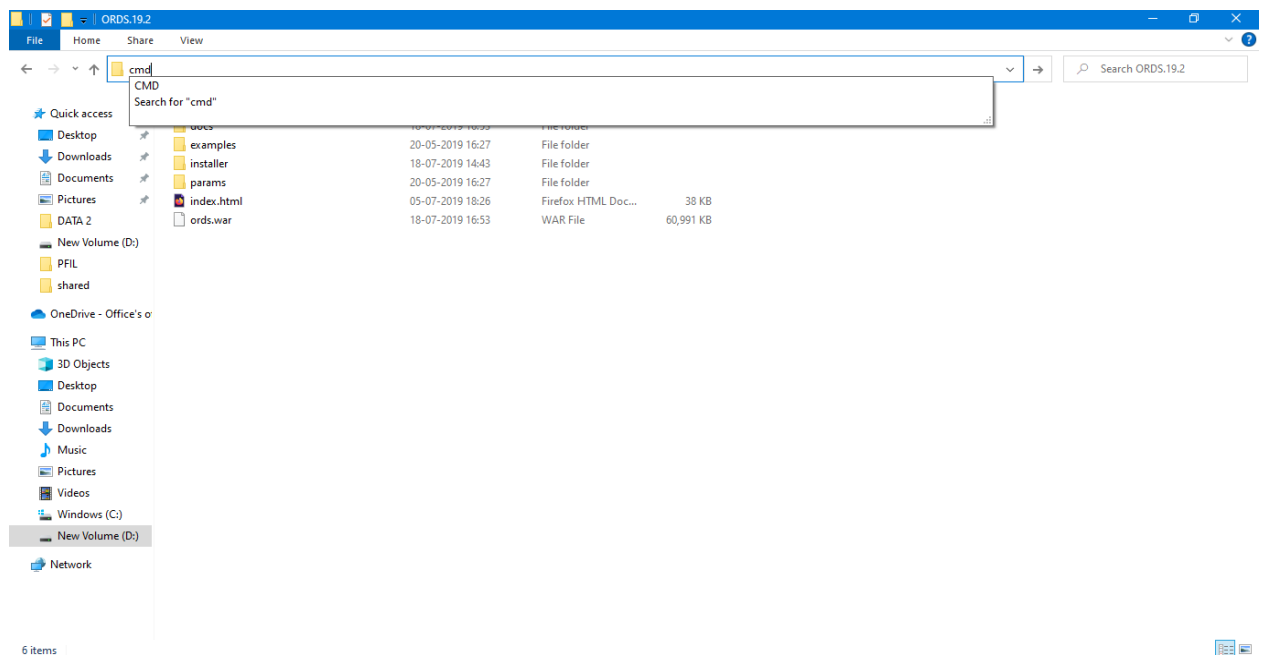
Now, you have successfully installed Oracle APEX 21.2 on your Oracle Database.



NOTE: - If you have database and application server on the different machines then please copy this “apex_21.2_en” folder in D drive from database server to D drive in the application server and then install ORDS on the application server to connect apex from database server to application server, please refer ORDS installation document for installation and configuration on application server.

Un-installation of ORDS 21.2: -

1. Go to D drive and go to ORDS.21.2 folder open the folder and in the address bar type cmd and hit enter and command prompt will open like this:



2. Now, in the command prompt window type
java -jar ords.war uninstall

NOTE: - For this version of ORDS jre 8 minimum is required, if on running above command JAVA is not detected make sure java is set in environment variable and if then also it does not work, then instead of the above command run:

“JAVA_PATH” -jar ords.war uninstall
(in JAVA_PATH enter java bin folder path)

3. Now, follow the same as installation and once the execution is complete delete the ORDS.21.2 folder from D drive for next installation or anyway.



HOW TO UNINSTALL 12C forms and 19c DB :-

Go to D:\db_home\deinstall \deinstall.bat

Click on “deinstall.bat” file.

Command prompt opens.

Follow the Instruction

Go to D:\app and Delete the “Administrator” Folder

Now go to Start→ All Programs→ Oracle→ Oracle Home1→ Uninstall Oracle Software (Press Enter).

Click on Uninstall

Press Next.

Press De-install.

Press Finish.

Restart the system.

Now again go to Start→ All Programs→ Oracle→ Oracle Home1→ Uninstall Oracle Software (Press Enter)

Press next

Press De-install

Press Next

Press Finish.

Restart the system.

Go to C:\ProgramData\Microsoft\Windows\Start Menu\Programs

Delete all oracle folders.

Go to D:\app\Oracle and delete all folders (except “JDK folder”, If you need it).

Go to C:\Program Files . Delete “Oracle folder”.

Restart the system.



LIST OF FREQUENT PROBLEMS:-

REP – 51002: Bind to report server <server name> failed.

SOLUTION:-

Go to D:\app and search for the file RWSERVER.CONF
Open all 4 files and make the following changes:-

BEFORE:-

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<discoveryService
  xmlns="http://xmlns.oracle.com/reports/network"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  >
  <multicast channel="228.5.6.7" port="14021" timeout="1000"/>
  <!--namingService name="Cos" host="%NAMING_HOST%" port="%
NAMING_PORT%"-->
</discoveryService>
```

AFTER:-

Comment

```
<!--multicast channel="228.5.6.7" port="14021" timeout="1000"/-->
```

And De-comment the below line and write the host name and port

```
<namingService name="Cos" host="%NAMING_HOST%" port="%NAMING_PORT%"/>
```

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<discoveryService
  xmlns="http://xmlns.oracle.com/reports/network"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  >
  <!--multicast channel="228.5.6.7" port="14021"
timeout="1000"/-->
  <namingService name="Cos" host="%NAMING_HOST%" port="%
NAMING_PORT%" />
</discoveryService>
```

Save all files and restart the server.